FIAP GRADUAÇÃO

TECNOLOGIA EM DESENVOLVIMENTO DE SISTEMAS

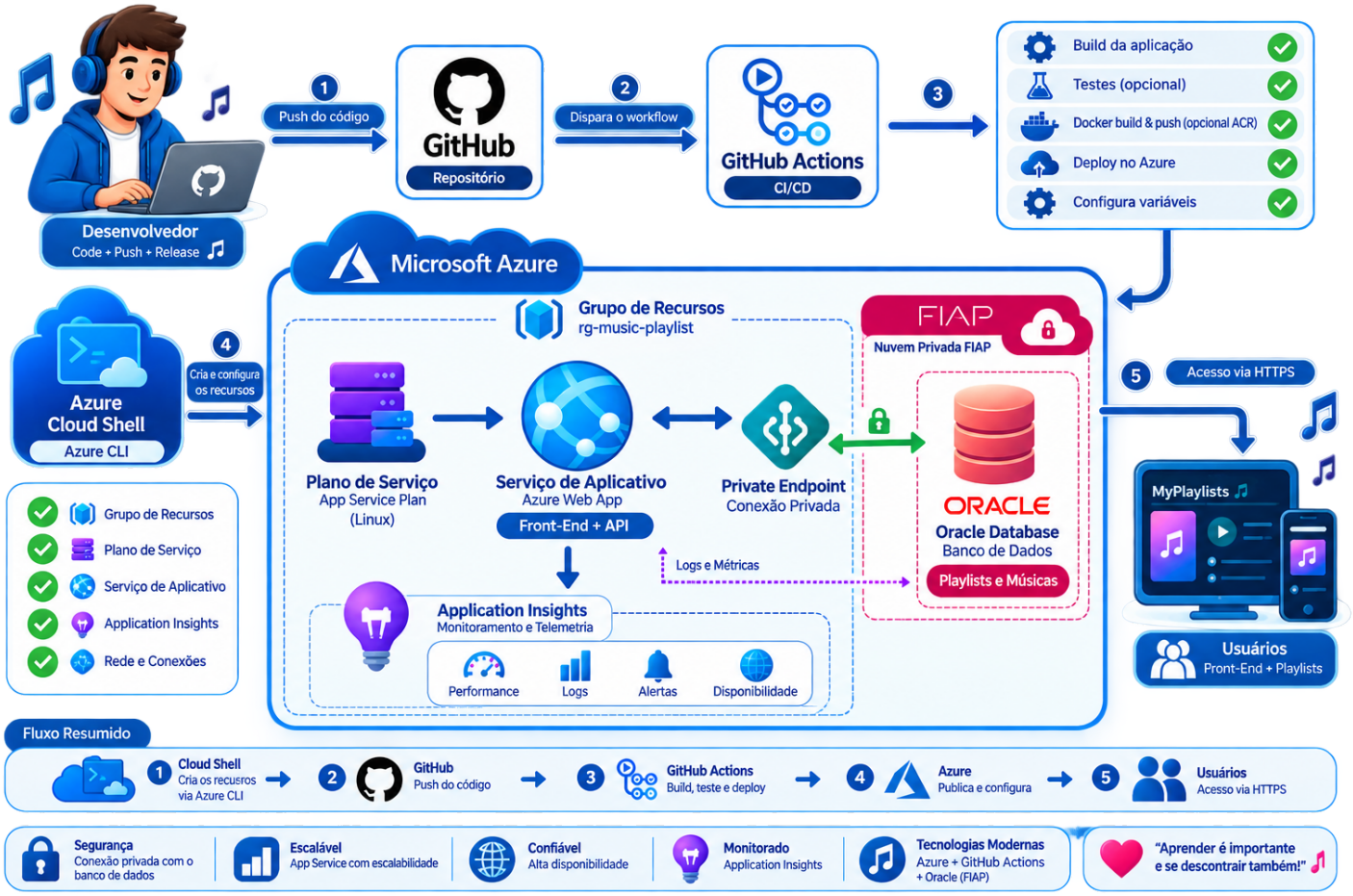
DevOps Tools & Cloud Computing

Azure Pipelines – Exercício Java x Oracle

PROF. João Menk profjoao.menk@fiap.com.br

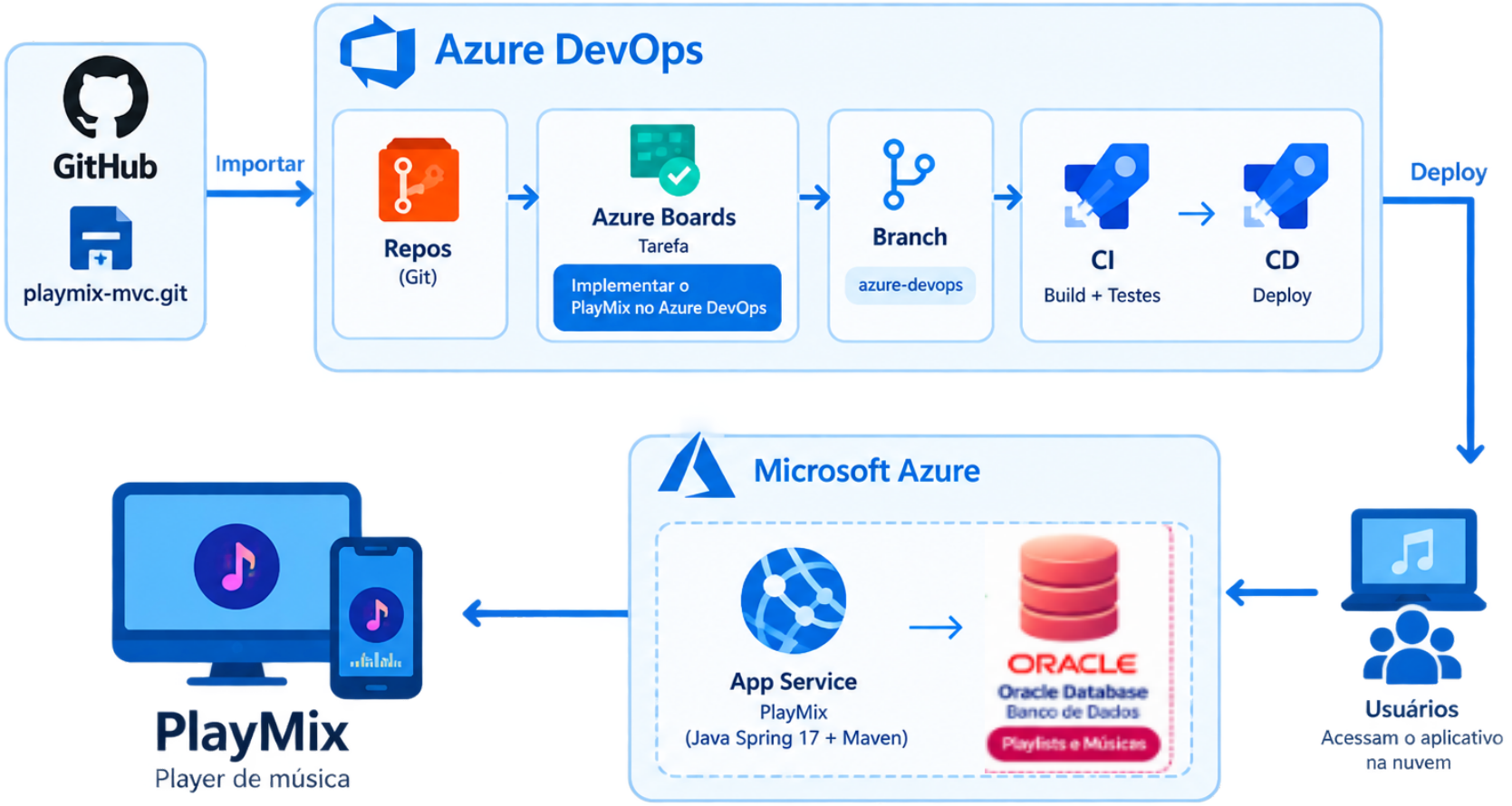
Azure Pipelines – Exercício Java x Oracle

Vamos realizar a implantação do App **PlayMix** que fizemos em nossa aula de Deploy de Serviços de Aplicativos



Azure Pipelines – Exercício Java x Oracle

Arquitetura macro do Projeto



Azure Pipelines – Exercício Java x Oracle

Vamos começar o Projeto abrindo o Cloud Shell no Portal da Azure para criar a infra necessária ao Projeto

No terminal Bash edite o Script **"Semana_19_02_criar-infra.sh"** fornecido pelo professor e altere seu RM, Usuário, Senha e sua Região

Salve o Script

Conceda o privilégio de execução nesse Script com o comando:

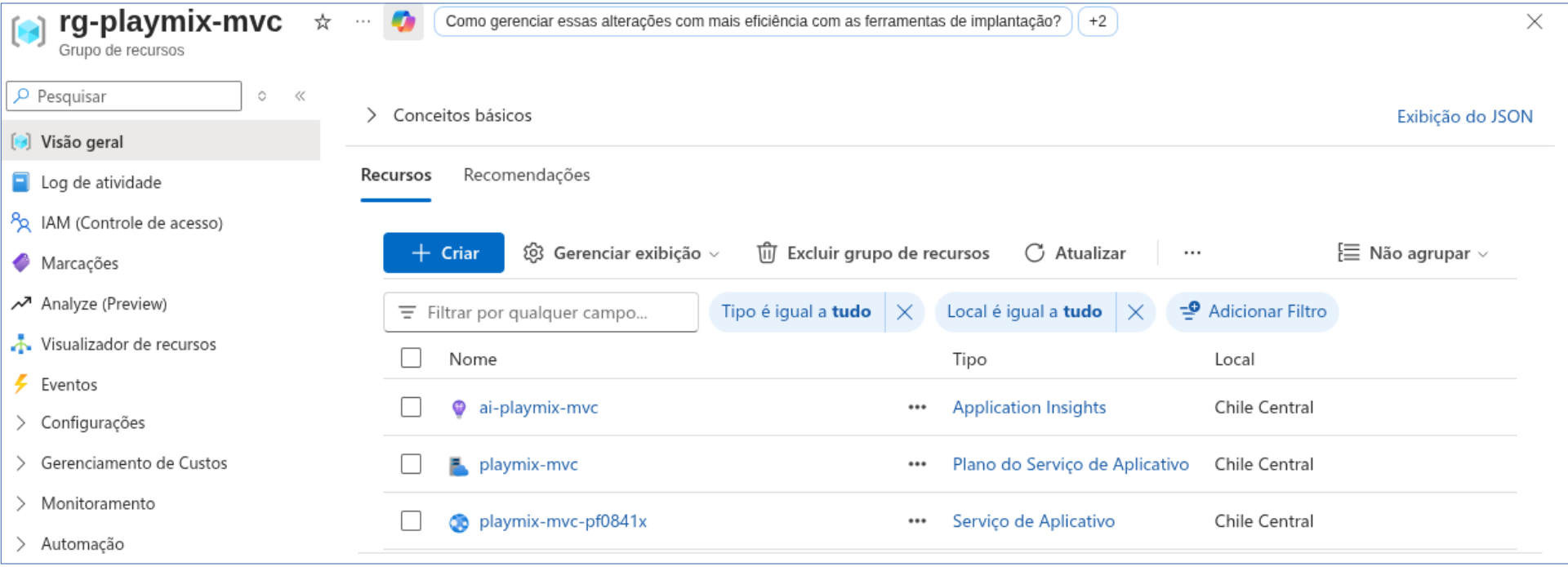
```
chmod +x Semana_19_02_criar-infra.sh
```

Execute o Script com o comando:

```
./Semana_19_02_criar-infra.sh
```

Azure Pipelines – Exercício Java x Oracle

Toda a Infra do Projeto criada, aguardando a Pipeline de CI/CD

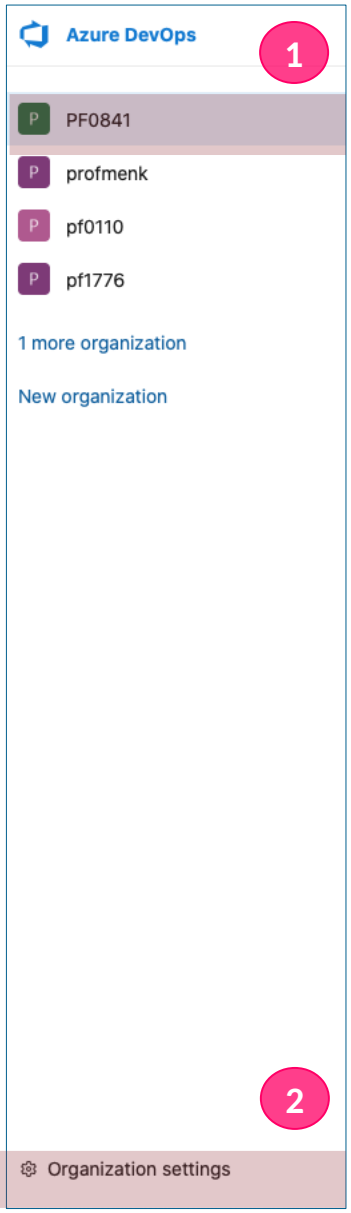


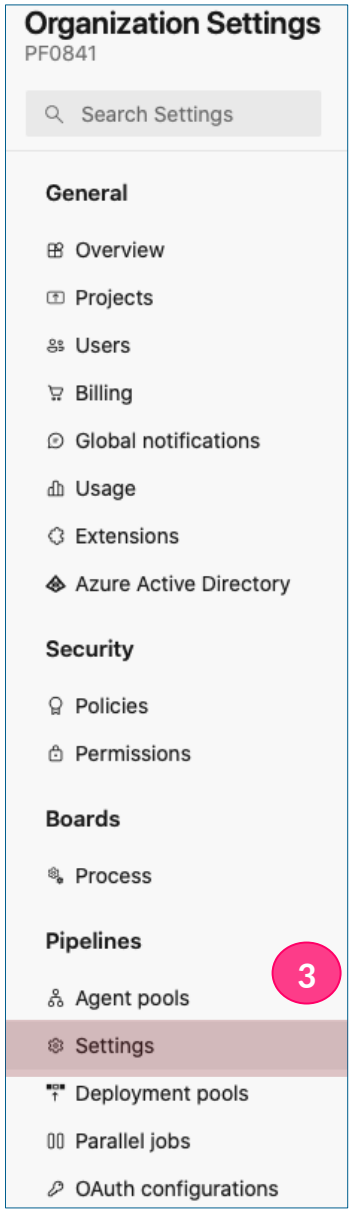
Azure Pipelines – Exercício Java x Oracle

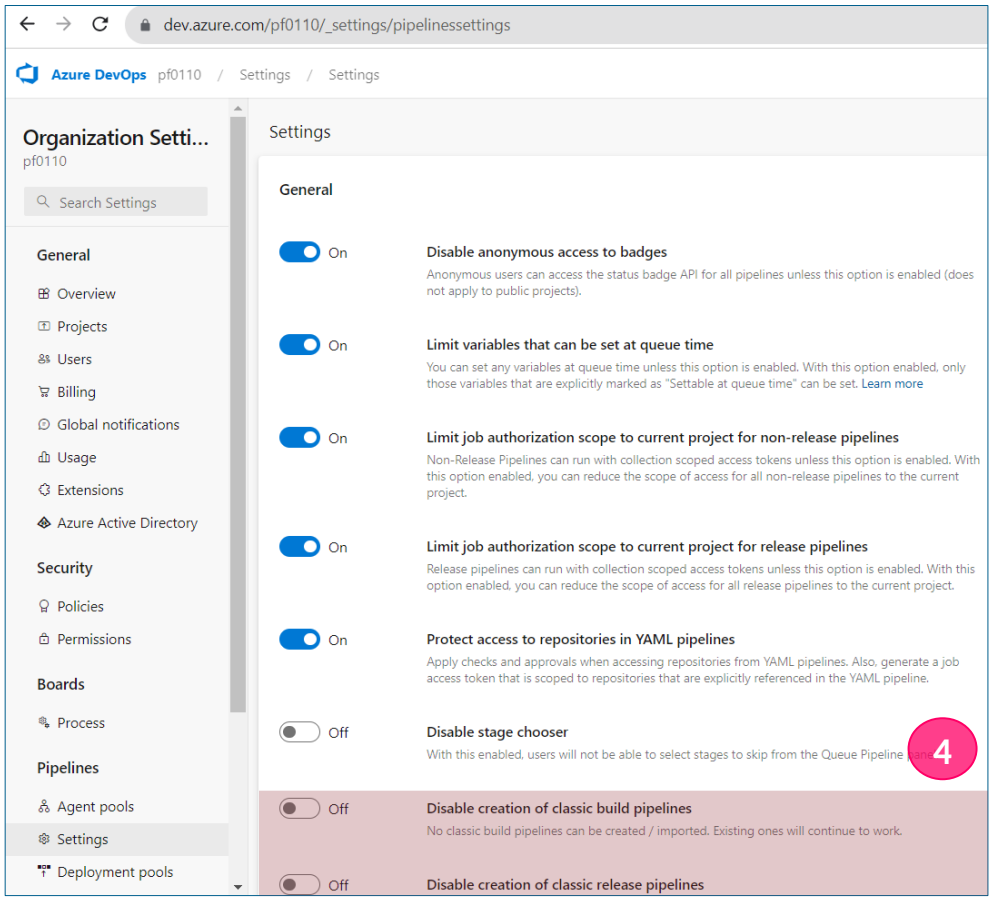
https://dev.azure.com/{suaOrganização}



Azure Pipelines – Exercício Java x Oracle





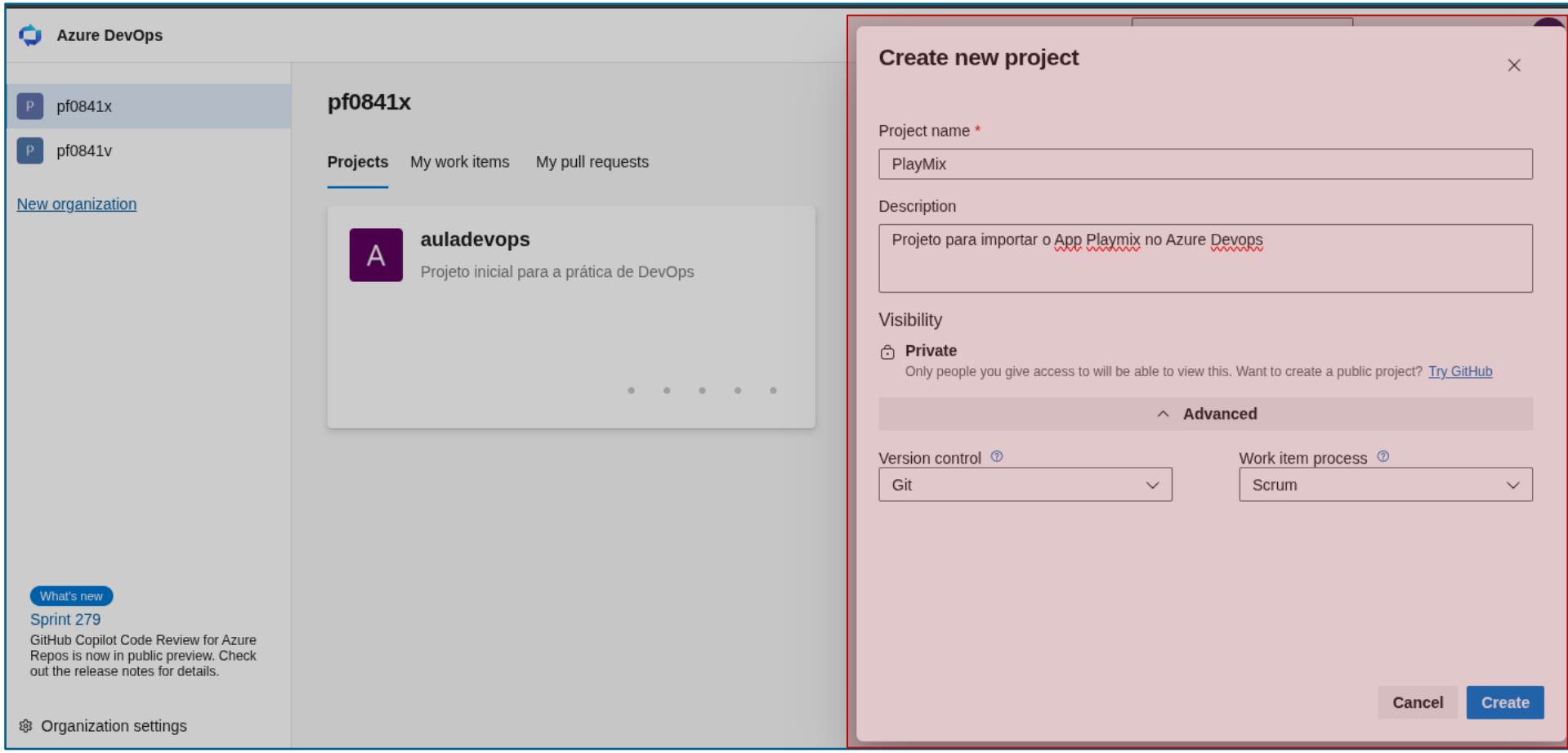


Deixe desligado



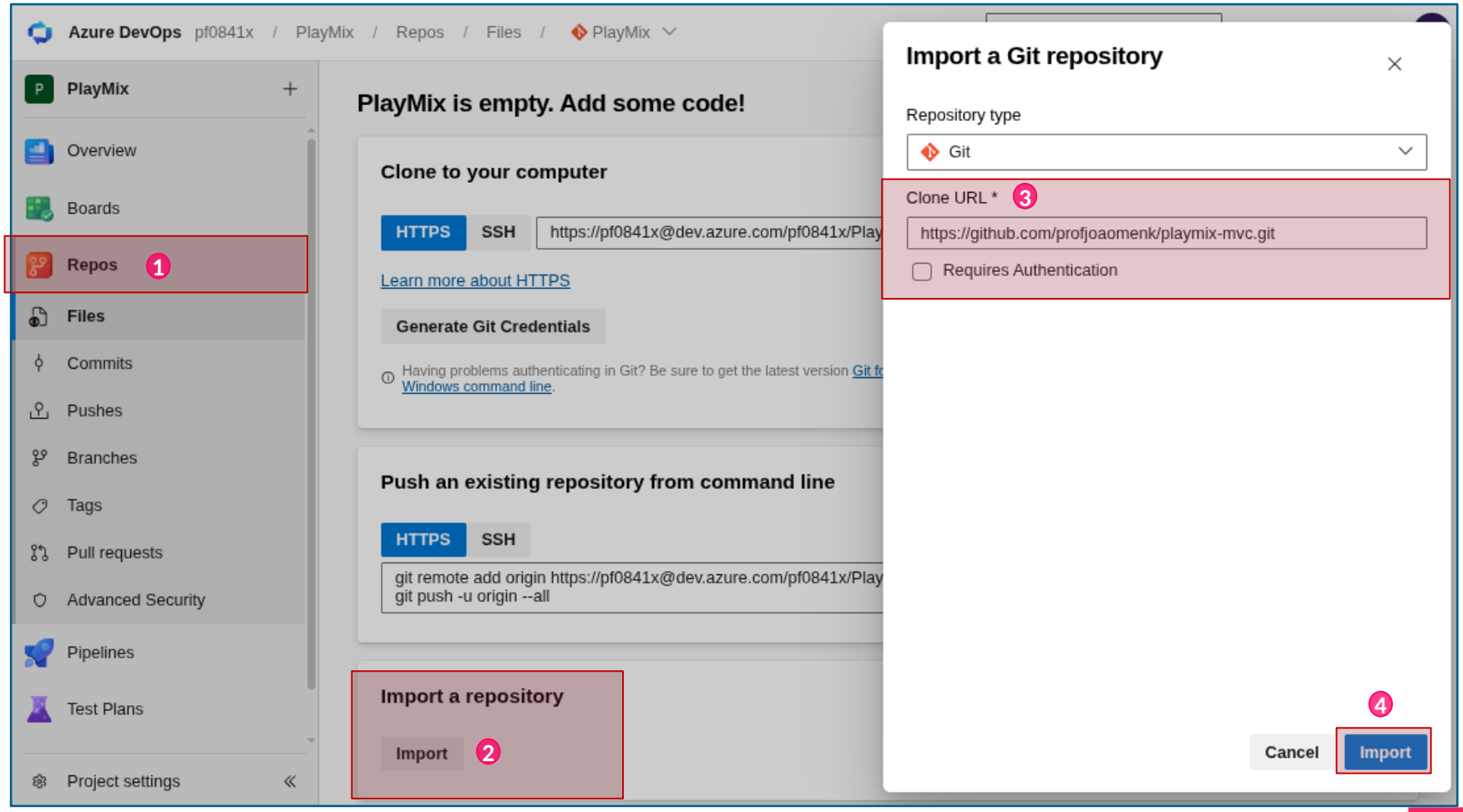
Azure Pipelines – Exercício Java x Oracle

Crie um Projeto com o nome **PlayMix**



Azure Pipelines – Exercício Java x Oracle

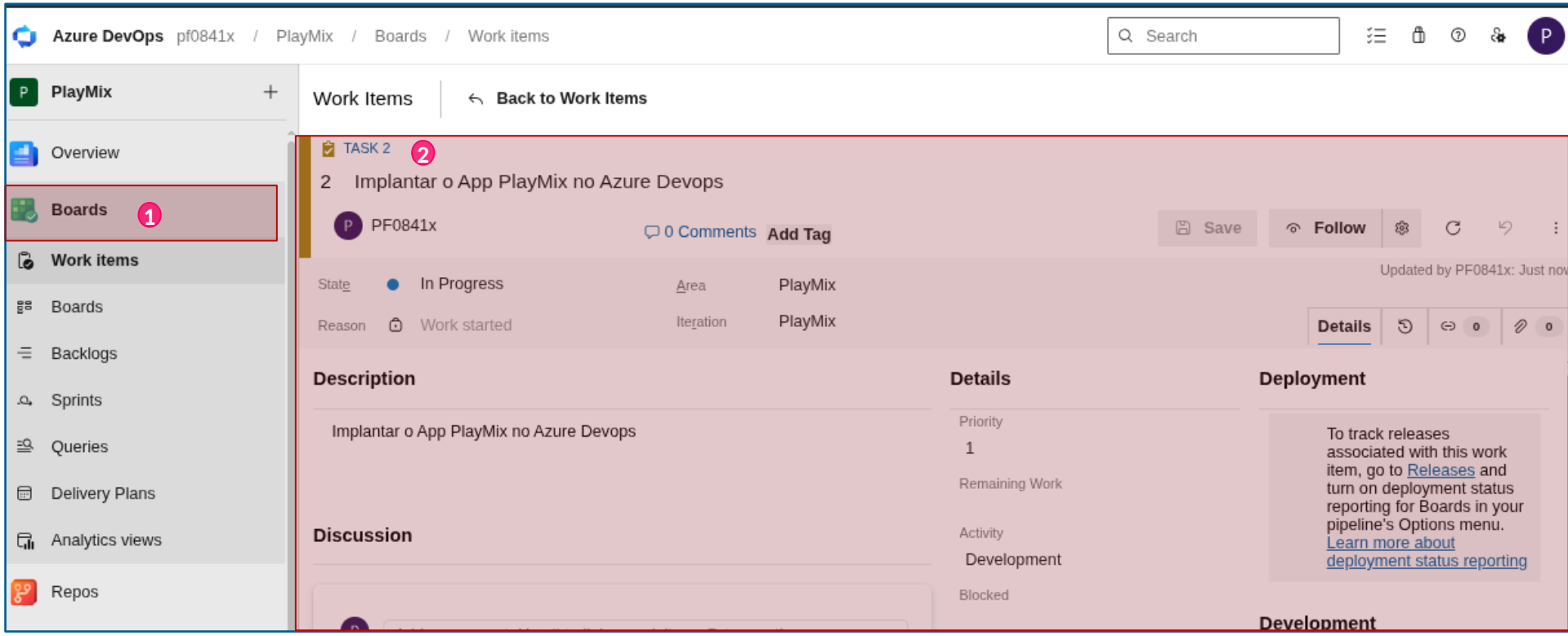
Importe o nosso Projeto do GitHub



https://github.com/profjoaomenk/playmix-mvc.git

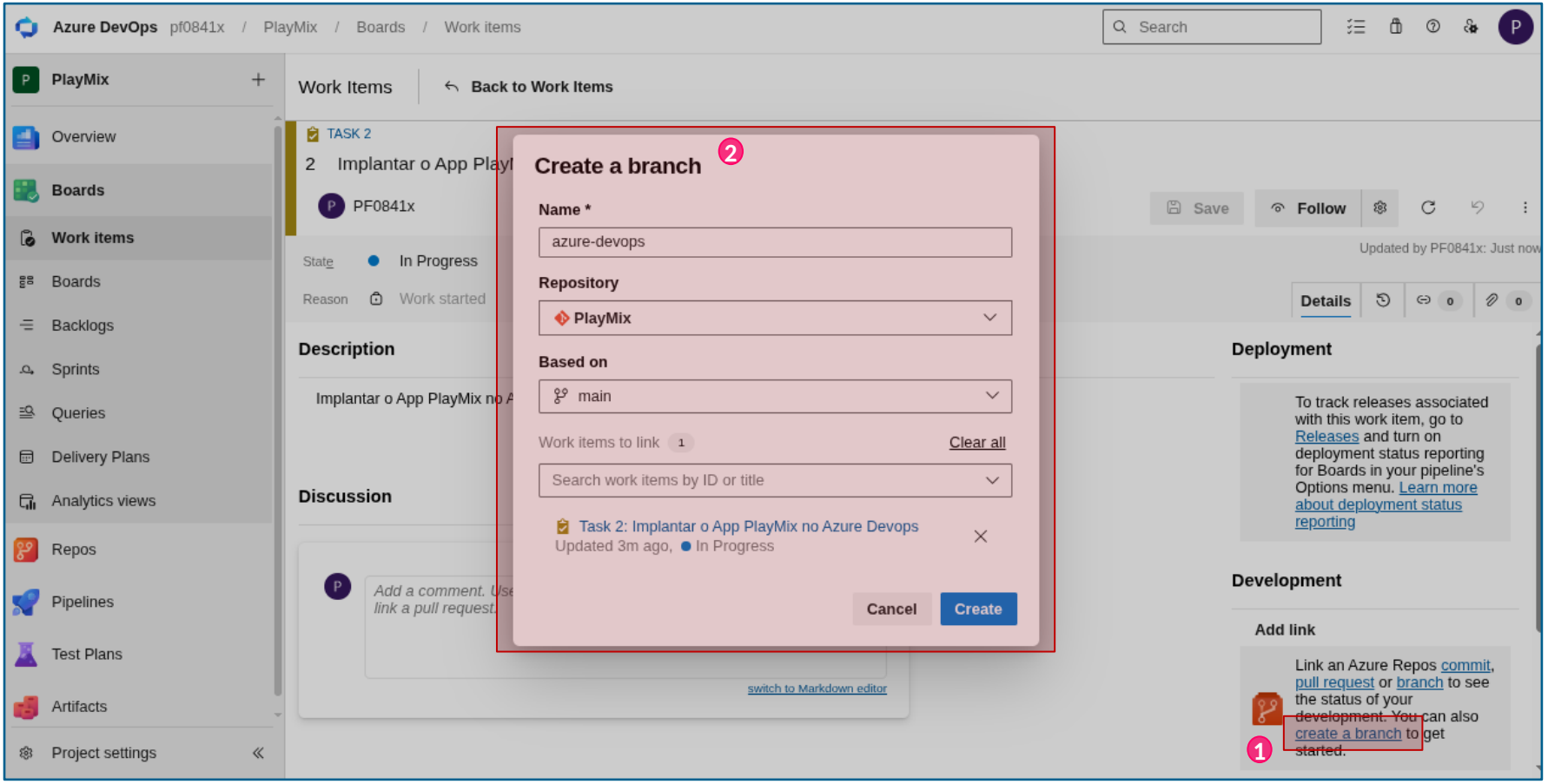
Azure Pipelines – Exercício Java x Oracle

Agora Vamos criar uma Tarefa no **Azure Board**s para termos o ciclo completo de vida de nosso Aplicativo



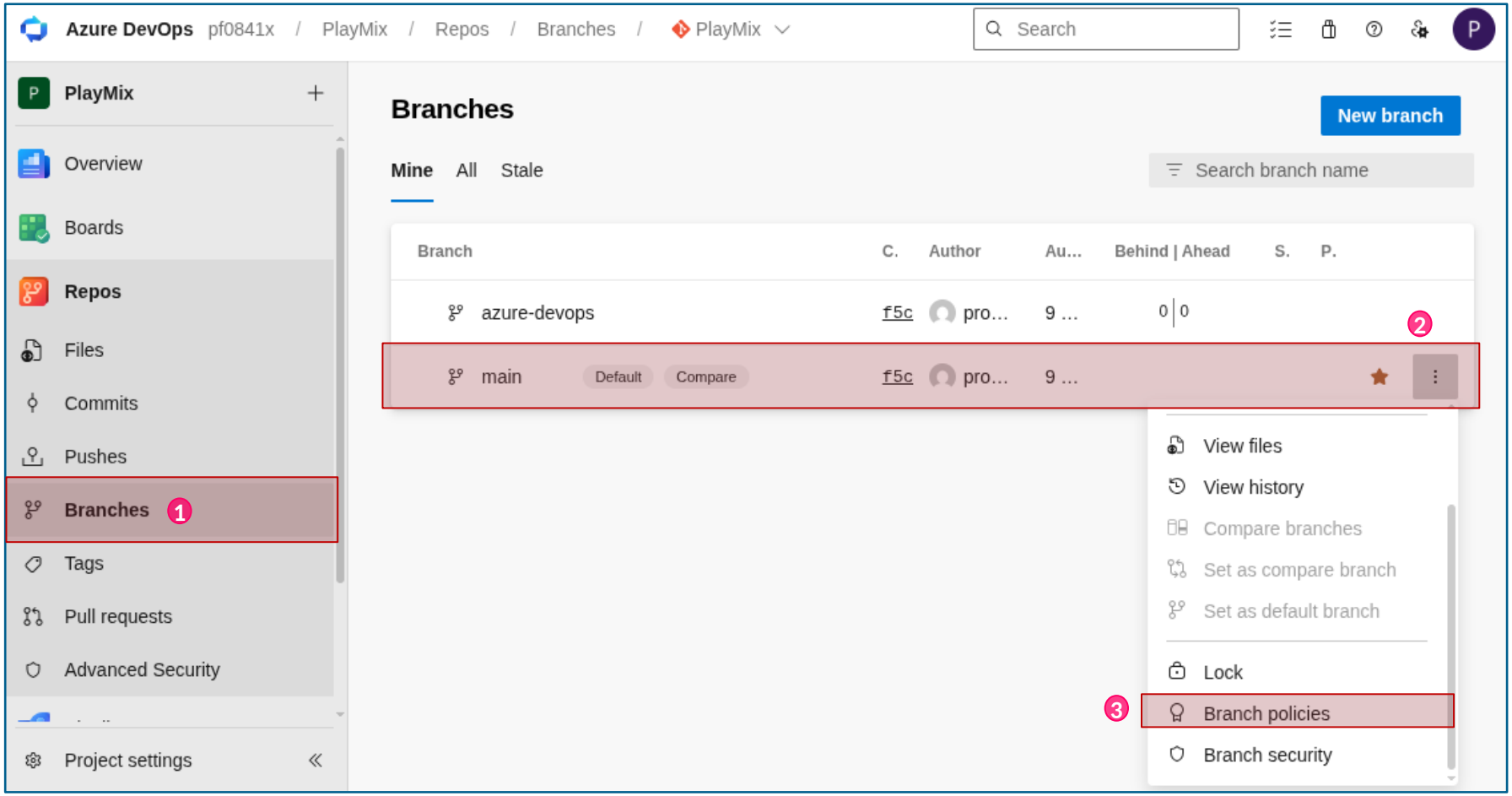
Azure Pipelines – Exercício Java x Oracle

Da Tarefa criada vamos **criar uma Branch** para realizar a solicitação

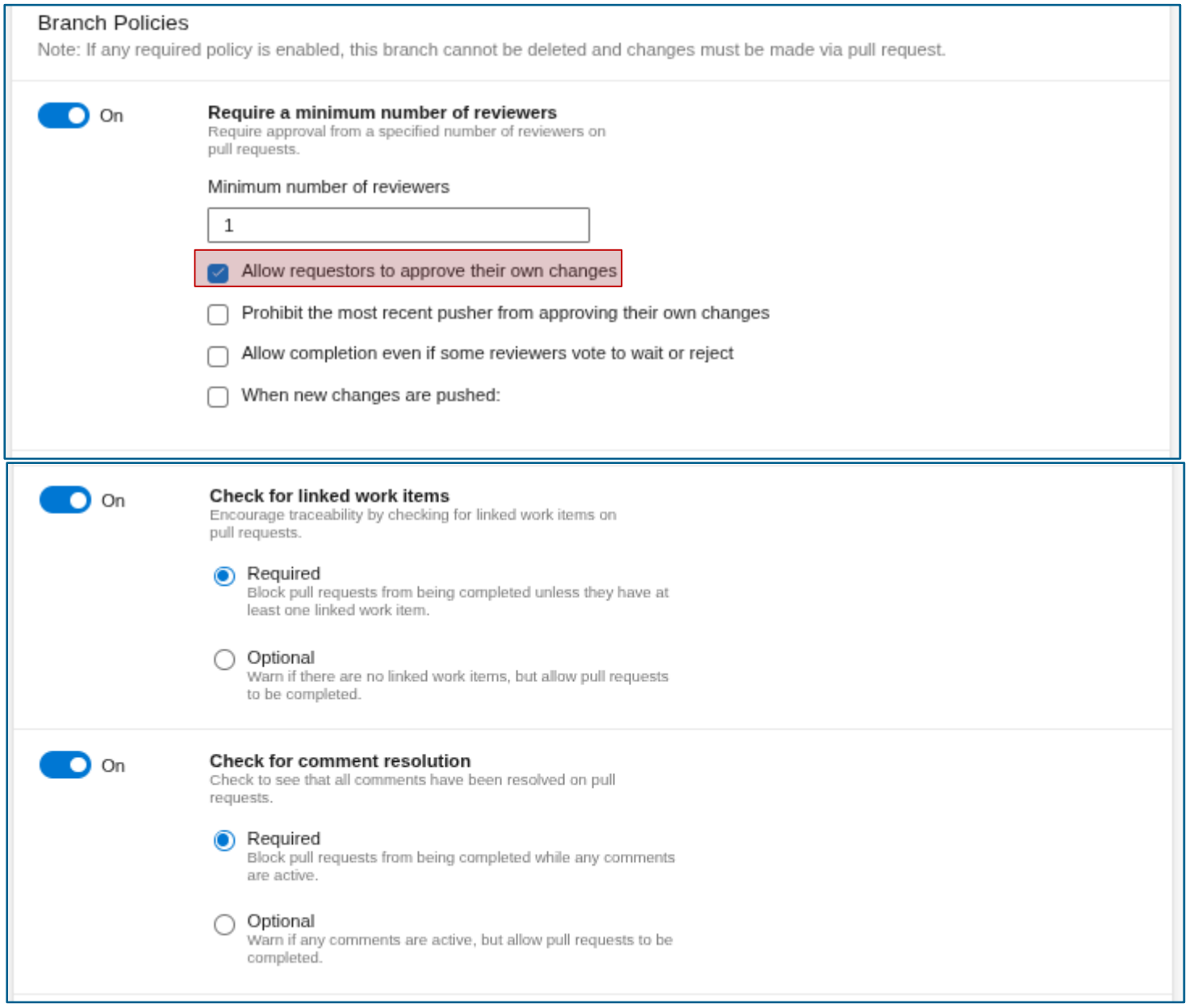


Azure Pipelines – Exercício Java x Oracle

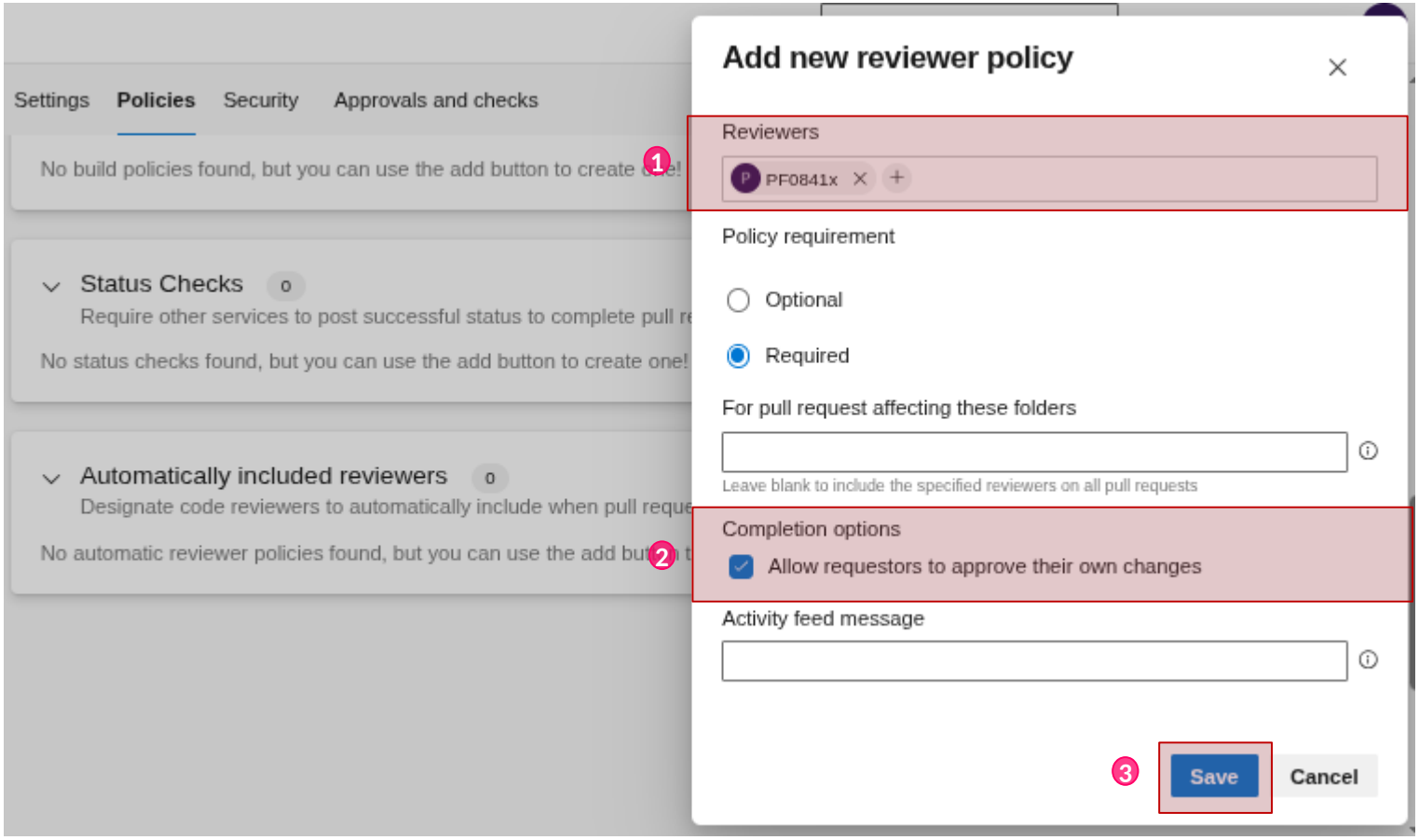
Agora vamos **proteger a nossa Branch Main**



Azure Pipelines – Exercício Java x Oracle

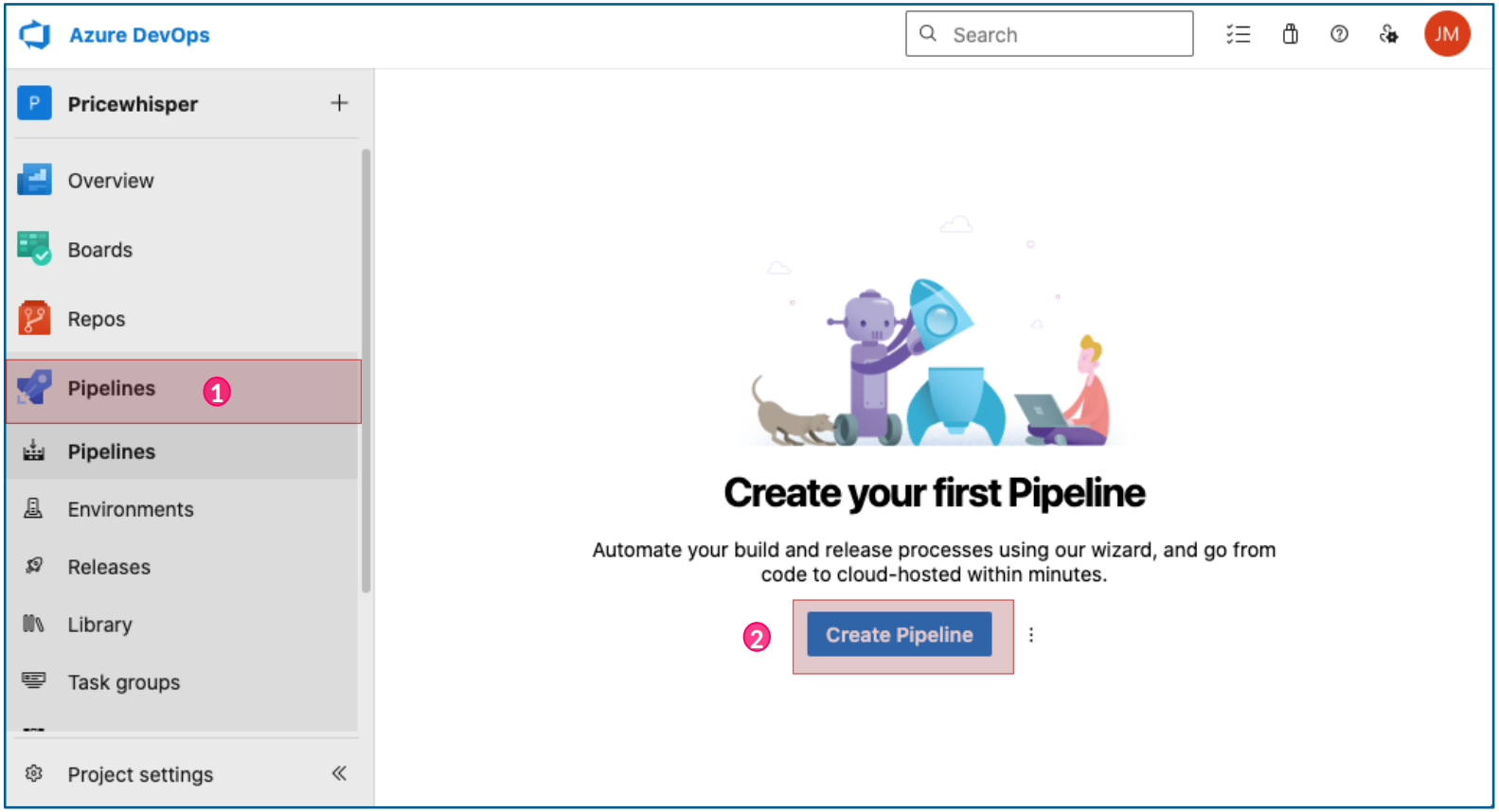


Azure Pipelines – Exercício Java x Oracle



Azure Pipelines – Exercício Java x Oracle

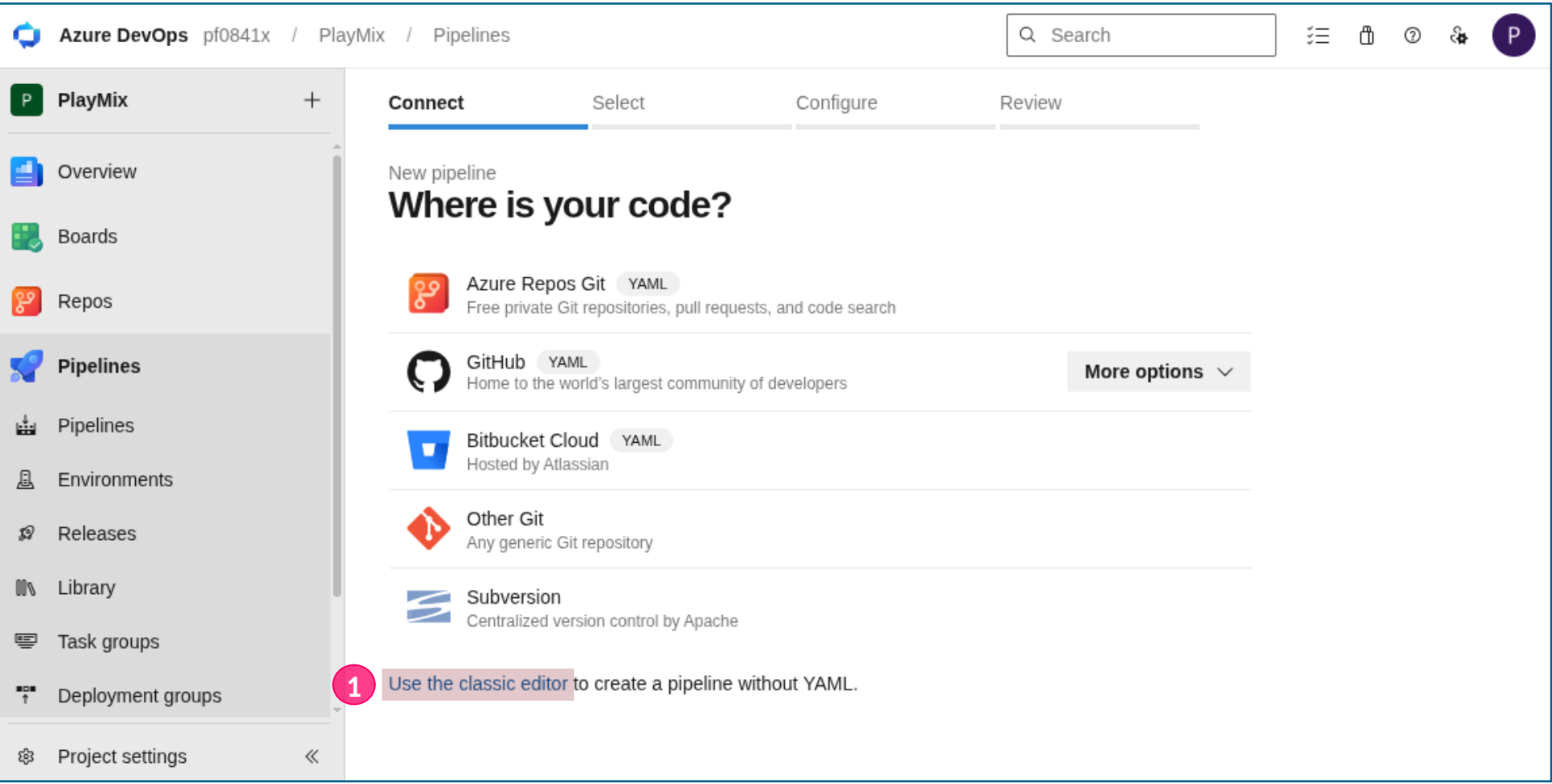
A integração do Boards com o Repos foi efetuada, começamos agora a criar a nossa Pipeline de CI



Azure Pipelines – Exercício Java x Oracle

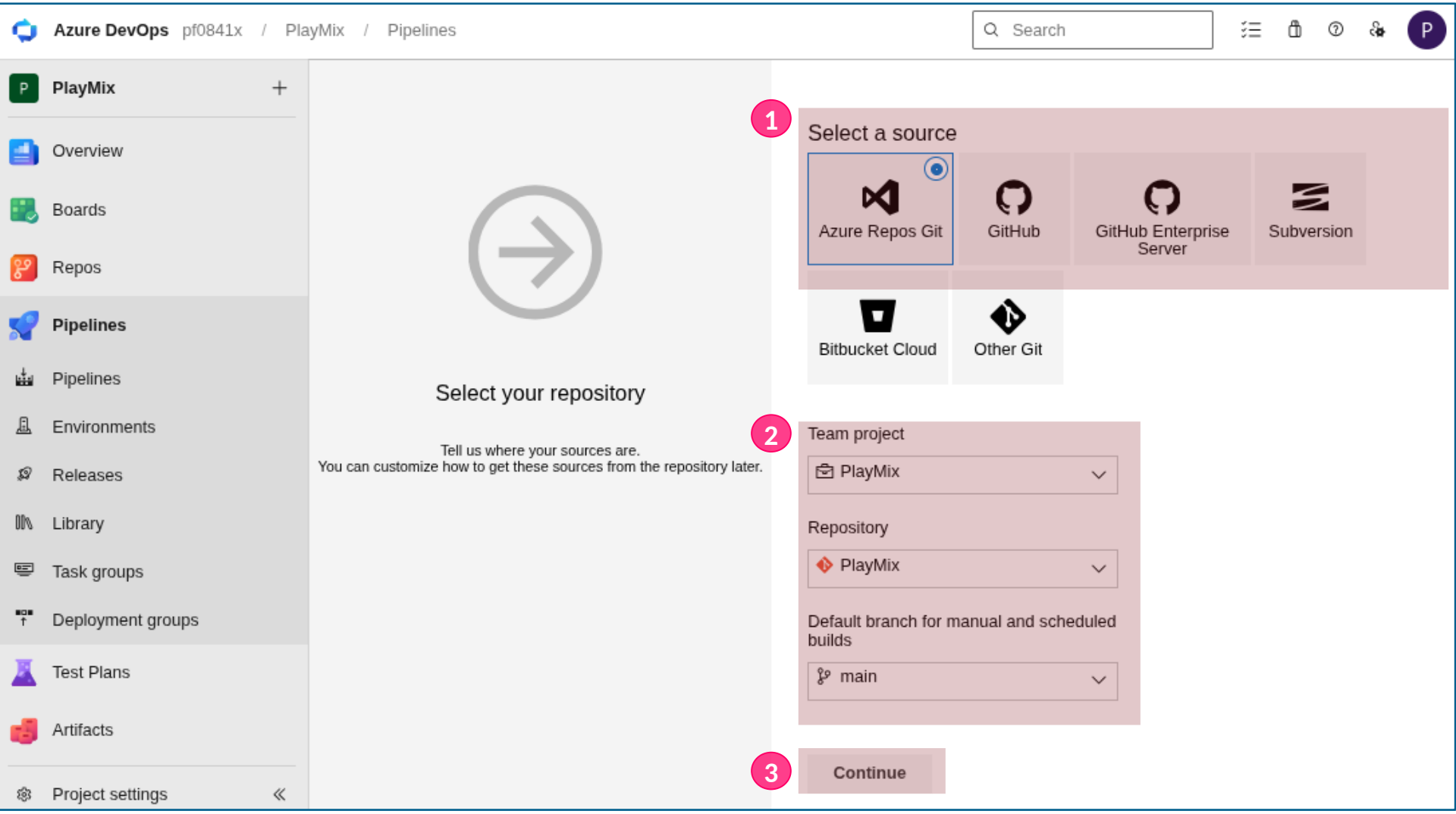
O primeiro passo é conectar a um **Sistema de Controle de Versão**

Em nossa primeira experiência iremos utilizar a *Interface Clássica* para criar nossa Pipeline. Clique em **Use the classic editor**



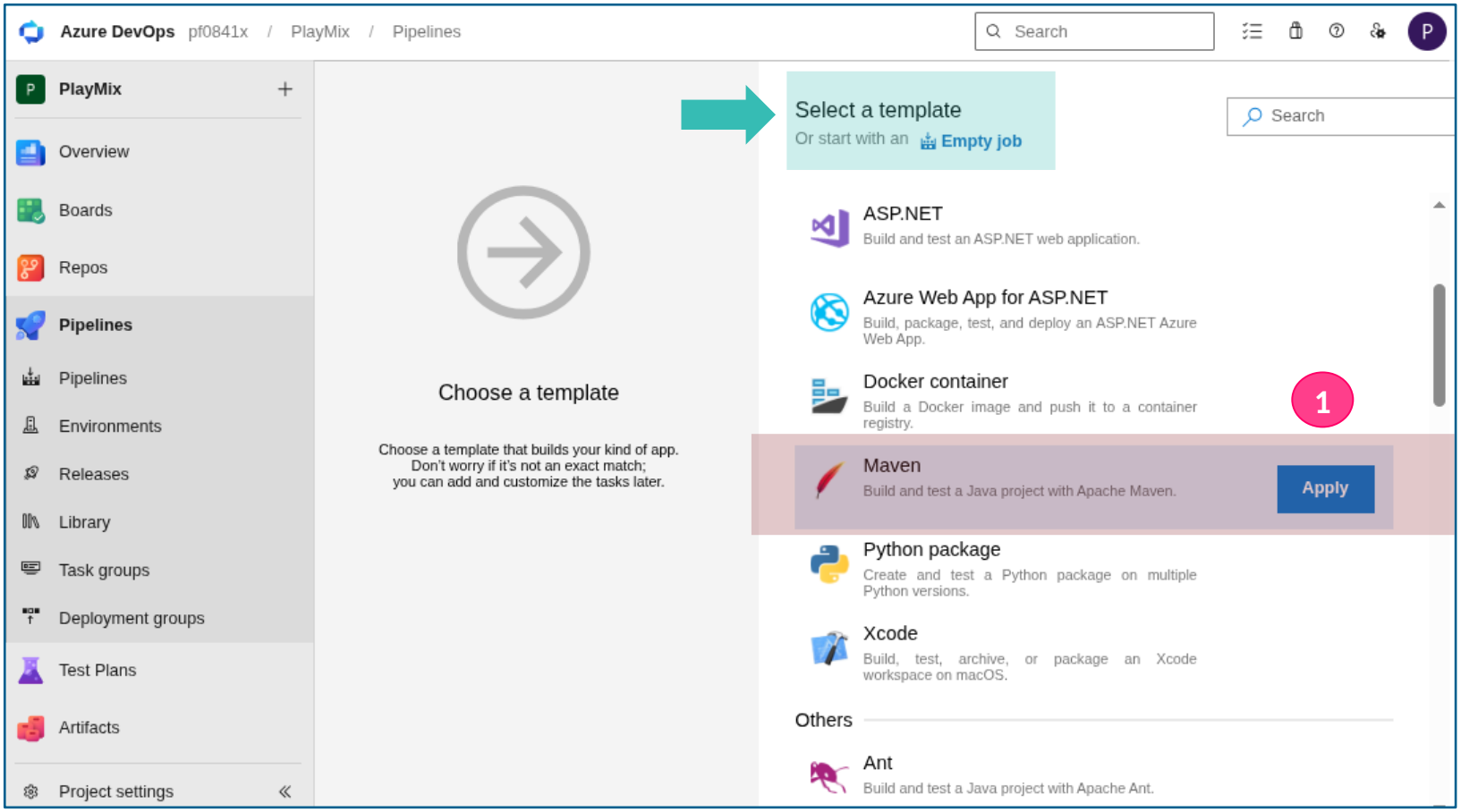
Azure Pipelines – Exercício Java x Oracle

Selecione o **local** que seu código está, informe o **Projeto**, o **Repositório**, e a **Branch** que irá trabalhar



Azure Pipelines – Exercício Java x Oracle

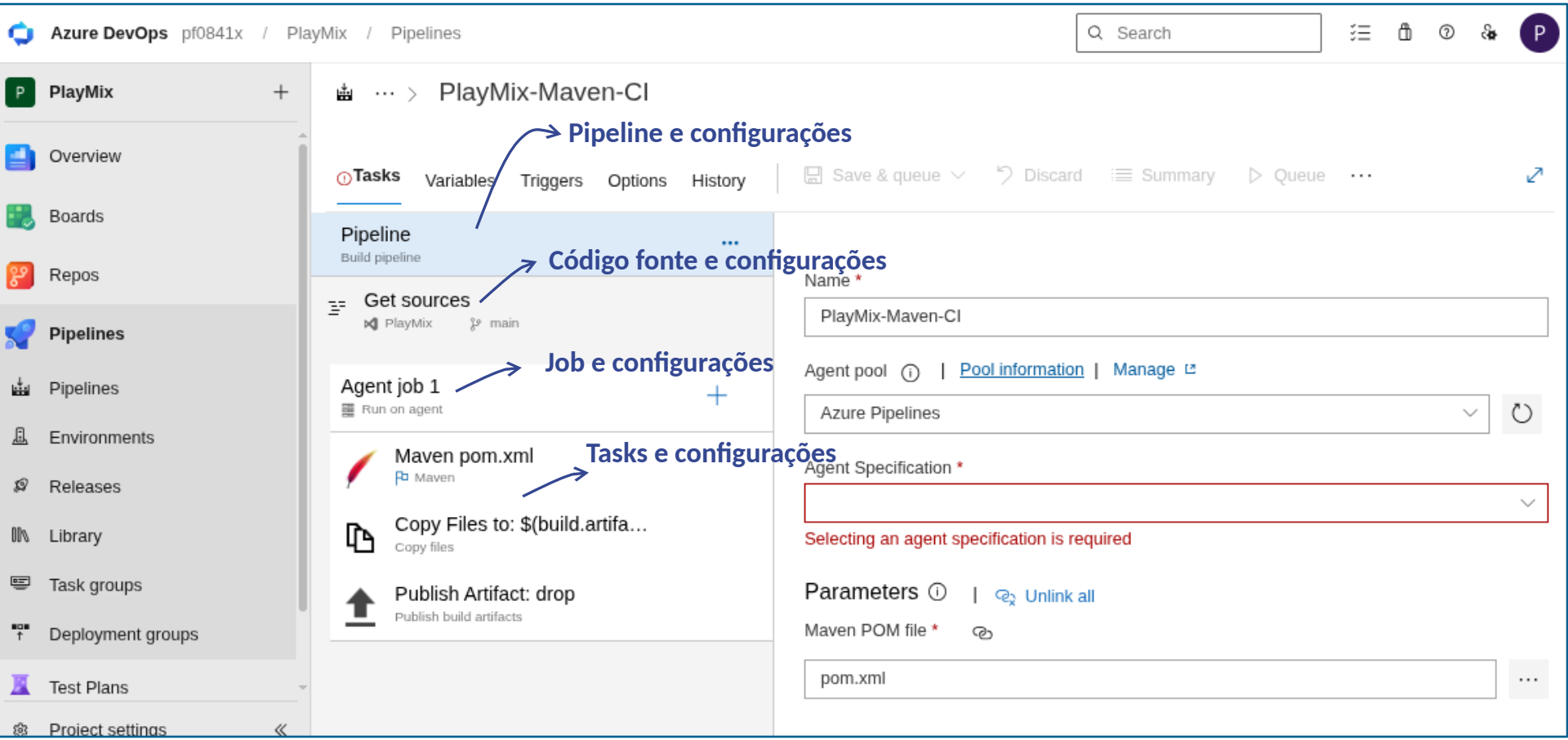
Podemos escolher criar uma Pipeline do zero **(Empty job)**, sem um modelo de criação, porém, como é nosso primeiro contato, iremos escolher um Template para criar a Pipeline



Esse projeto utiliza o Maven para o Build

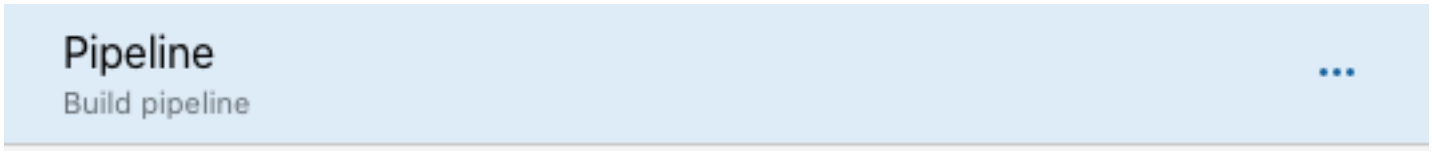
Azure Pipelines – Exercício Java x Oracle

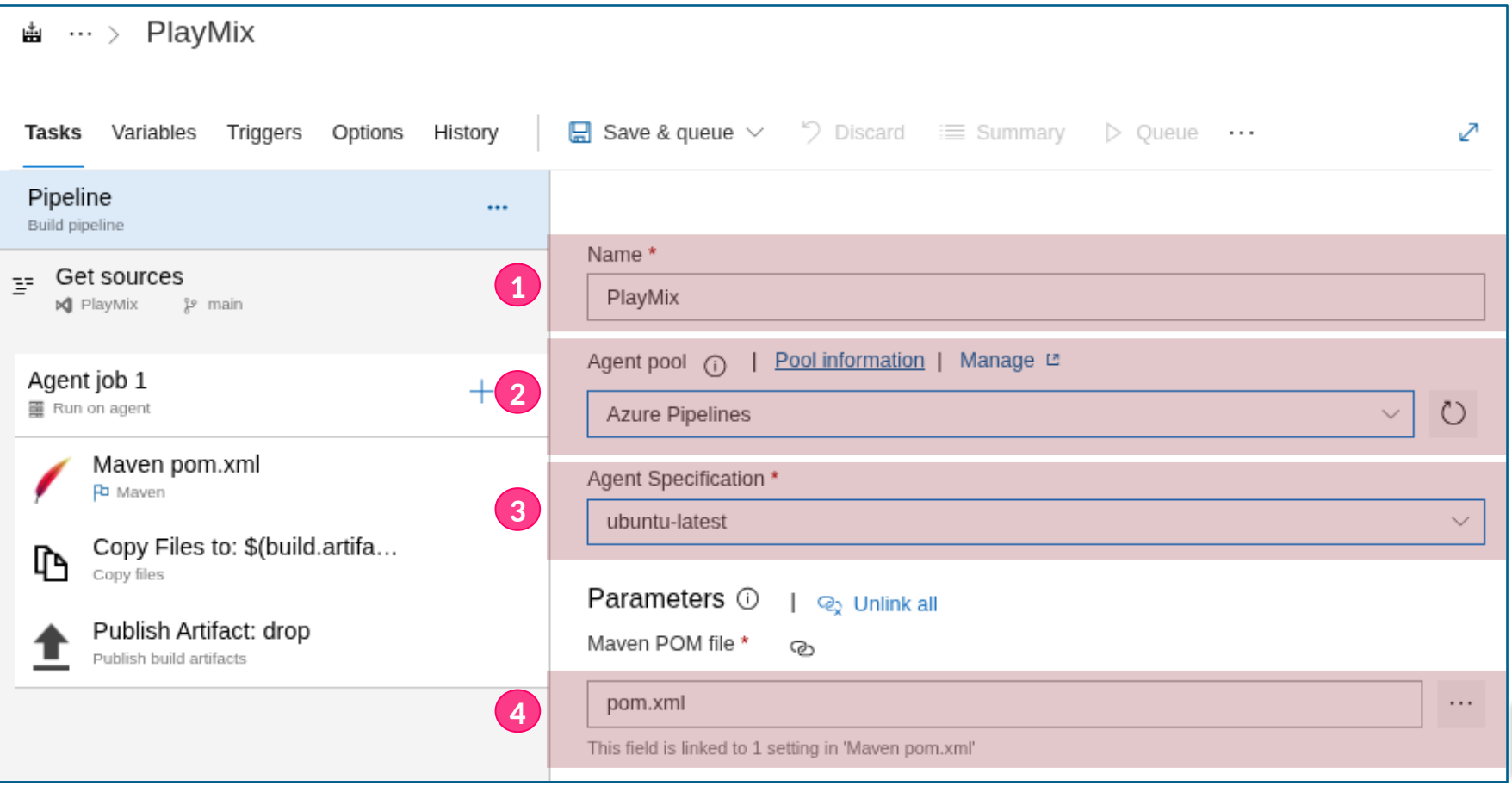
Tela inicial da Pipeline criada



Azure Pipelines – Exercício Java x Oracle

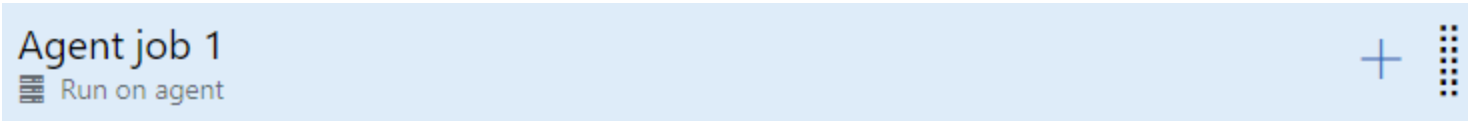
Vamos configurar a **Pipeline** criada. Siga as imagens abaixo

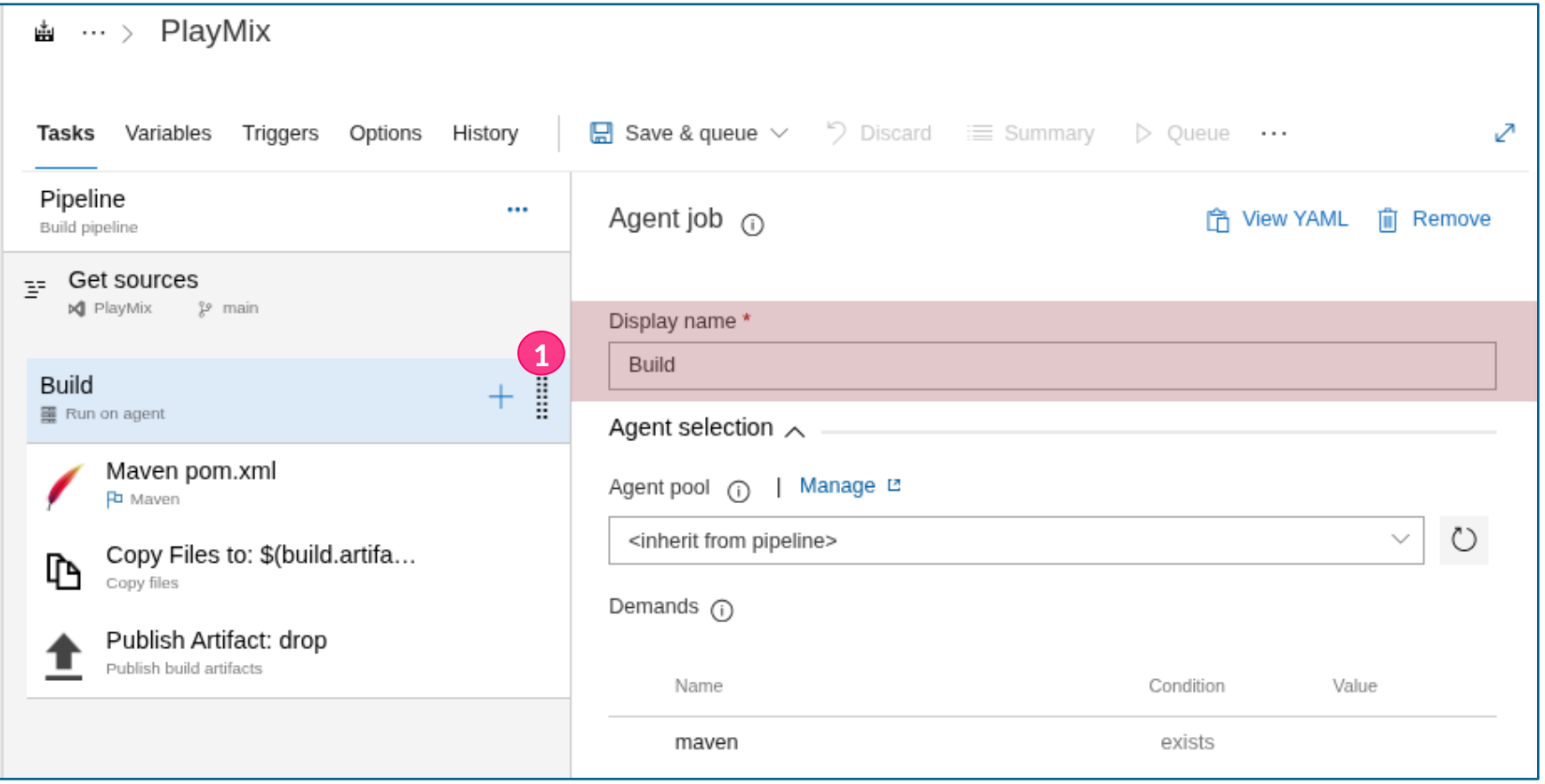




Azure Pipelines – Exercício Java x Oracle

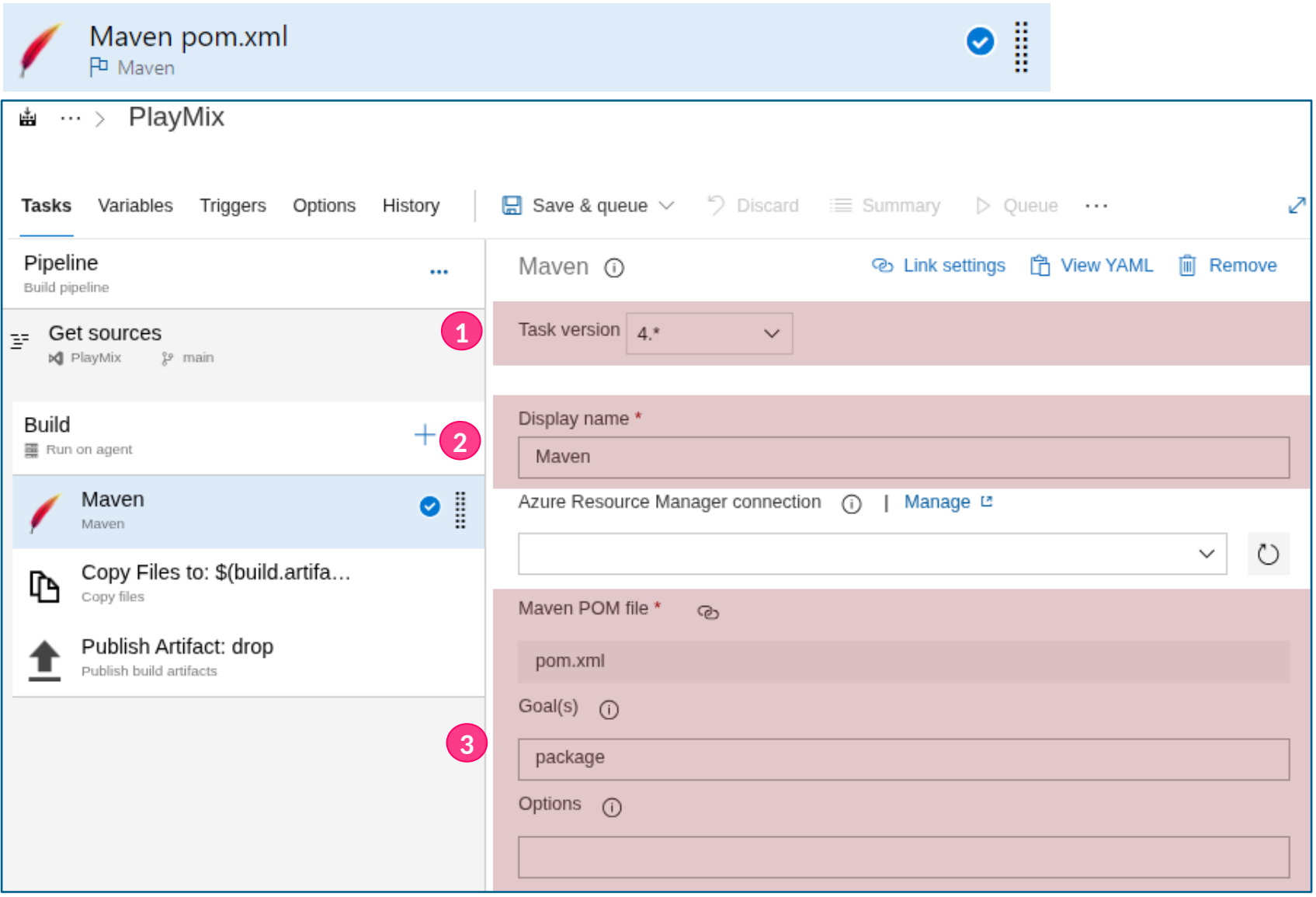
Vamos configurar agora a **Etapa** (**Agent**) criada. Selecione a opção **Agent job 1** e siga as imagens abaixo





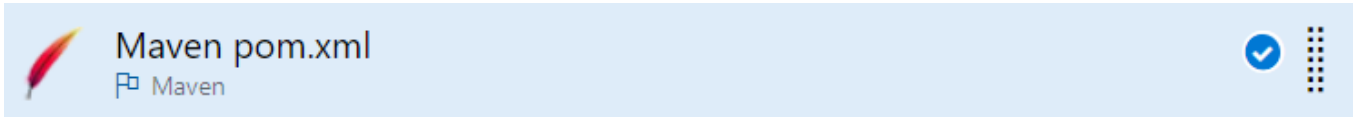
Azure Pipelines – Exercício Java x Oracle

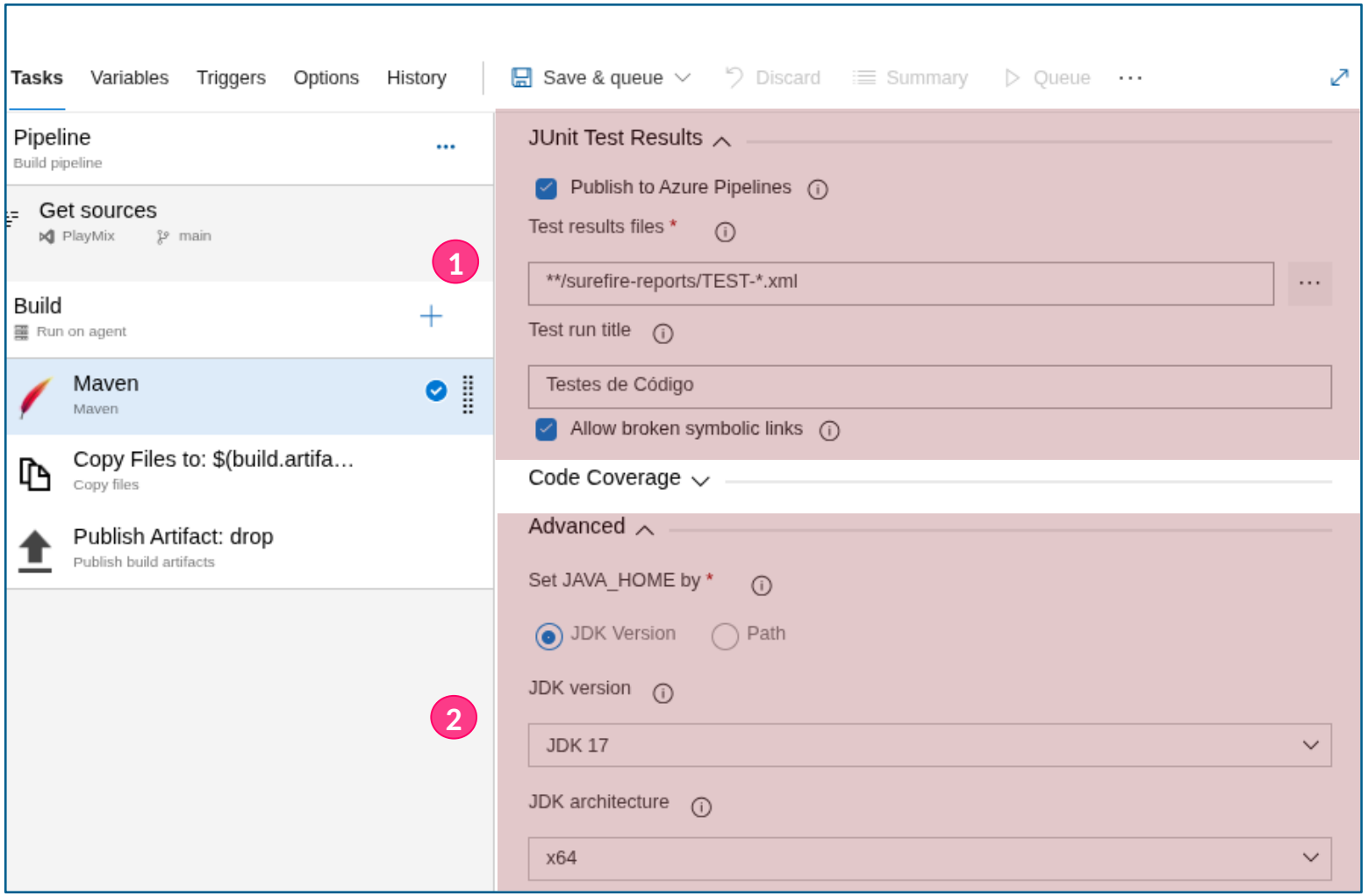
Agora iremos configurar as **Tarefas** (**Steps**). Siga as imagens abaixo



Azure Pipelines – Exercício Java x Oracle

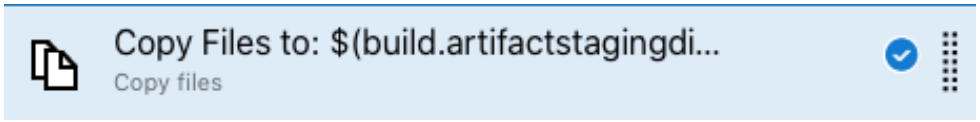
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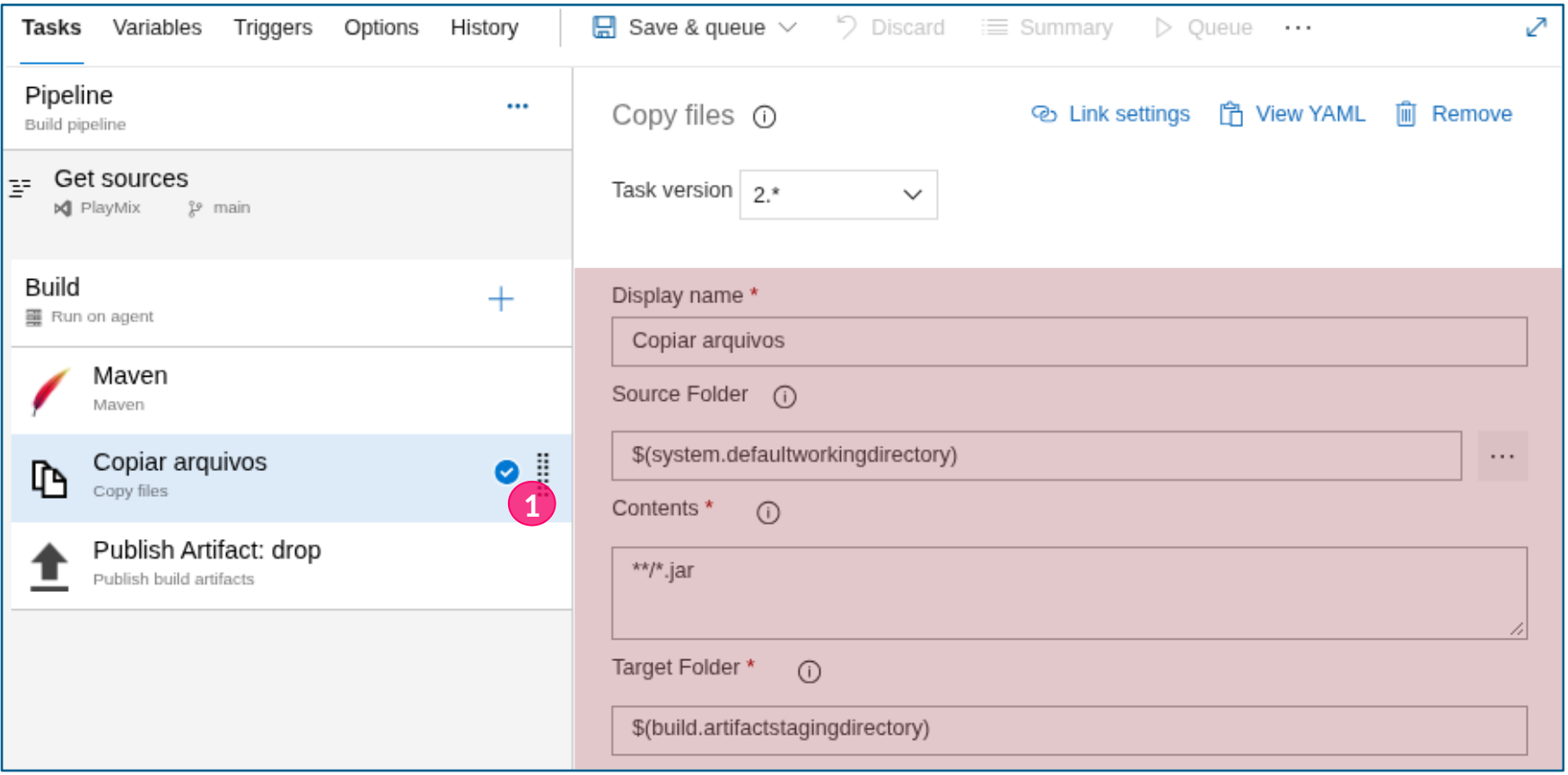




Azure Pipelines – Exercício Java x Oracle

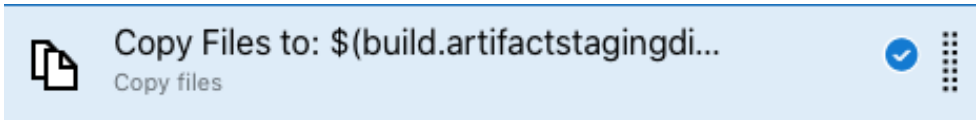
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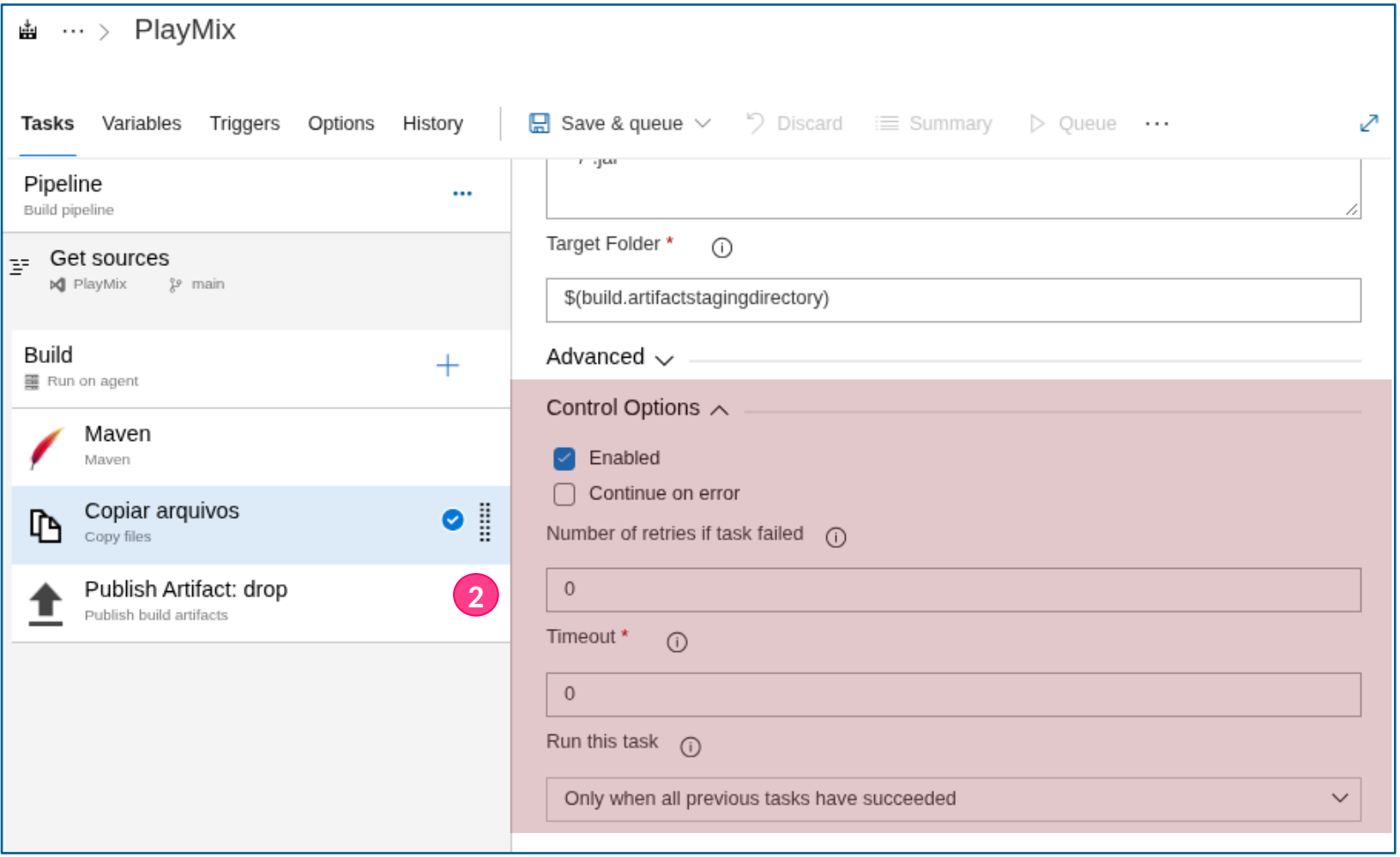




Azure Pipelines – Exercício Java x Oracle

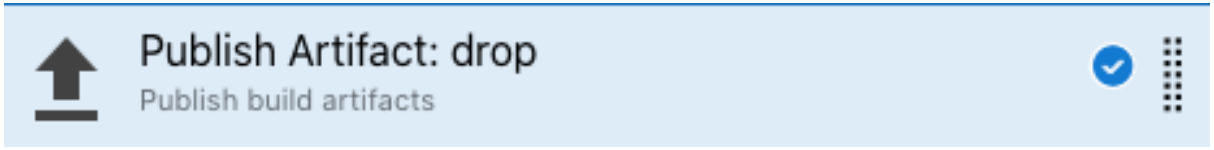
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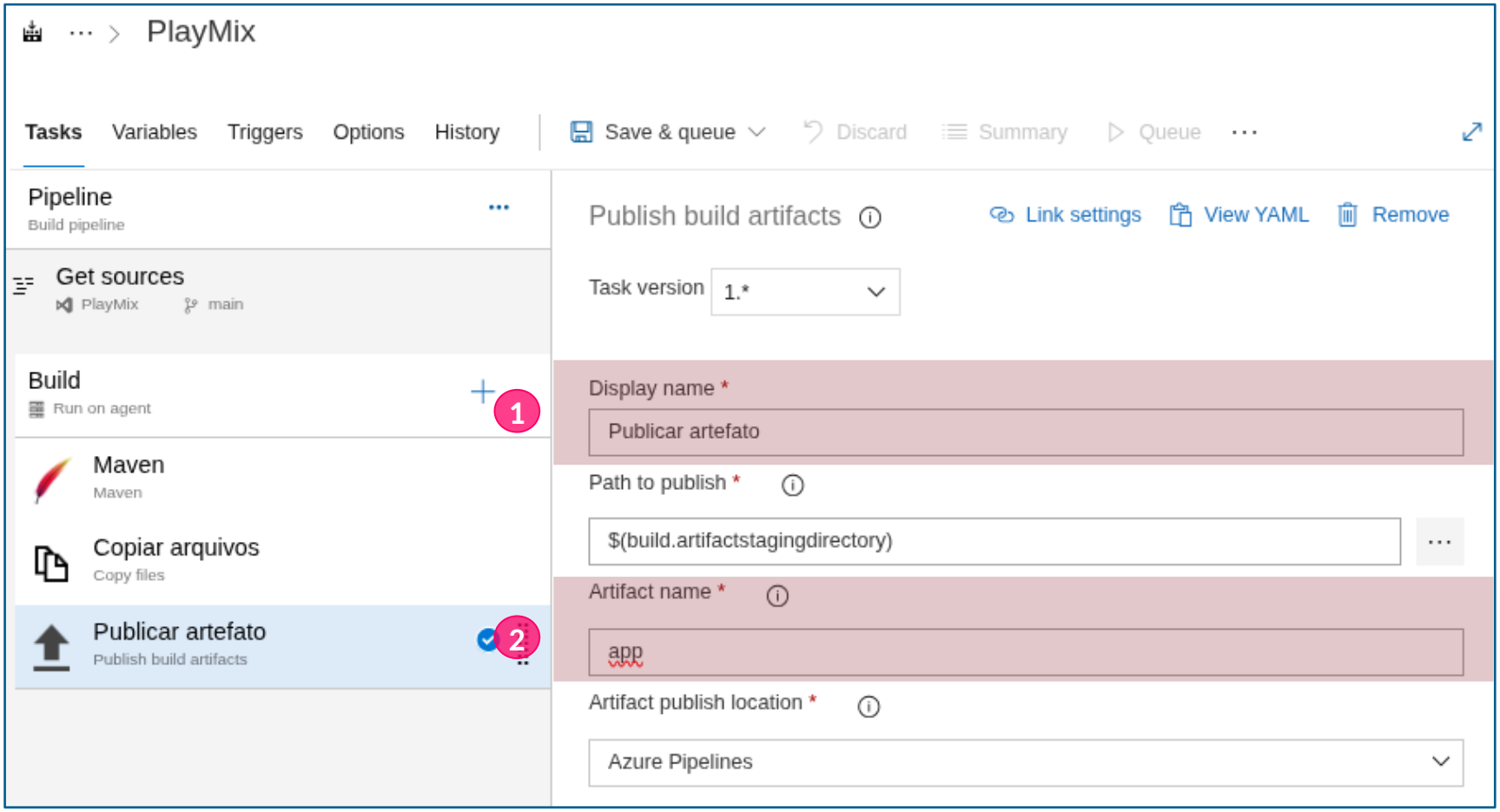




Azure Pipelines – Exercício Java x Oracle

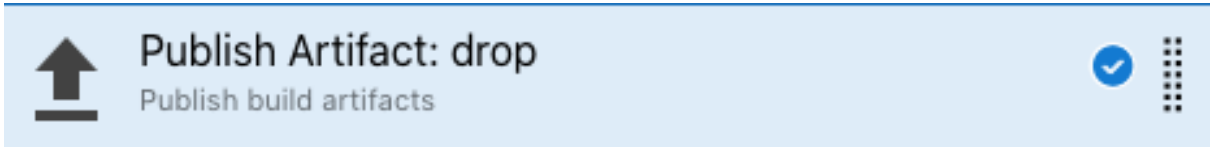
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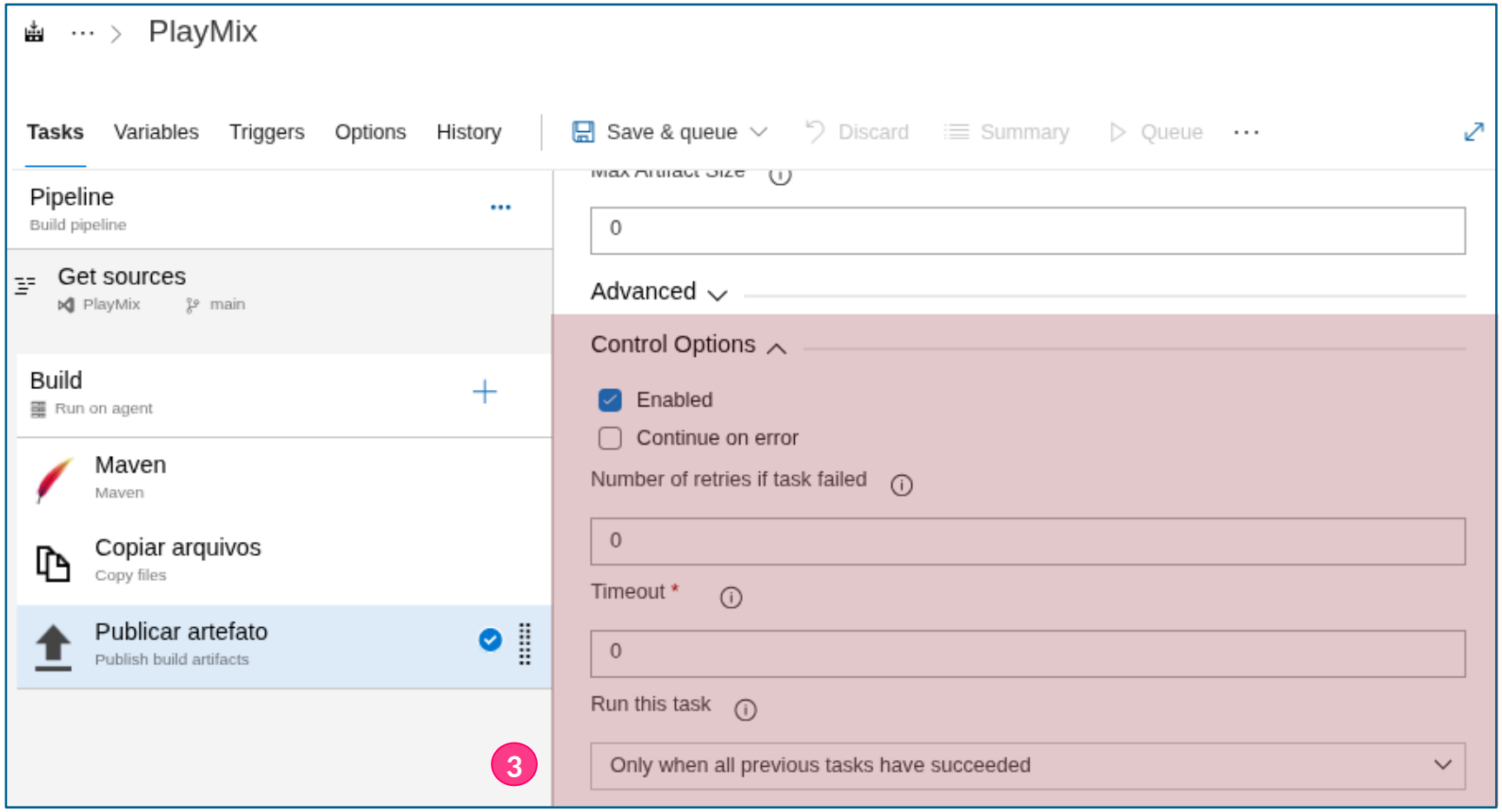




Azure Pipelines – Exercício Java x Oracle

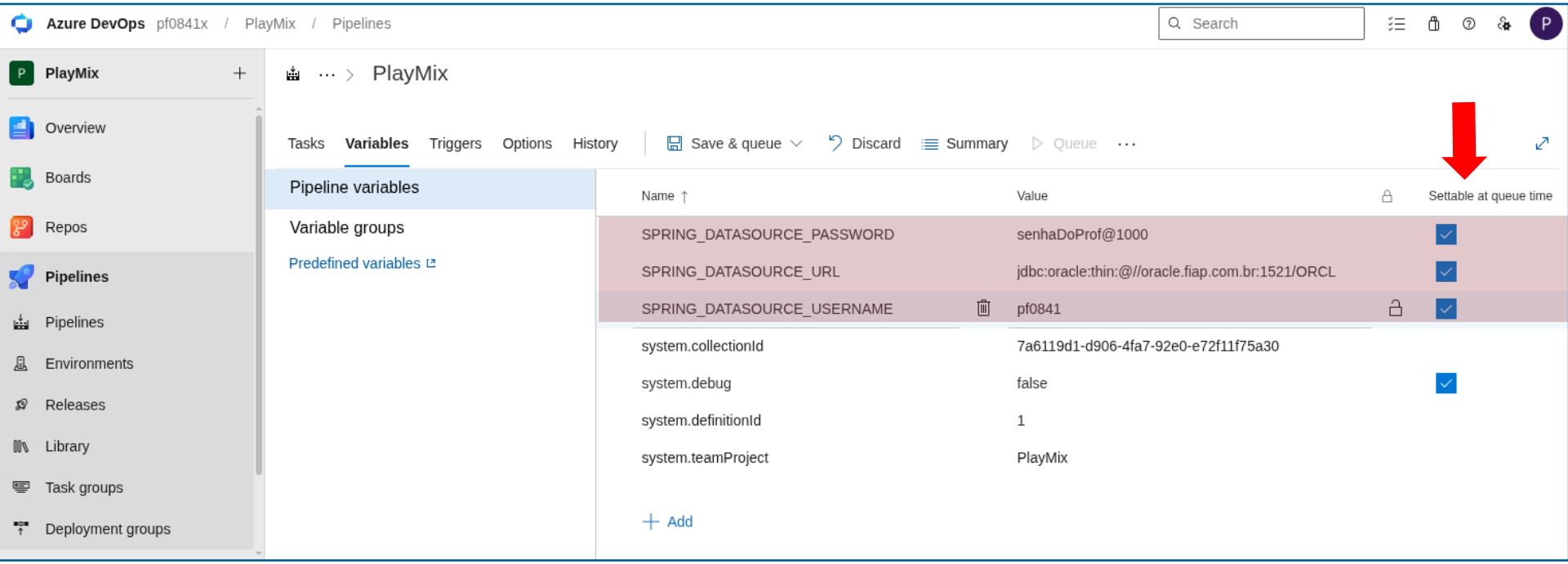
Agora iremos configurar as **Tarefas** (**Steps**). Siga as imagens abaixo





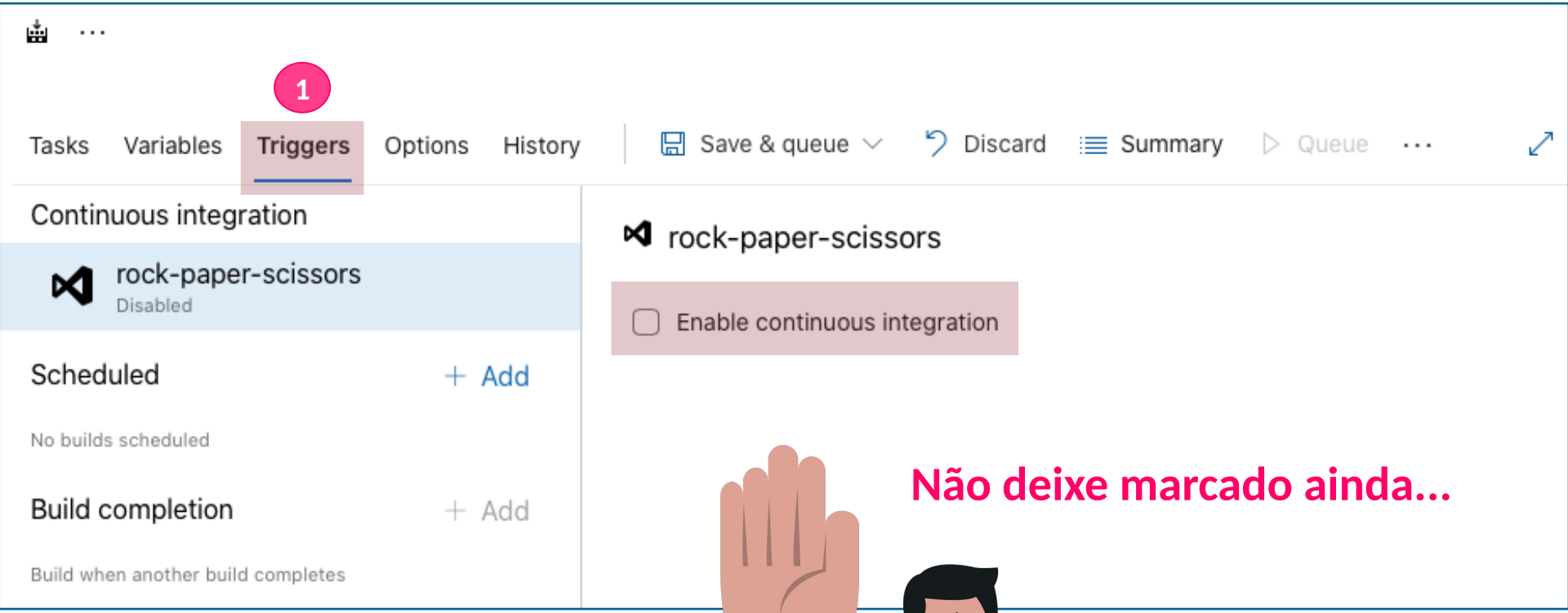
Azure Pipelines – Exercício Java x Oracle

Na aba **Variables** existe a possibilidade de incluir variáveis para nossa Pipeline



Azure Pipelines – Exercício Java x Oracle

Na aba **Triggers** existe a possibilidade de realizar o **Continuous integration**

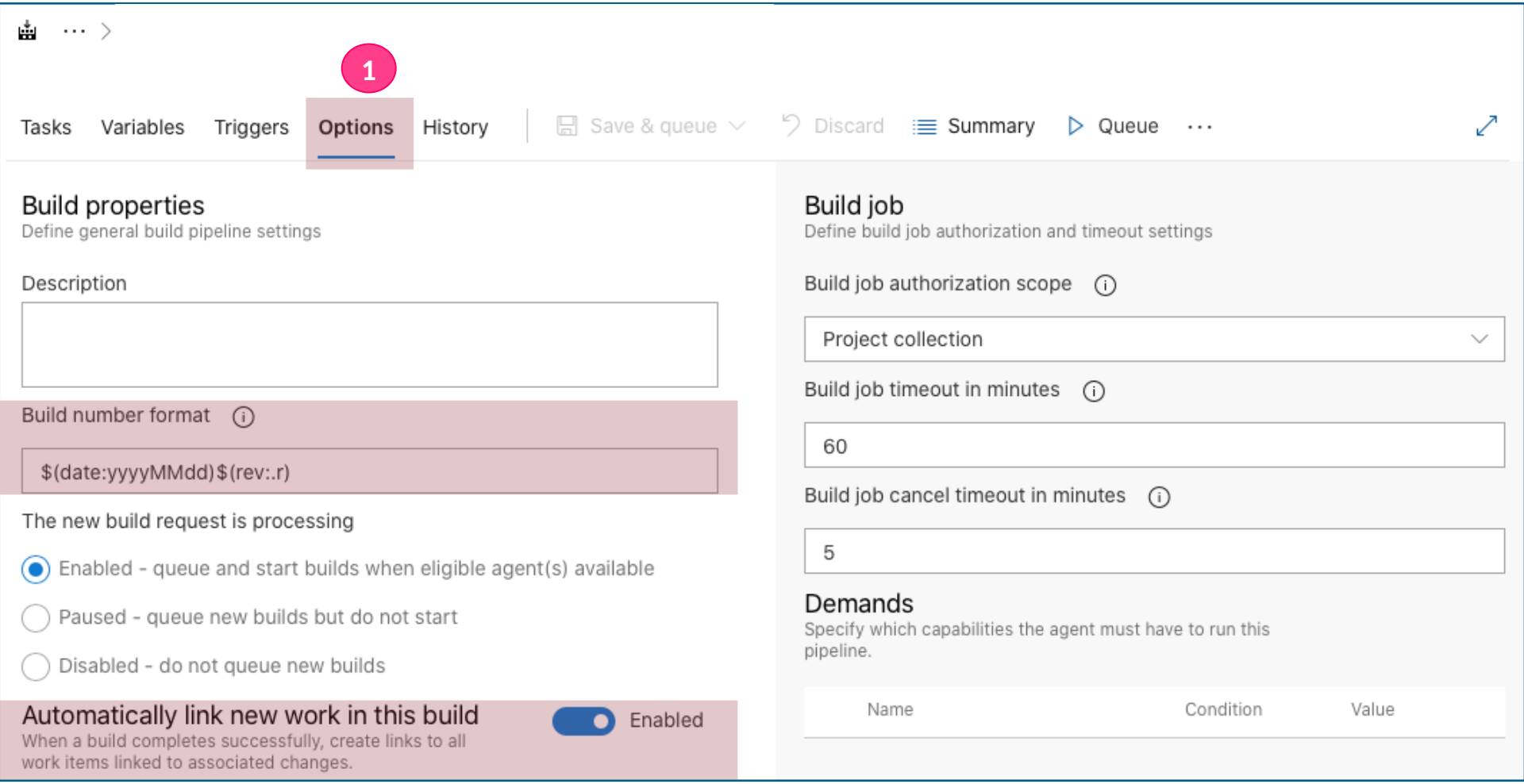




Não deixe marcado ainda...

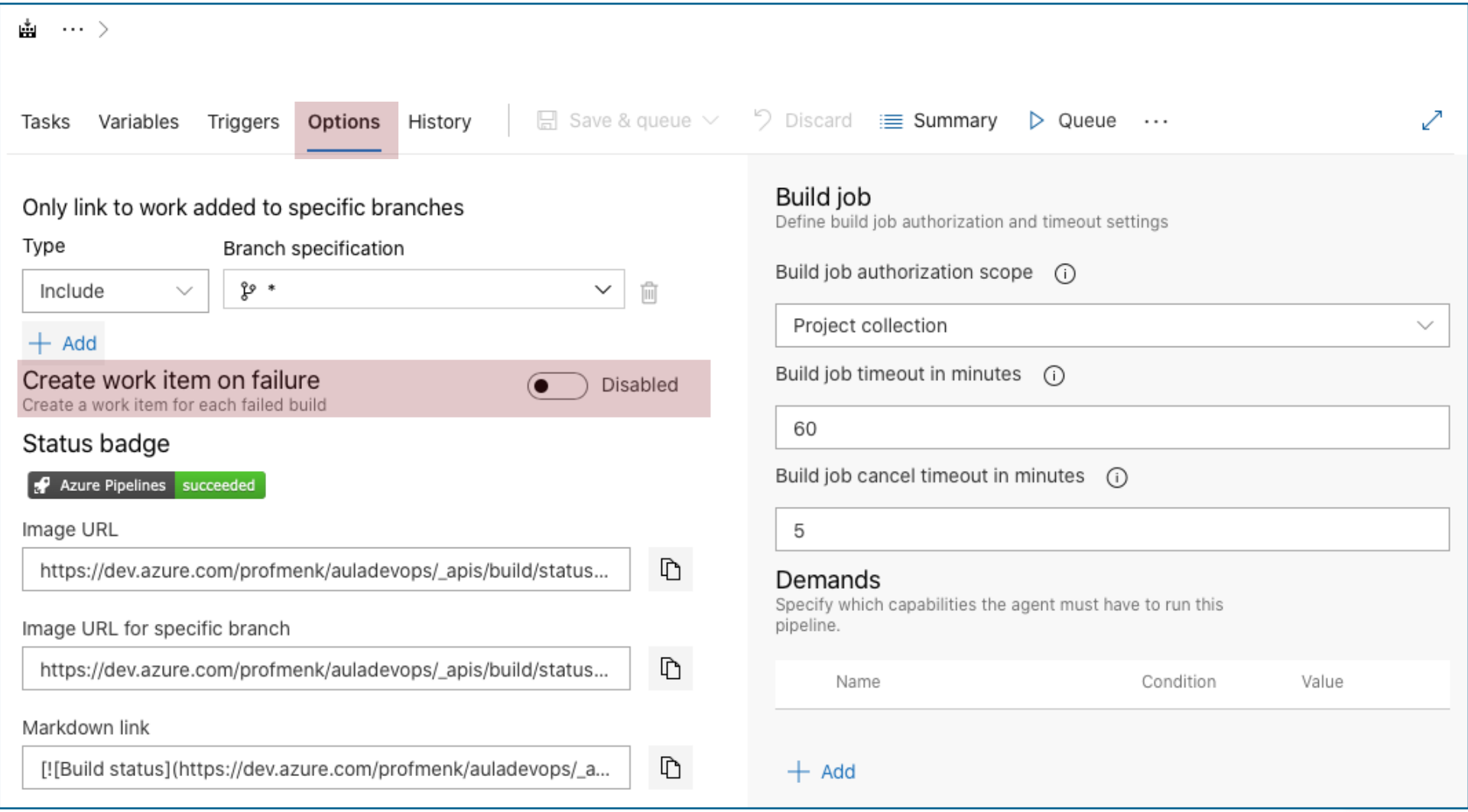
Azure Pipelines – Exercício Java x Oracle

Na aba **Options** existe a possibilidade de alterar as configurações do Build



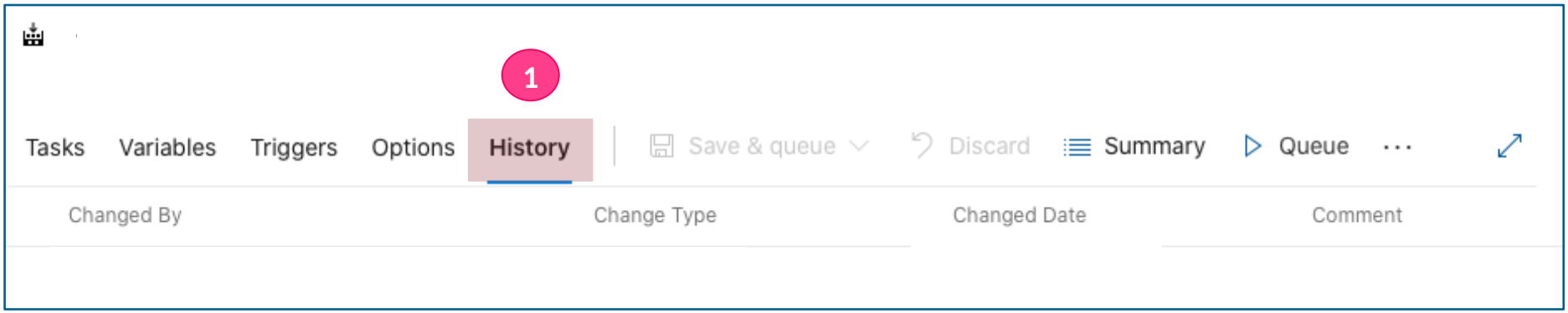
Azure Pipelines – Exercício Java x Oracle

Na aba **Options** existe a possibilidade de alterar as configurações do Build



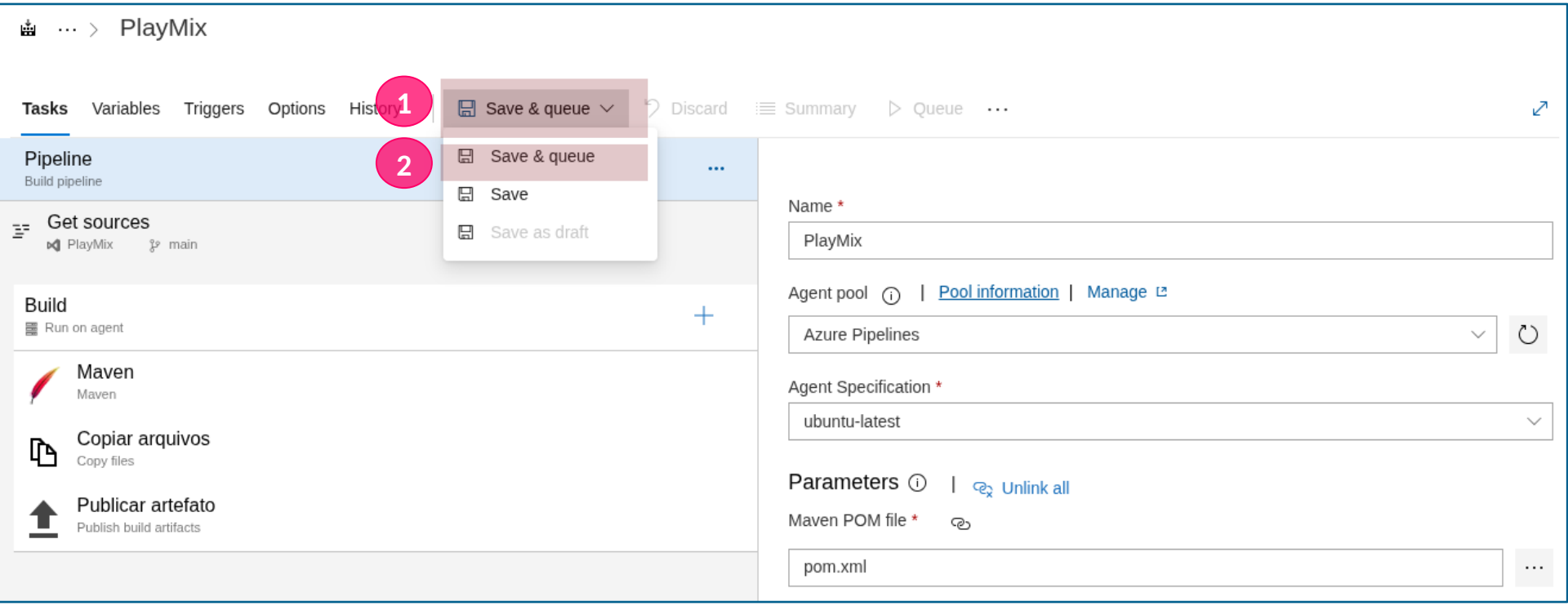
Azure Pipelines – Exercício Java x Oracle

Na aba **History** existe a possibilidade de visualizar tudo o que ocorreu com a Pipeline



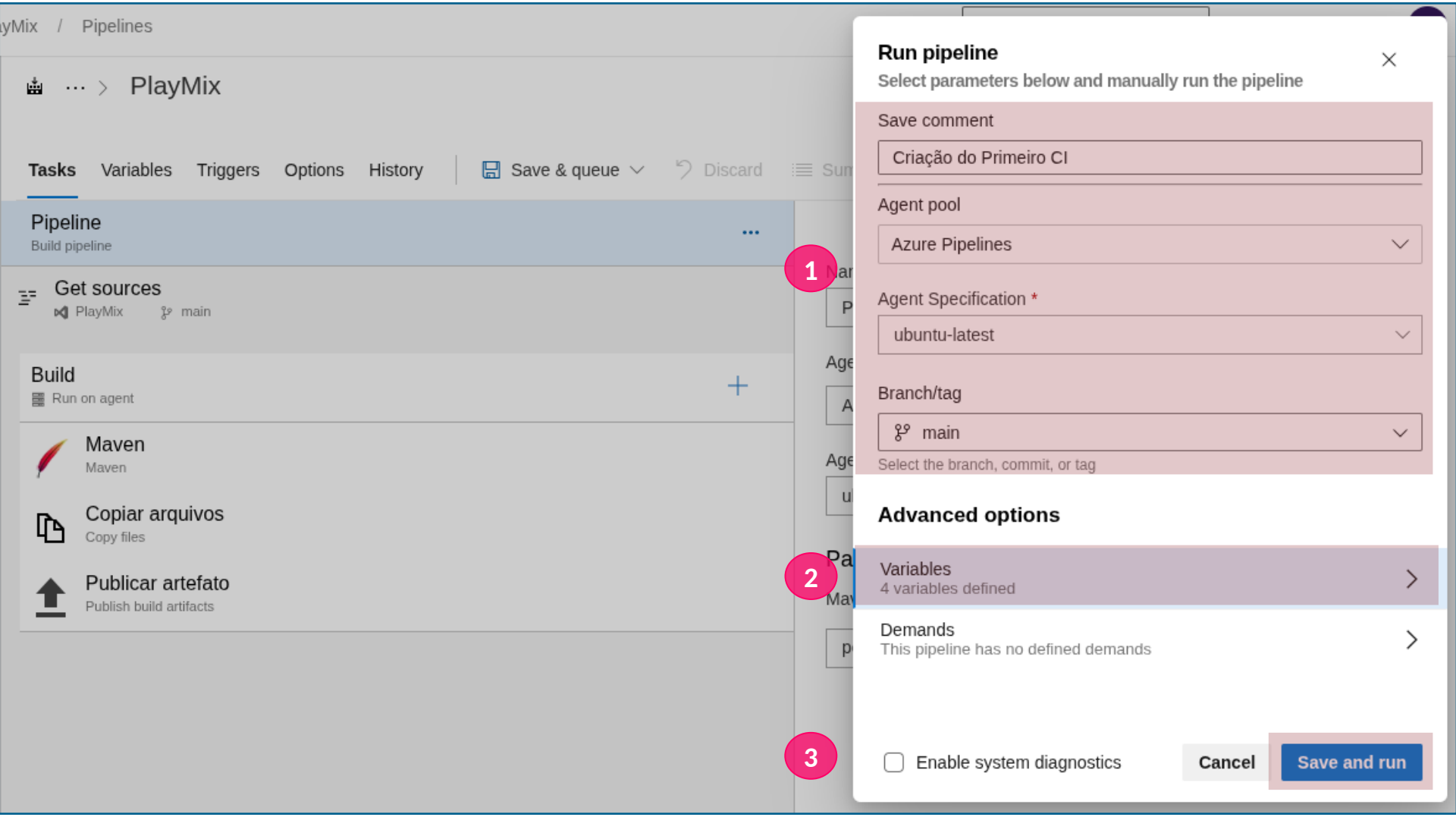
Azure Pipelines – Exercício Java x Oracle

Agora que as adaptações foram efetuadas, clique em **Save & queue**



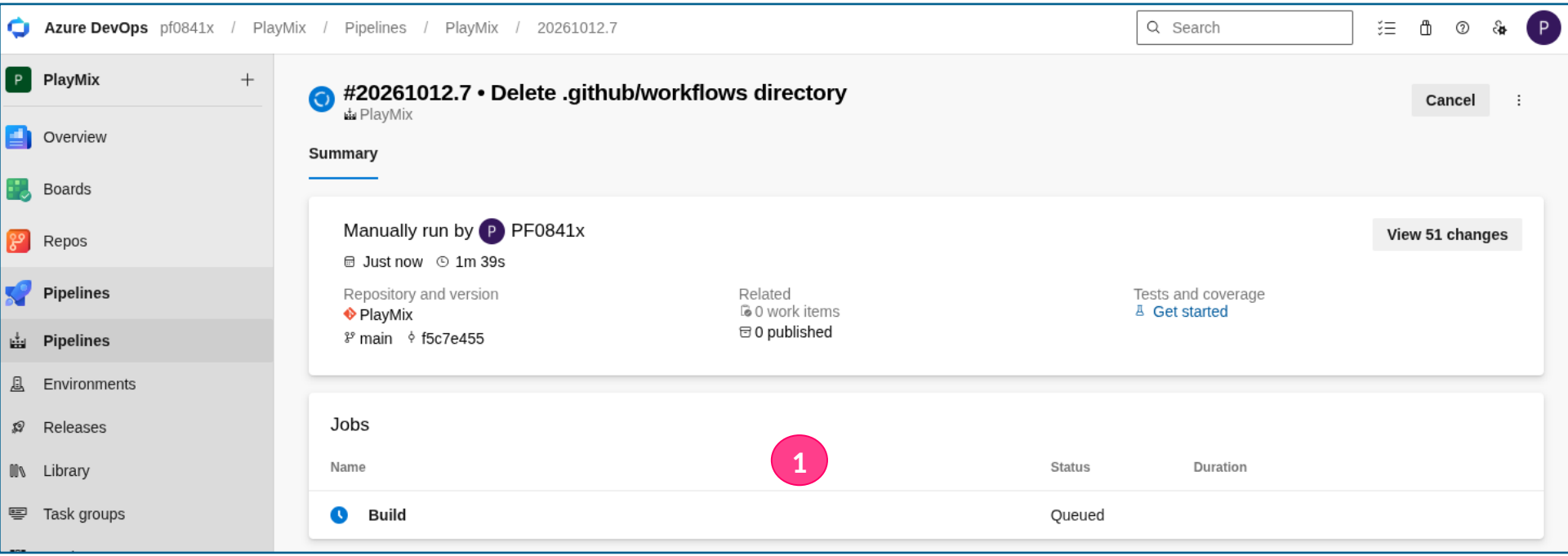
Azure Pipelines – Exercício Java x Oracle

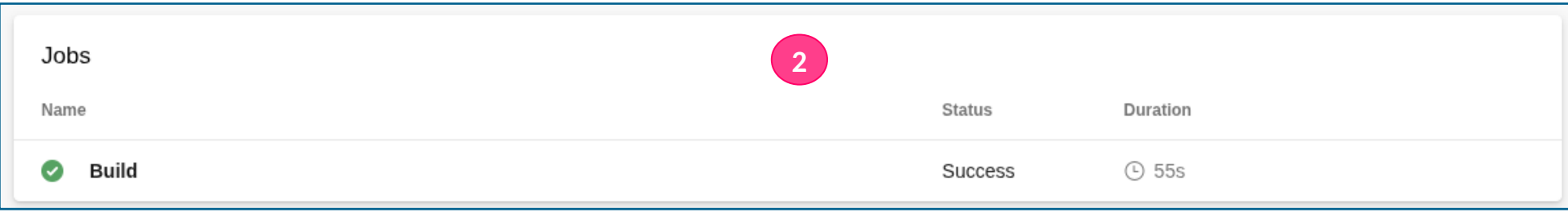
Antes de Salvar e rodar você ainda pode alterar algumas propriedades na execução (conforme configurado)



Azure Pipelines – Exercício Java x Oracle

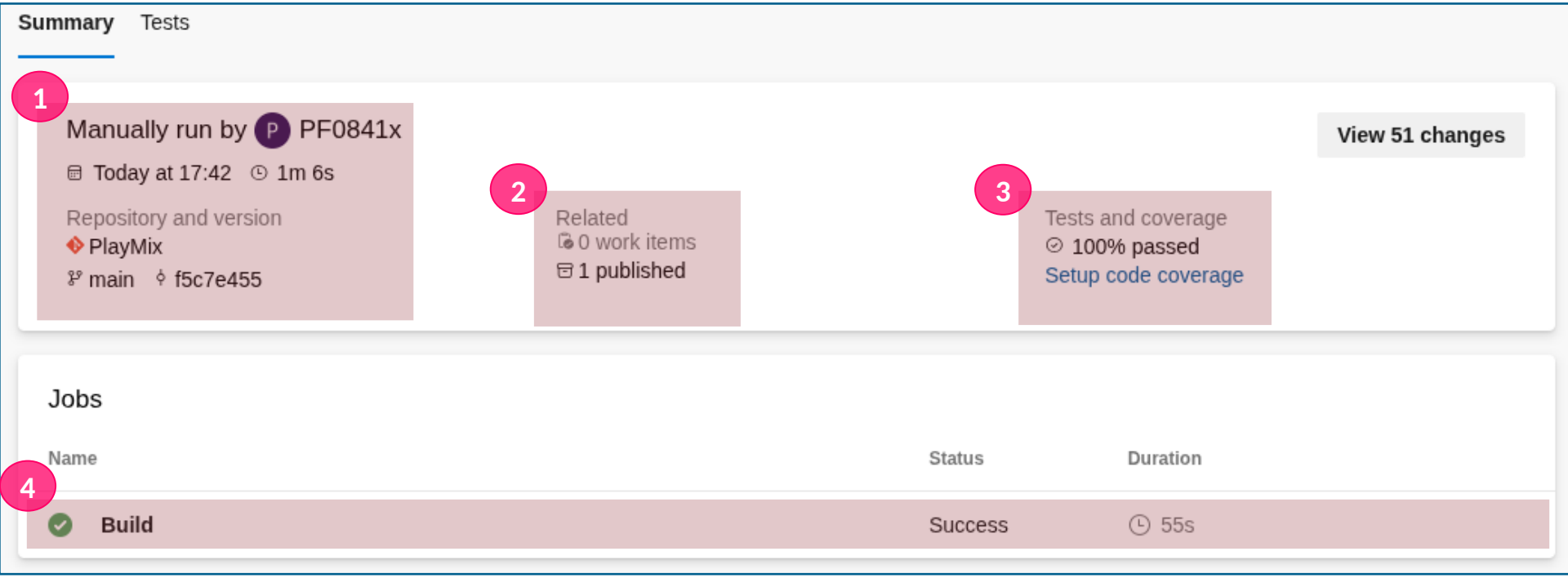
Aguarde a execução da Pipeline





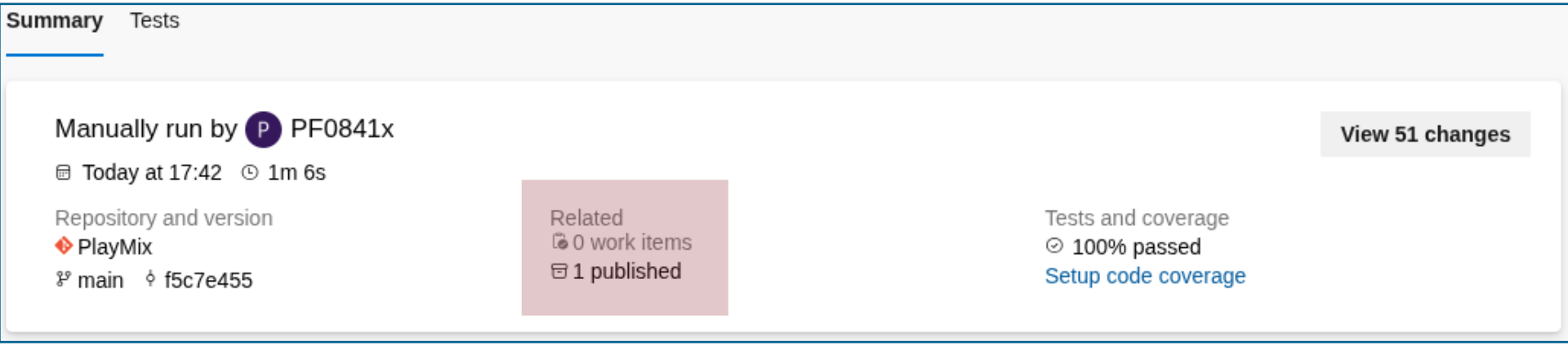
Azure Pipelines – Exercício Java x Oracle

Investigue a Pipeline criada e executada examinando o log de saída



Azure Pipelines – Exercício Java x Oracle

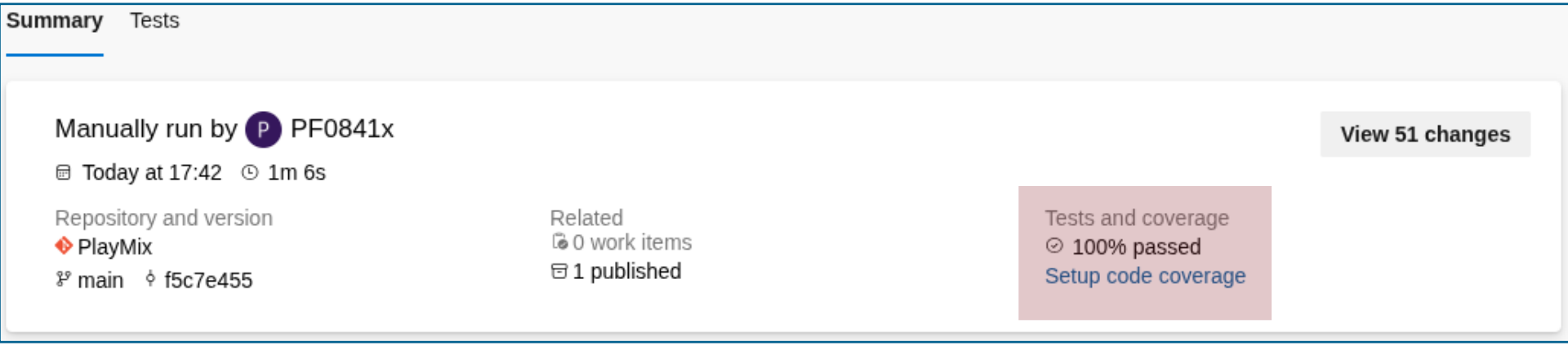
Investigue a Pipeline criada e executada examinando o log de saída

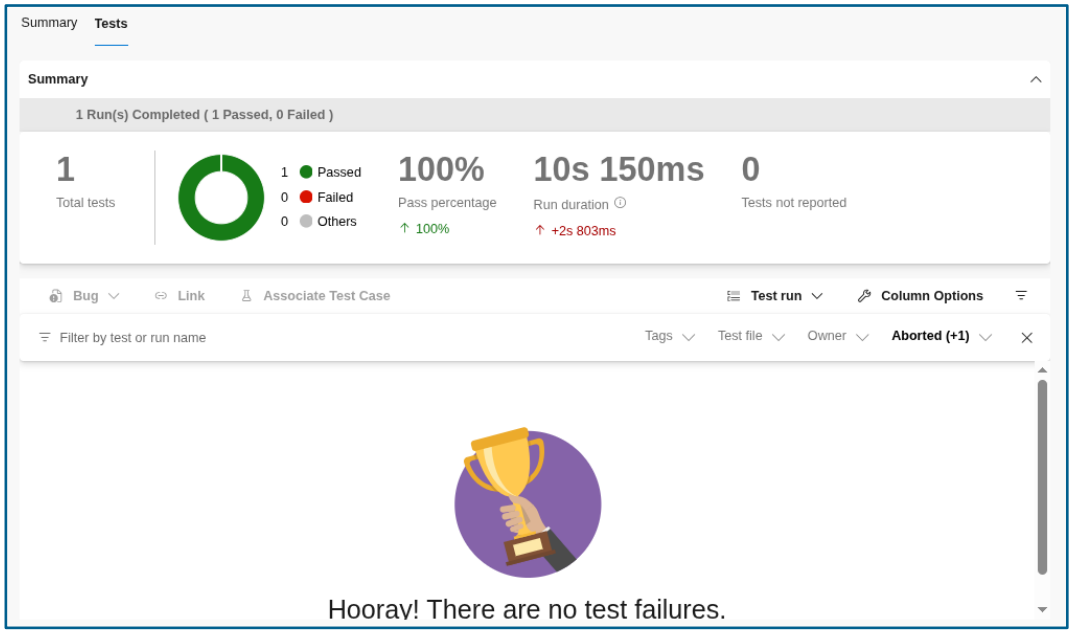




Azure Pipelines – Exercício Java x Oracle

Investigue a Pipeline criada e executada examinando o log de saída



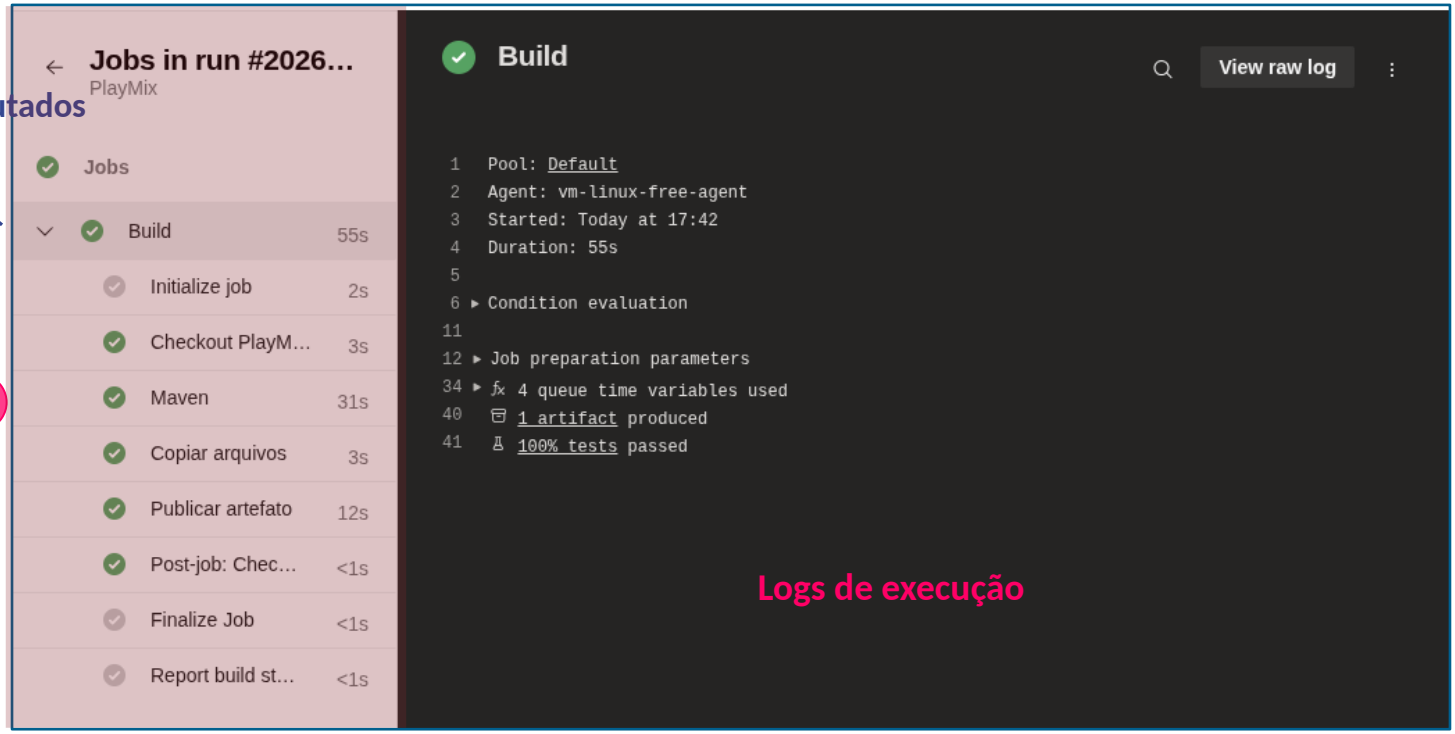


Azure Pipelines – Exercício Java x Oracle

Investigue a Pipeline criada e executada examinando o log de saída



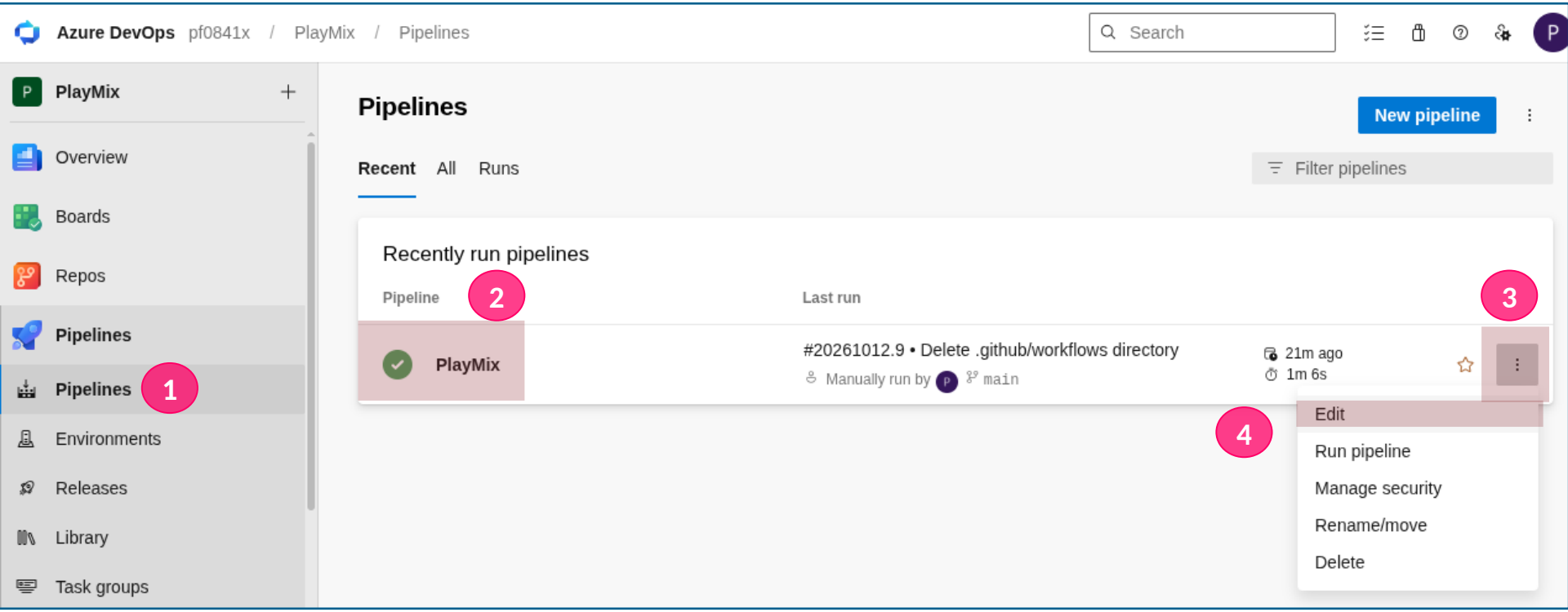
Jobs executados



2

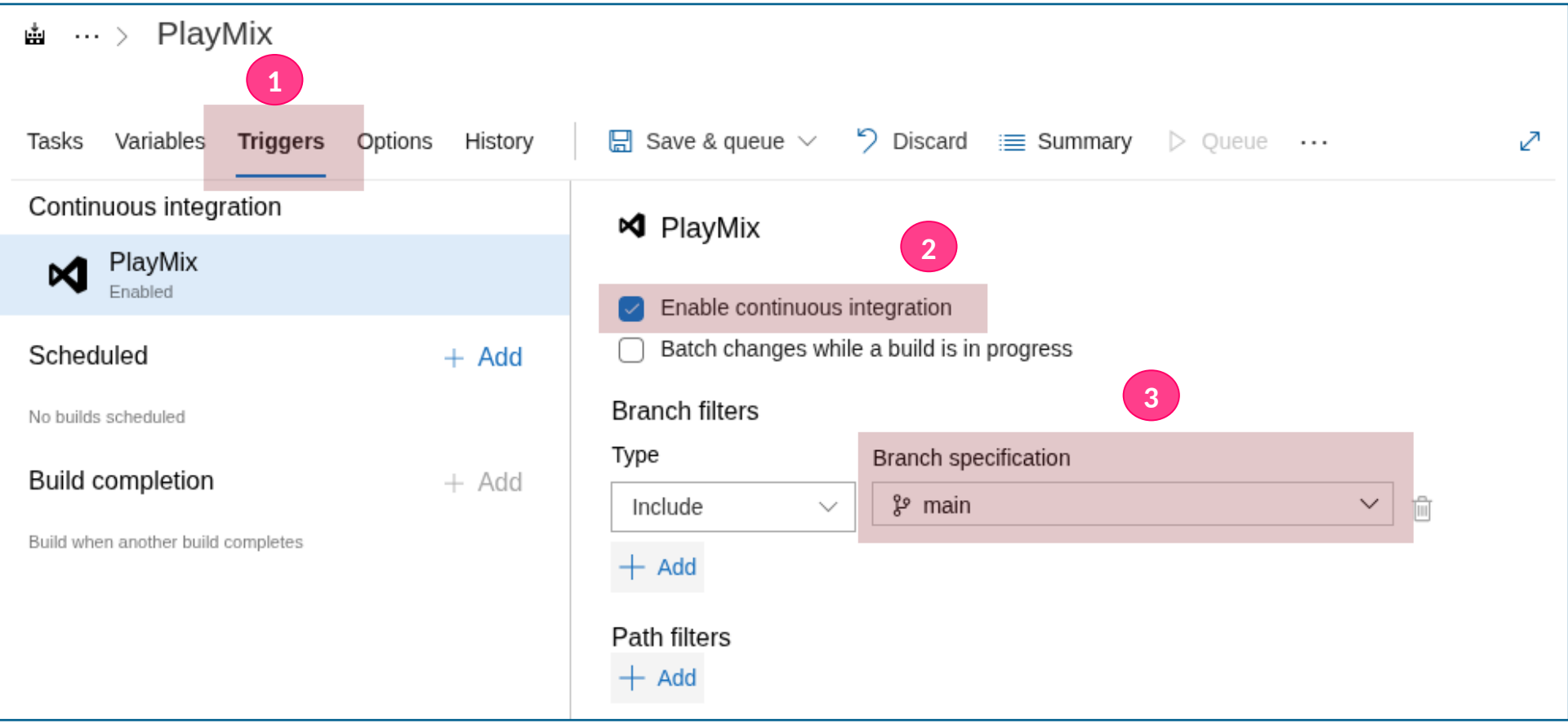
Azure Pipelines – Exercício Java x Oracle

Estando tudo funcionando, edite sua Pipeline para habilitar o **CI**



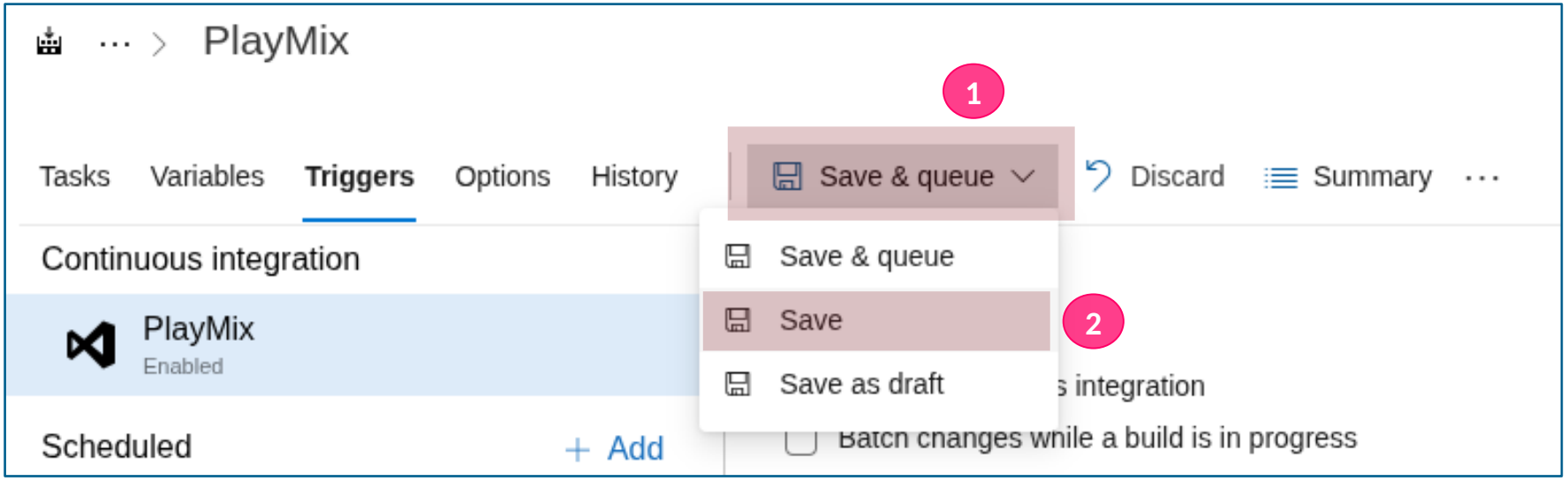
Azure Pipelines – Exercício Java x Oracle

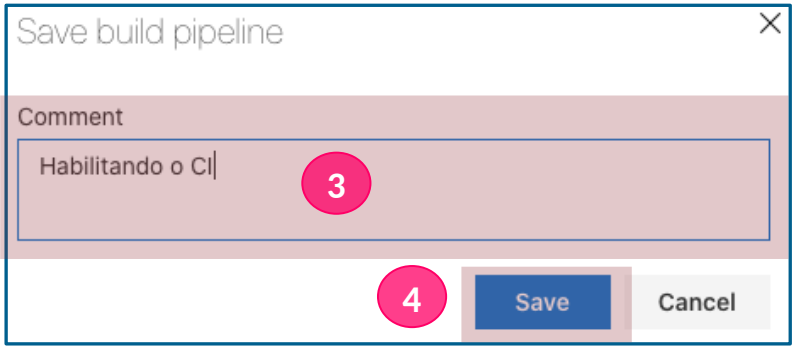
Na aba **Triggers** deixe checado **Enable continuous integration**



Azure Pipelines – Exercício Java x Oracle

Clique em **Save & queue** e depois em **Save**



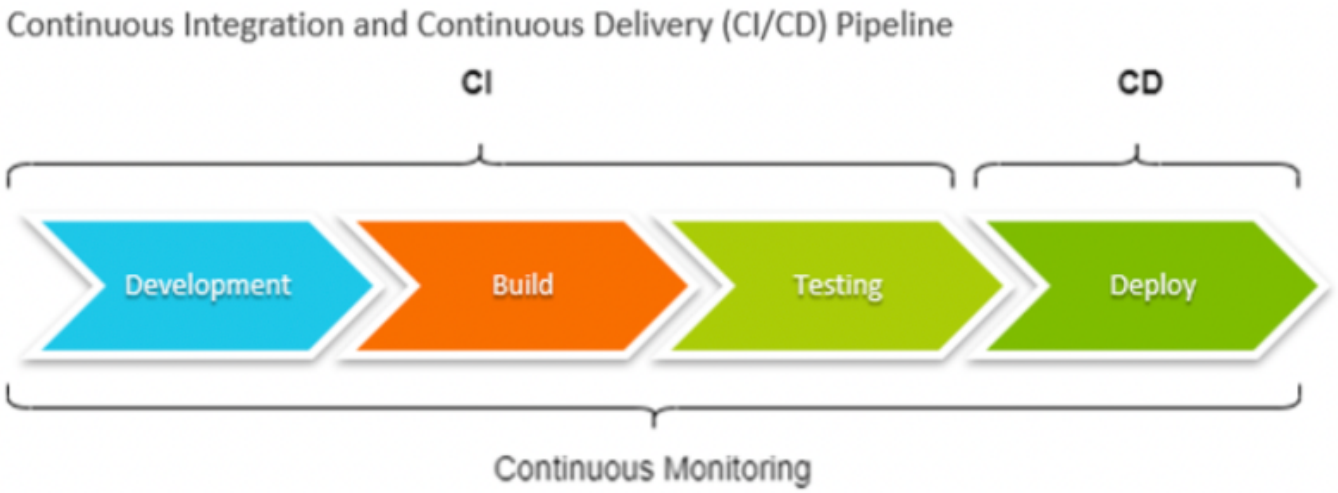


Azure Pipelines – Exercício Java x Oracle

- Agora que ativamos o **Continuos Integration (CI)**, chegou o momento de integrar o build automático com a parte de Deploy Automático, denominada **Continuos Delivery (CD)**

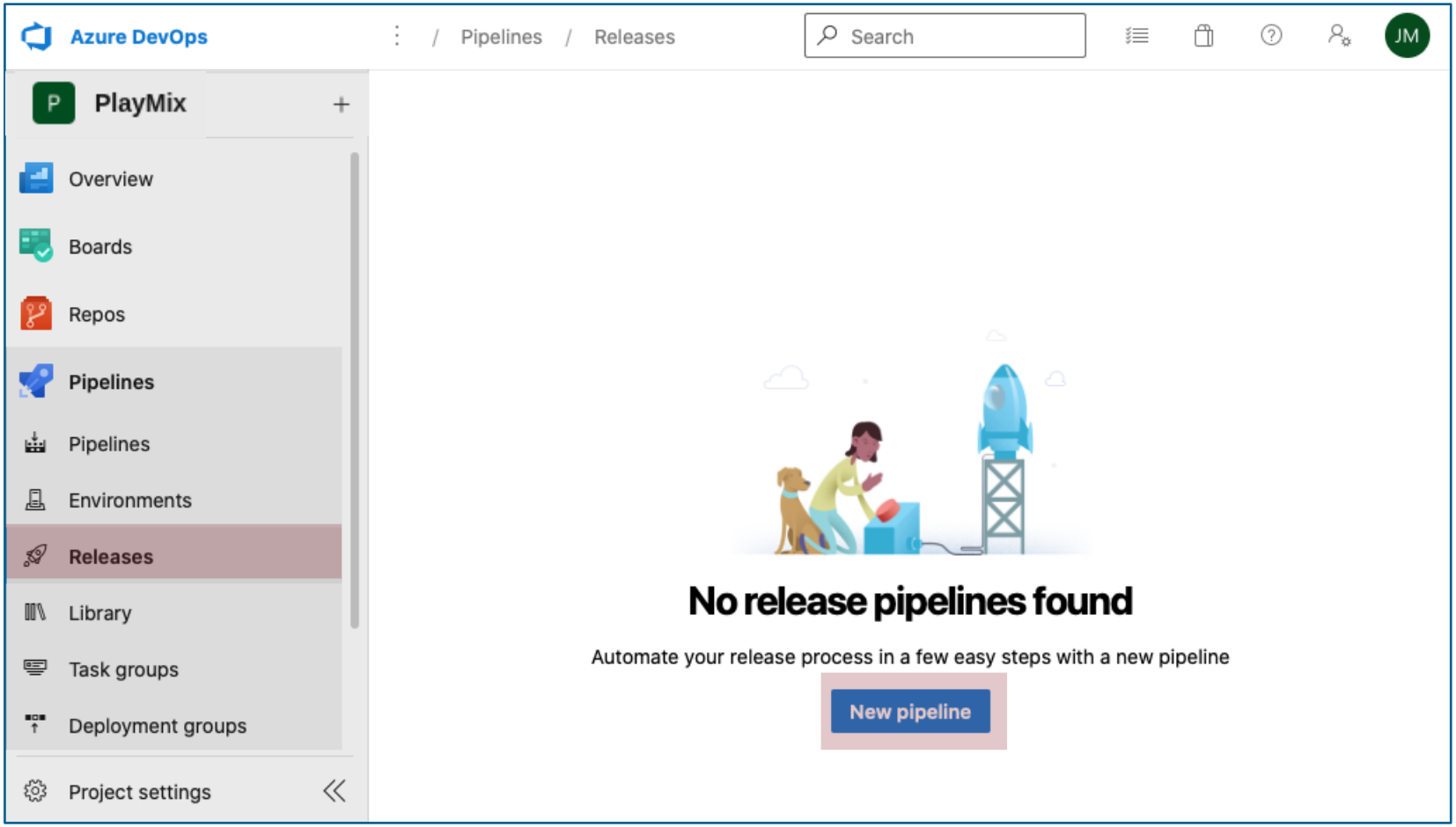
- A ideia é que após realizar um commit atualizando o código fonte dentro do repositório, o Azure Pipelines identifique essa mudança e acione o Deploy Automático, garantindo assim que a nova versão seja atualizada para o usuário final

- Siga os passos em conjunto com seu professor



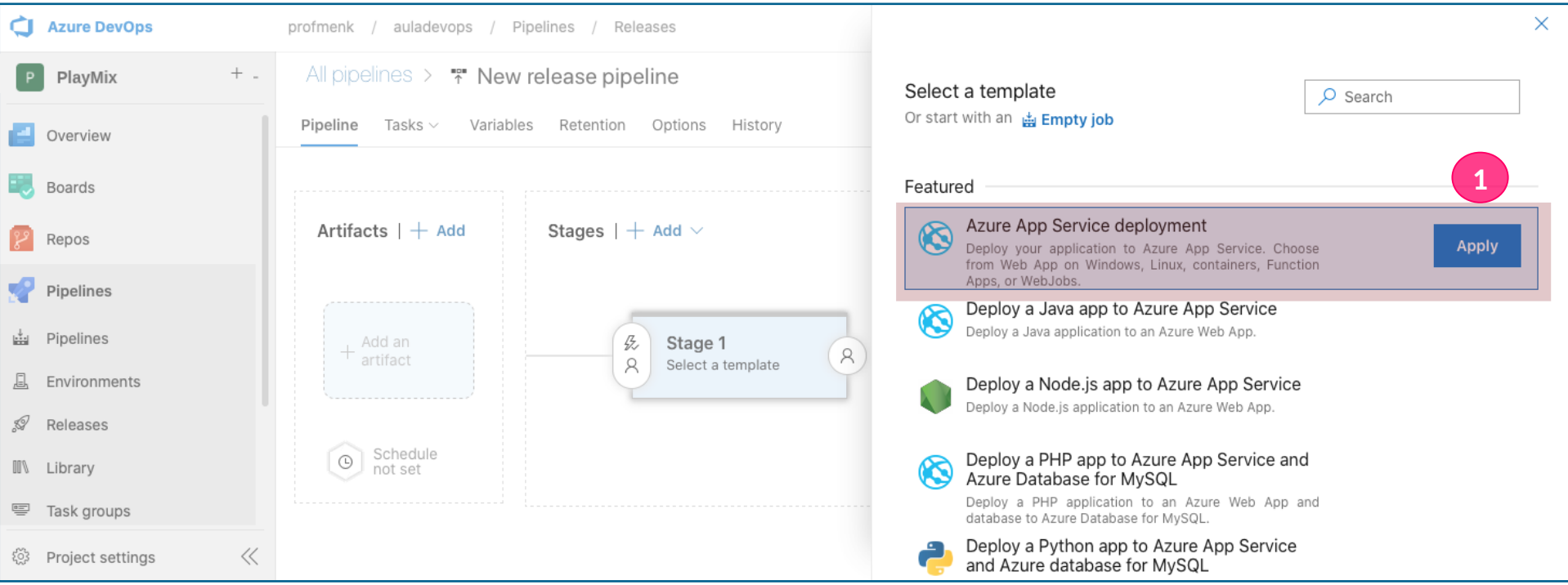
Azure Pipelines – Exercício Java x Oracle

Agora que temos a Build completa, configurada e testada, vamos criar um Release para o Deploy do App no ambiente de **Desenvolvimento**. Clique em **Releases** e depois em **New Pipeline**



Azure Pipelines – Exercício Java x Oracle

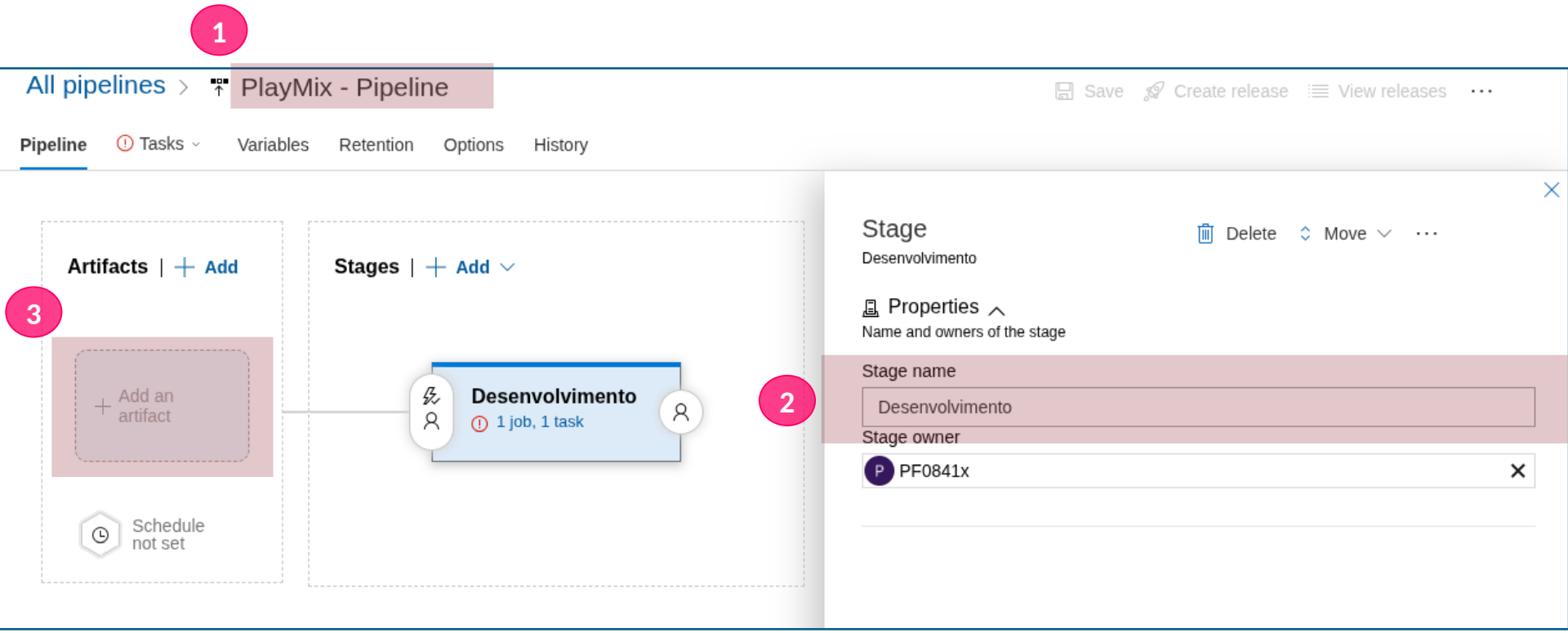
Escolha **Azure App Service deployment** como template e clique em **Apply**



Azure Pipelines – Exercício Java x Oracle

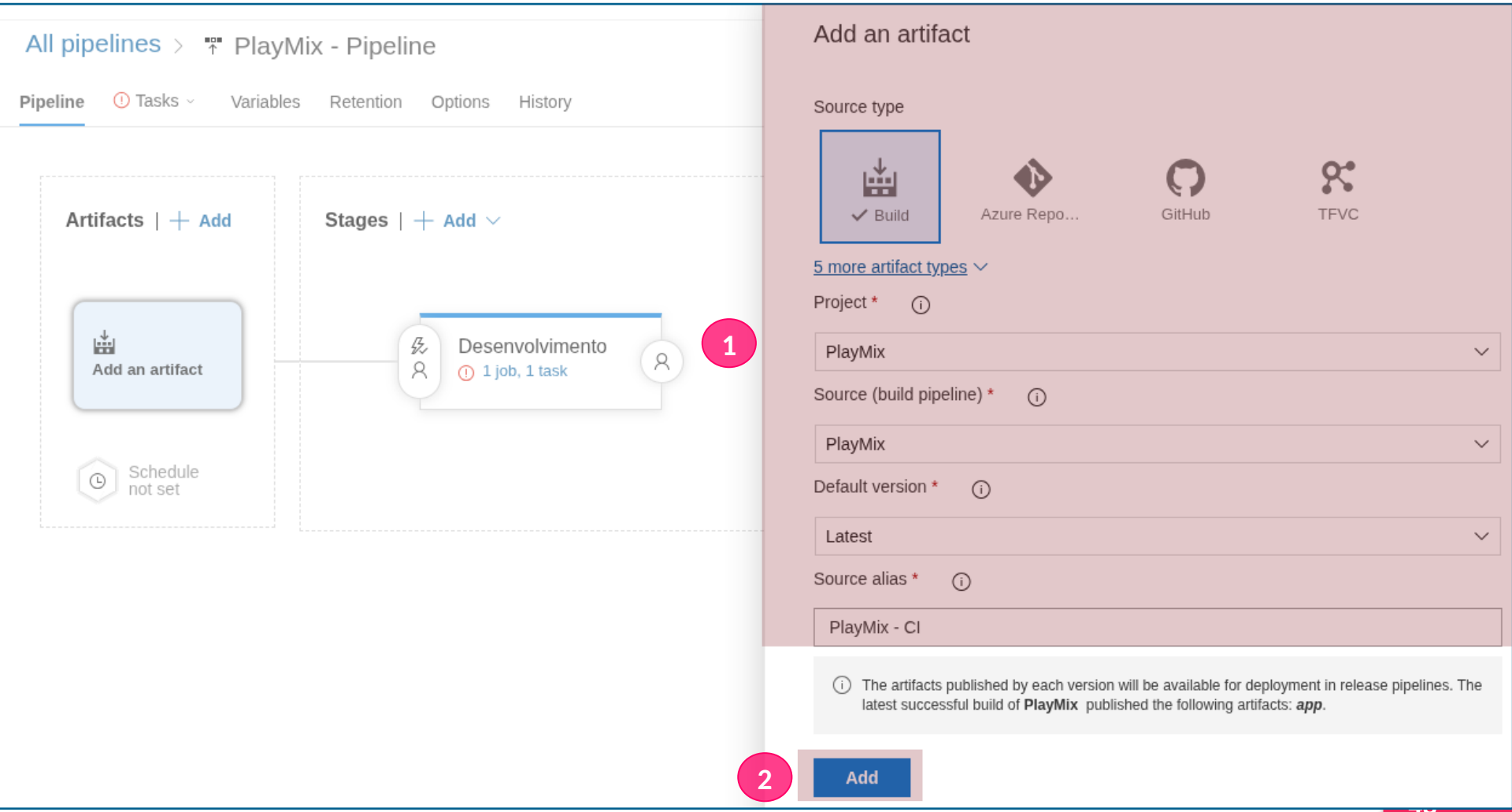
Altere o nome da Pipeline e o nome do Stage (1 e 2)

Depois de alterar os nomes, clique em **Add an artifact** (3)



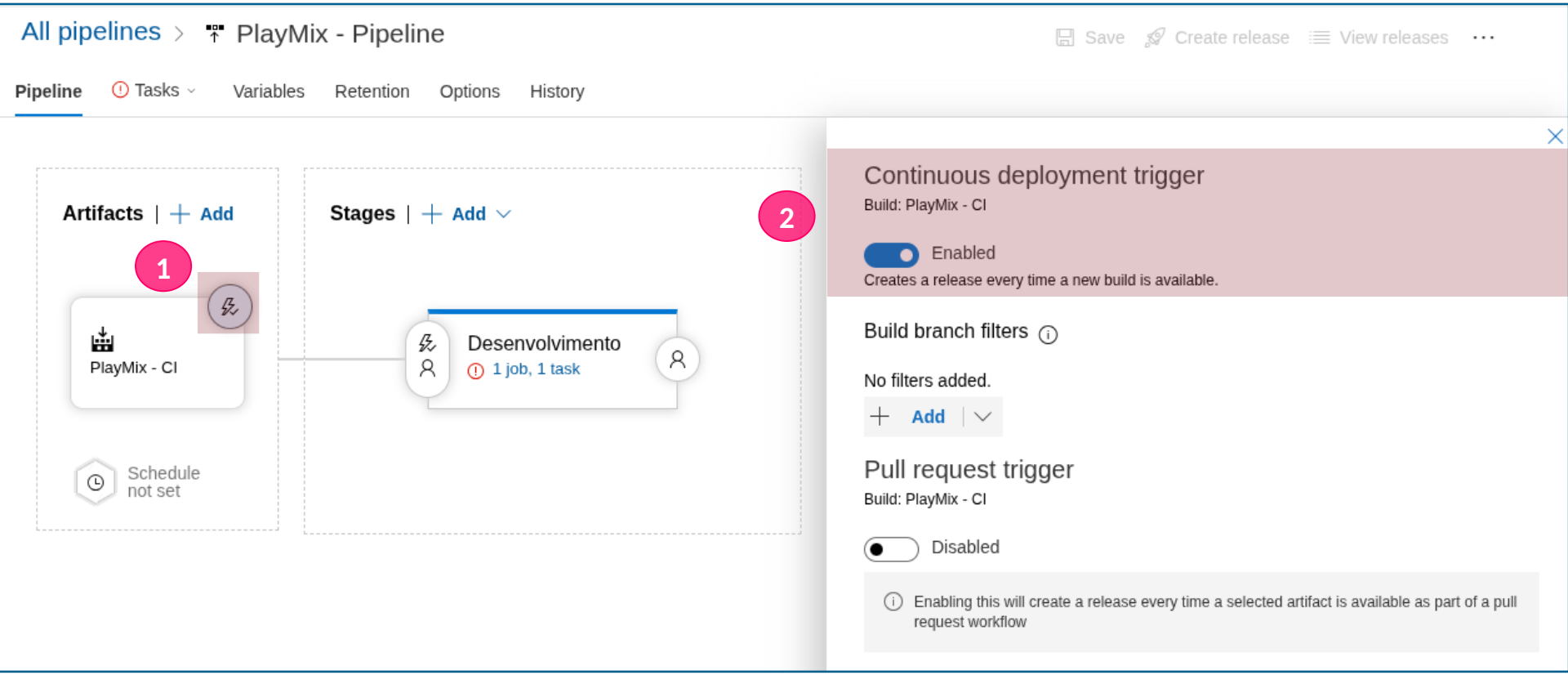
Azure Pipelines – Exercício Java x Oracle

Realize as configurações conforme imagem abaixo



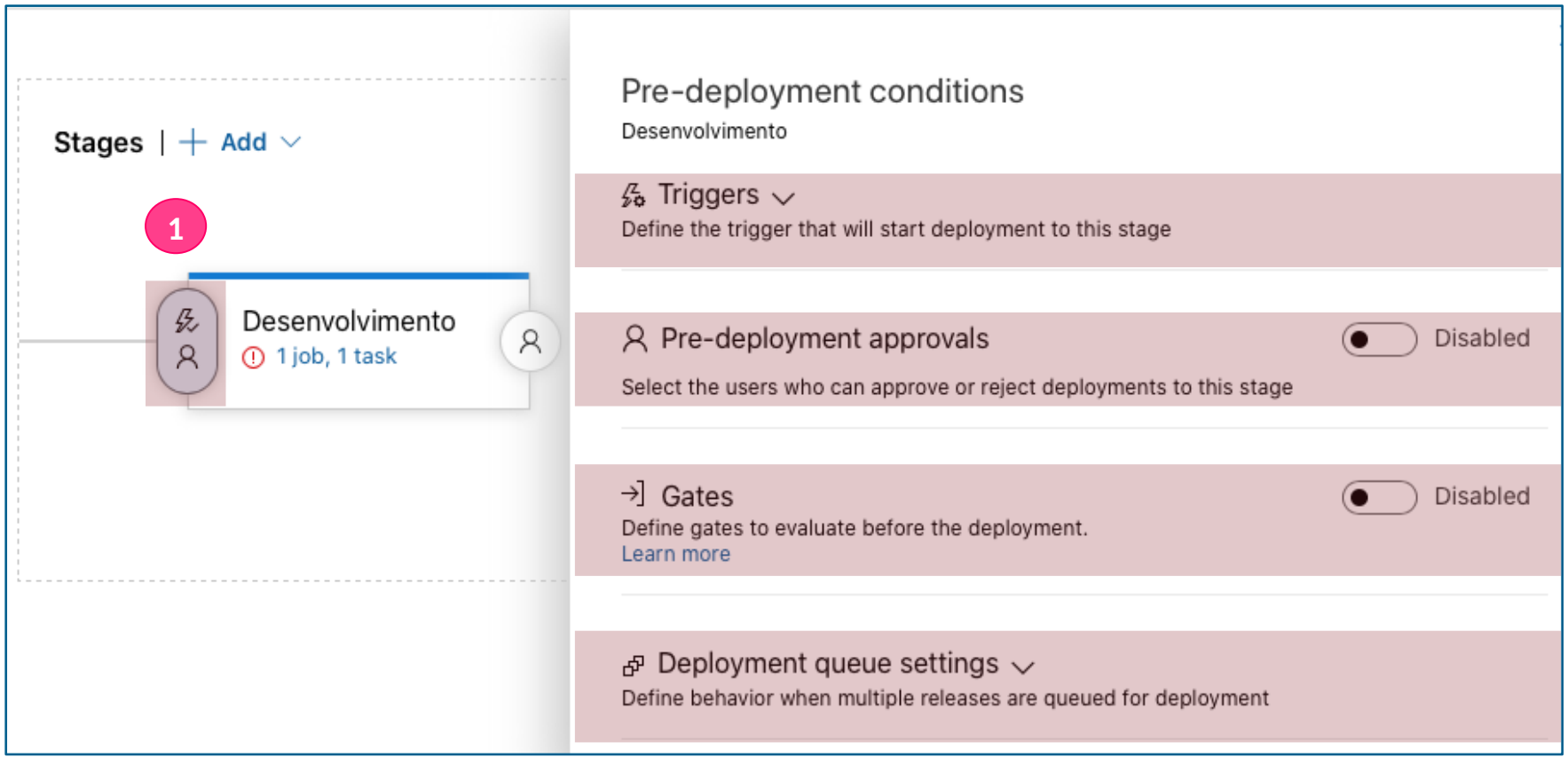
Azure Pipelines – Exercício Java x Oracle

Clique no símbolo do raio no artefato e configure o Continuous Deployment conforme figura abaixo



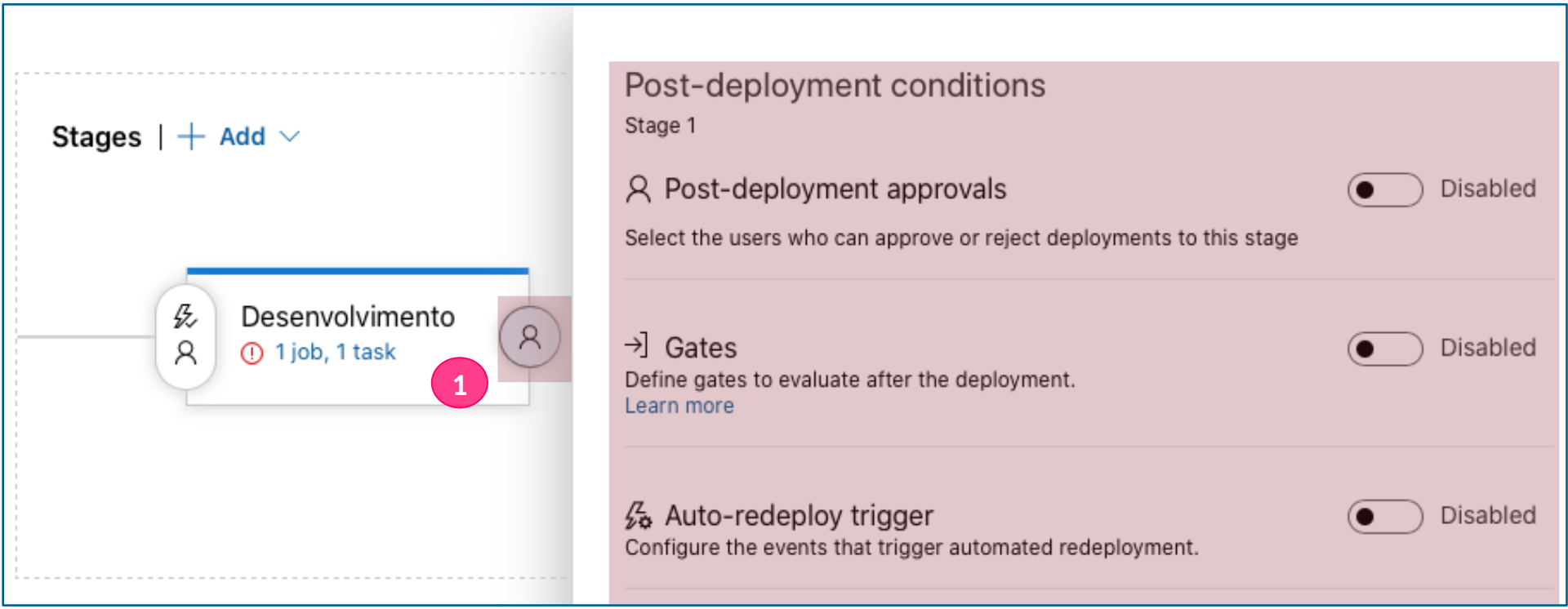
Azure Pipelines – Exercício Java x Oracle

No **Stage Desenvolvimento** clique no ícone  aonde você pode configurar as condições de **Pré-implantação** *(nesse momento opcional)*



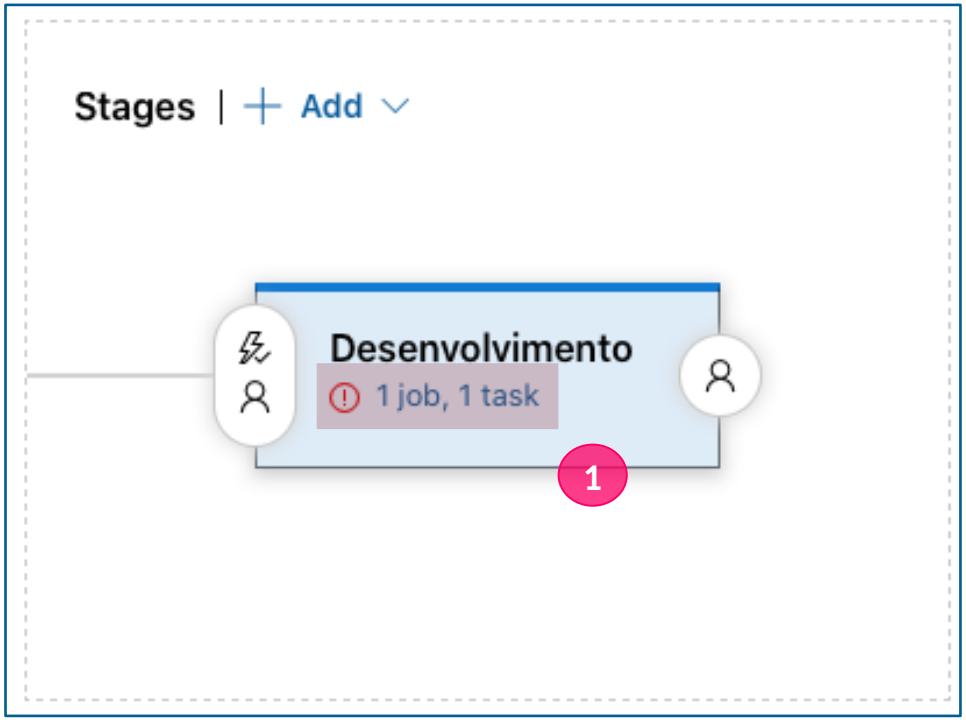
Azure Pipelines – Exercício Java x Oracle

No mesmo **Stage** clique no ícone para configurar as condições de **Pós-implantação** *(nesse momento opcional)*



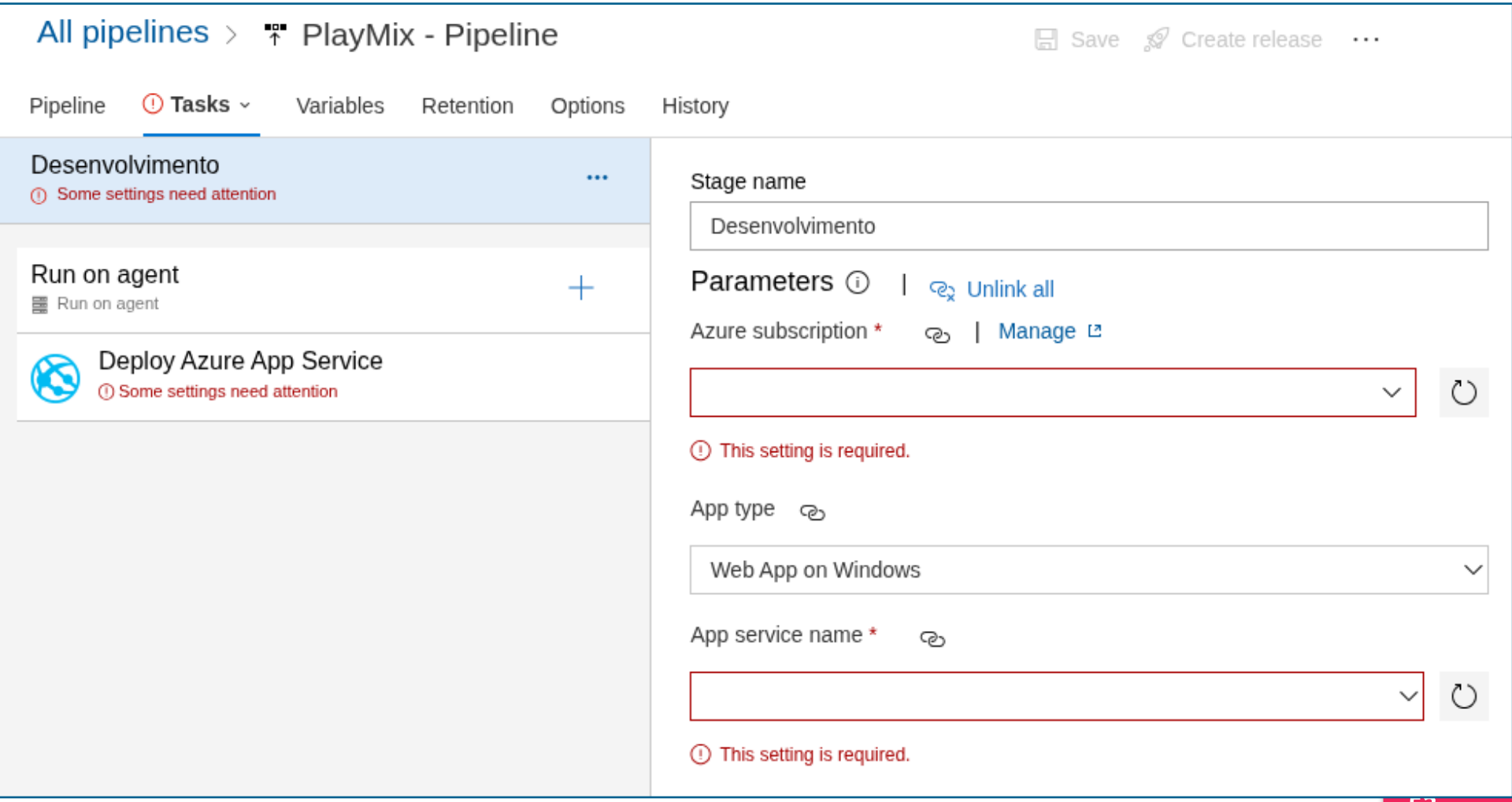
Azure Pipelines – Exercício Java x Oracle

Clique em **Task** para acessar o Agent e suas Tarefas



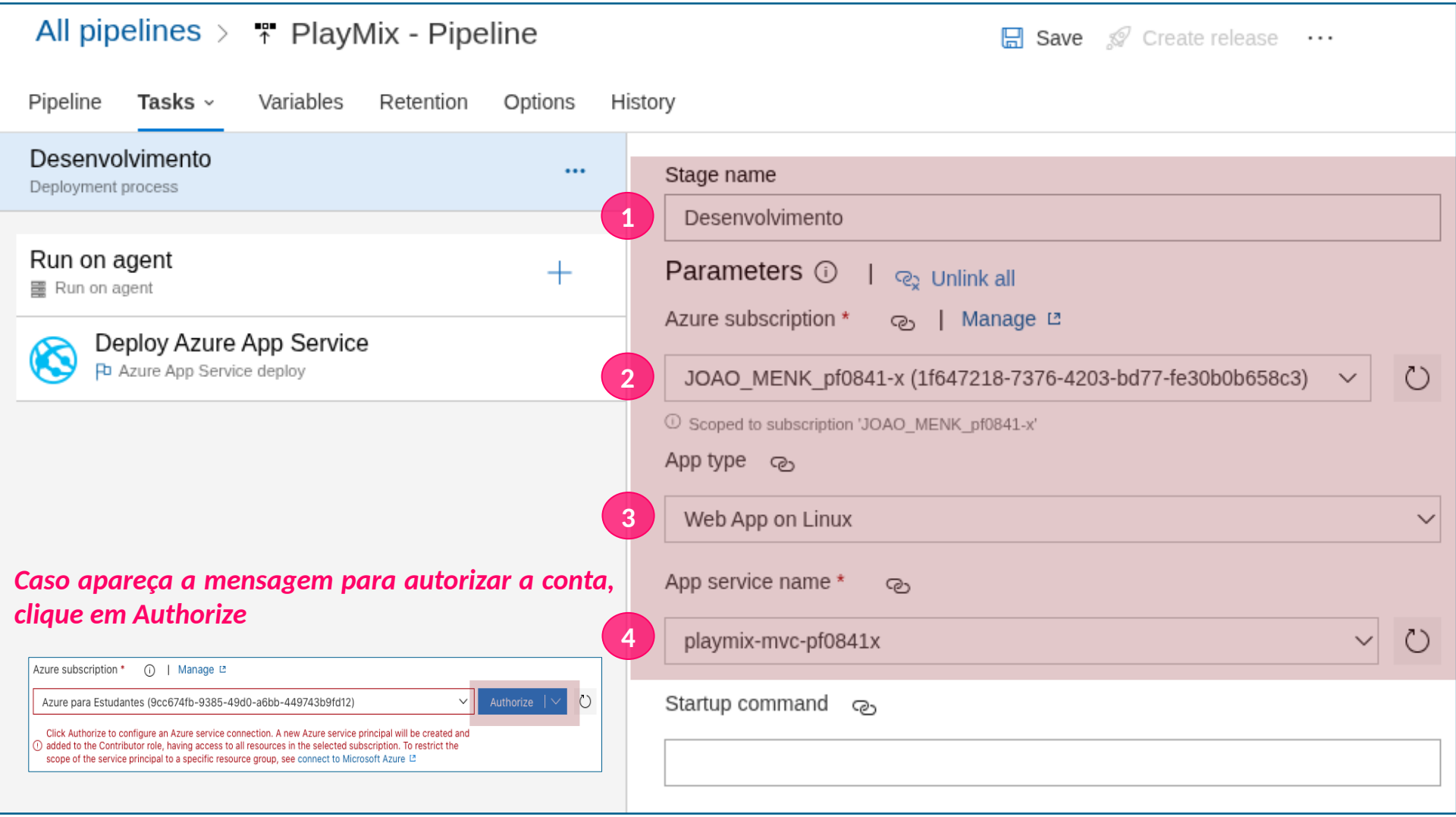
Azure Pipelines – Exercício Java x Oracle

A seguinte tela irá ser exibida, **vamos realizar as configurações do Deploy**



Azure Pipelines – Exercício Java x Oracle

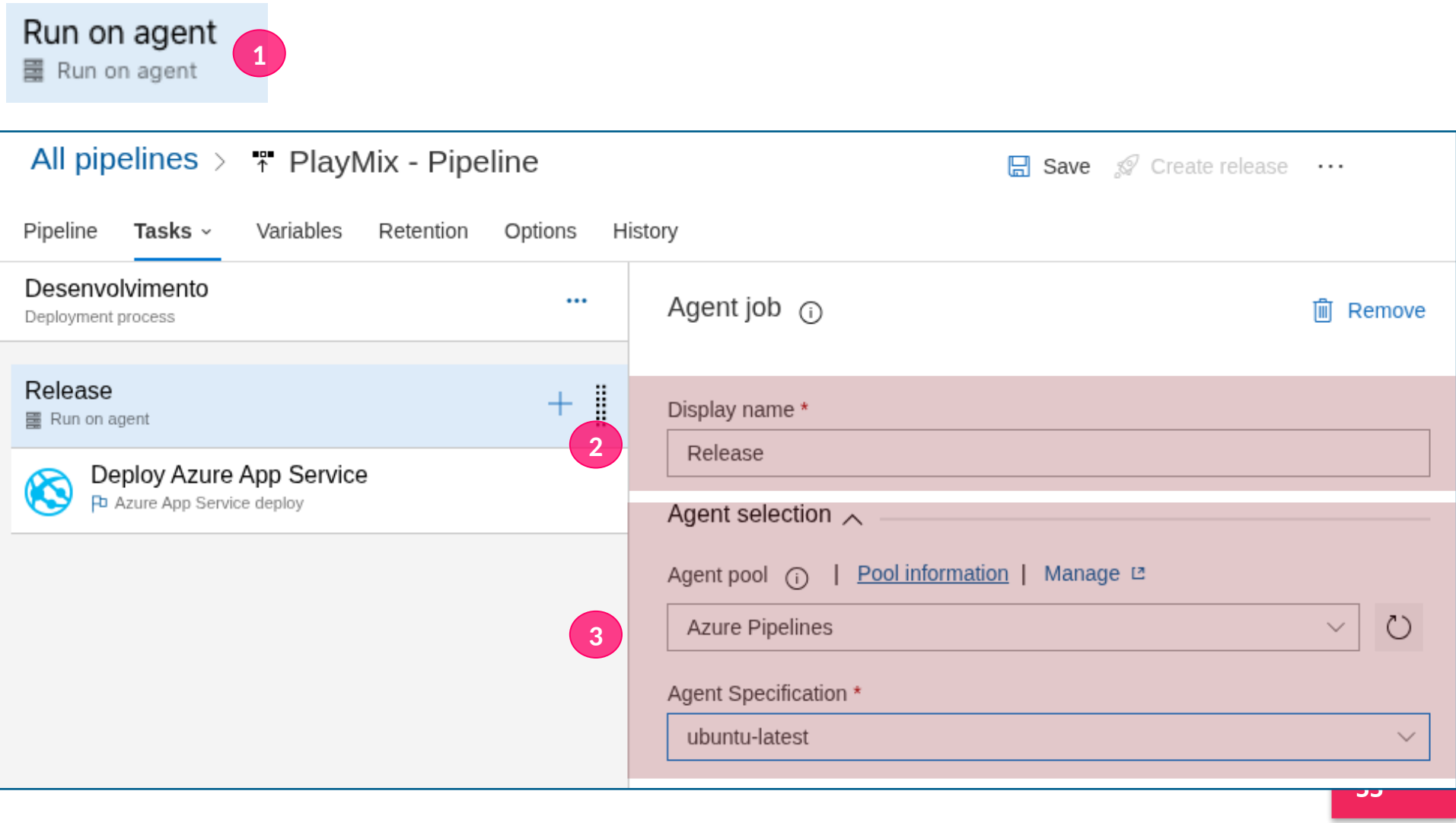
Altere as propriedades da seguinte forma



Caso apareça a mensagem para autorizar a conta, clique em Authorize

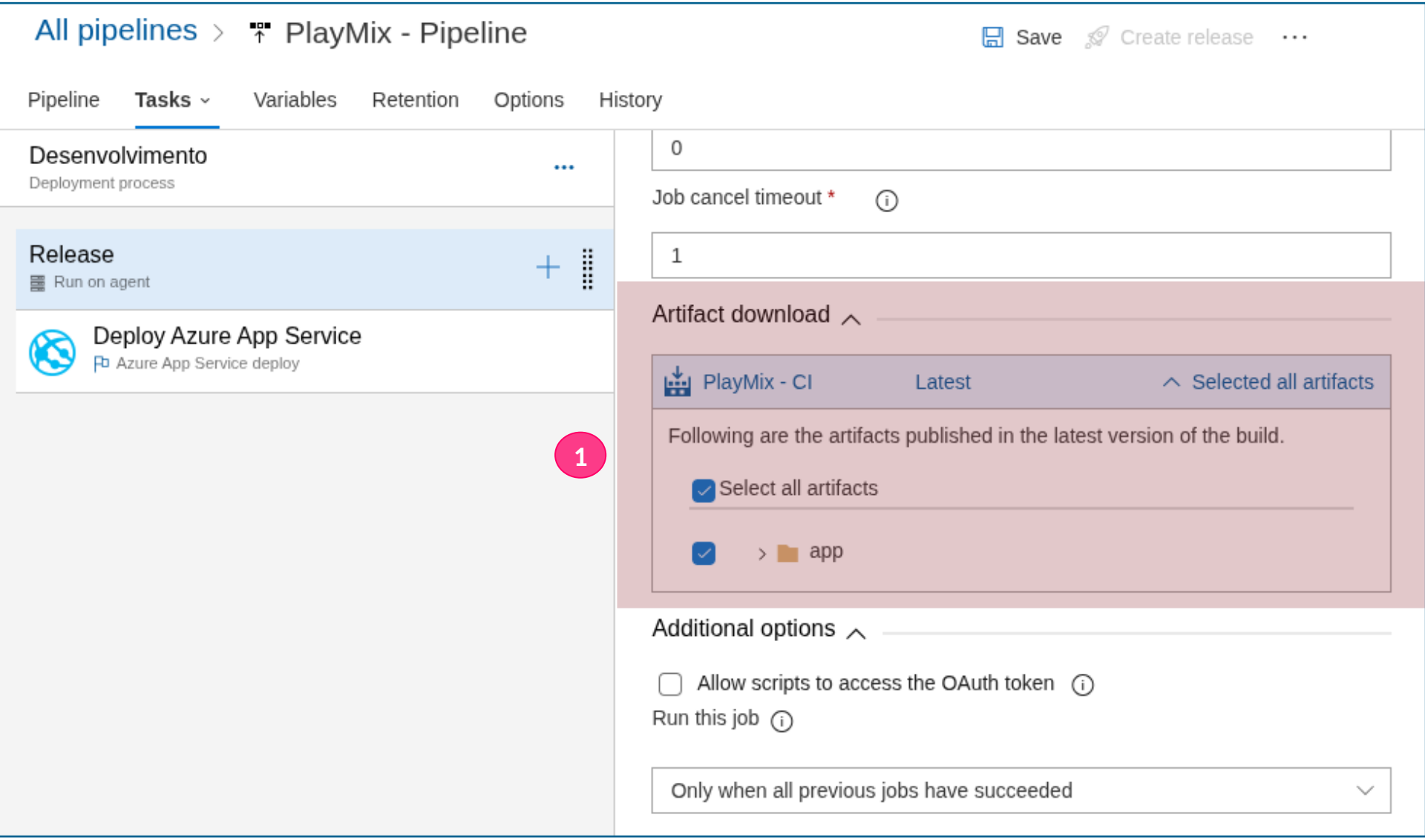
Azure Pipelines – Exercício Java x Oracle

Altere as propriedades da seguinte forma



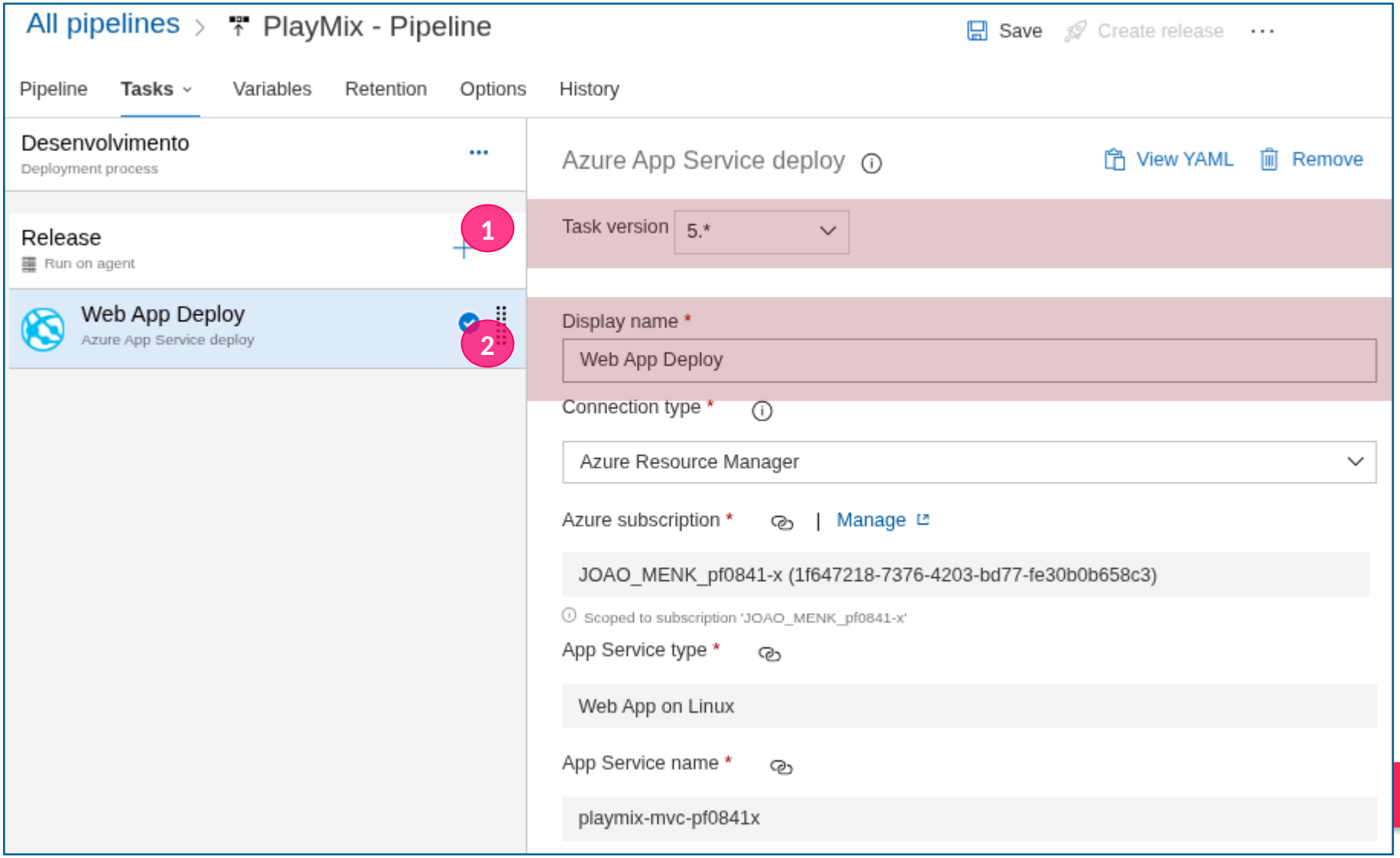
Azure Pipelines – Exercício Java x Oracle

Verifique a fonte do Deploy



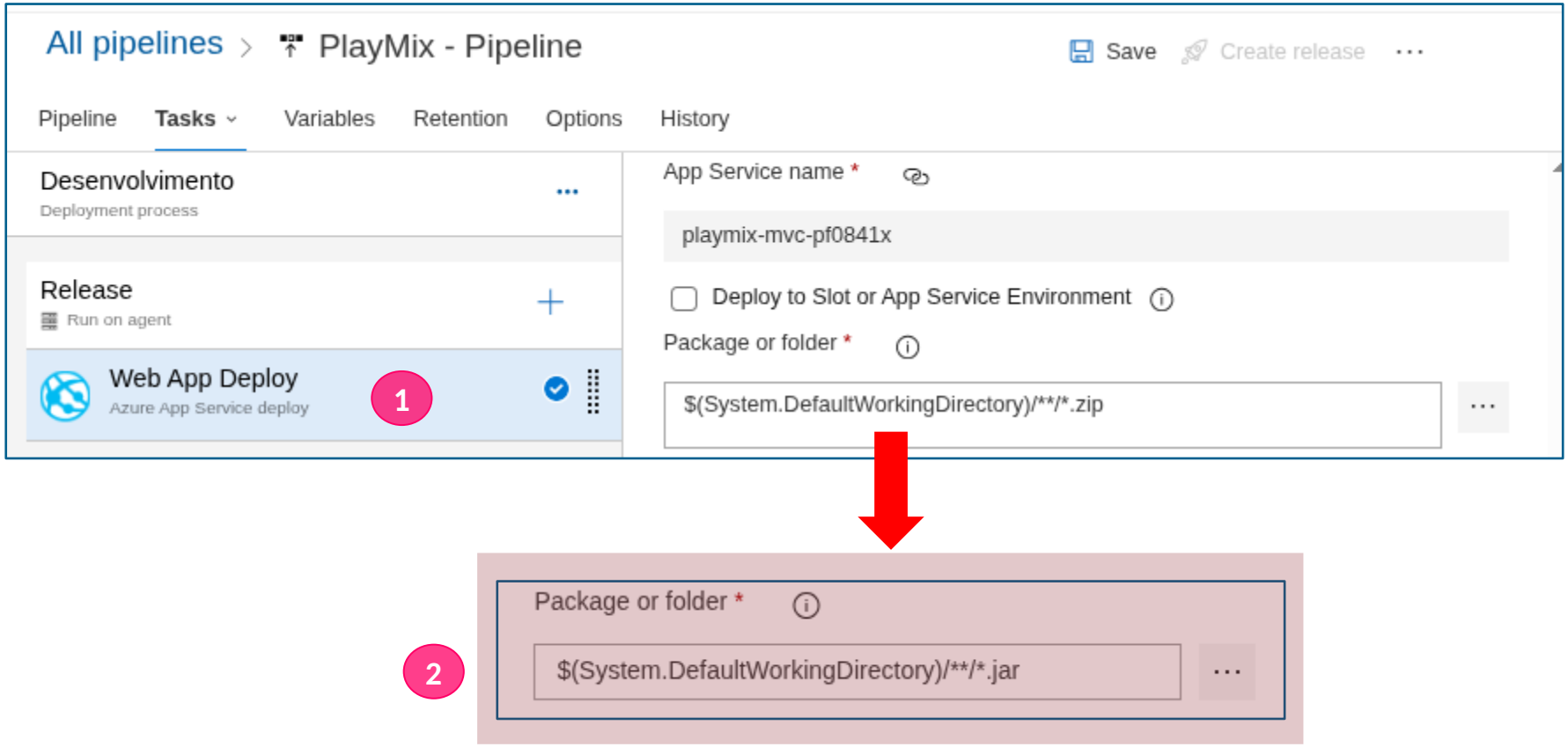
Azure Pipelines – Exercício Java x Oracle

Altere as propriedades da seguinte forma



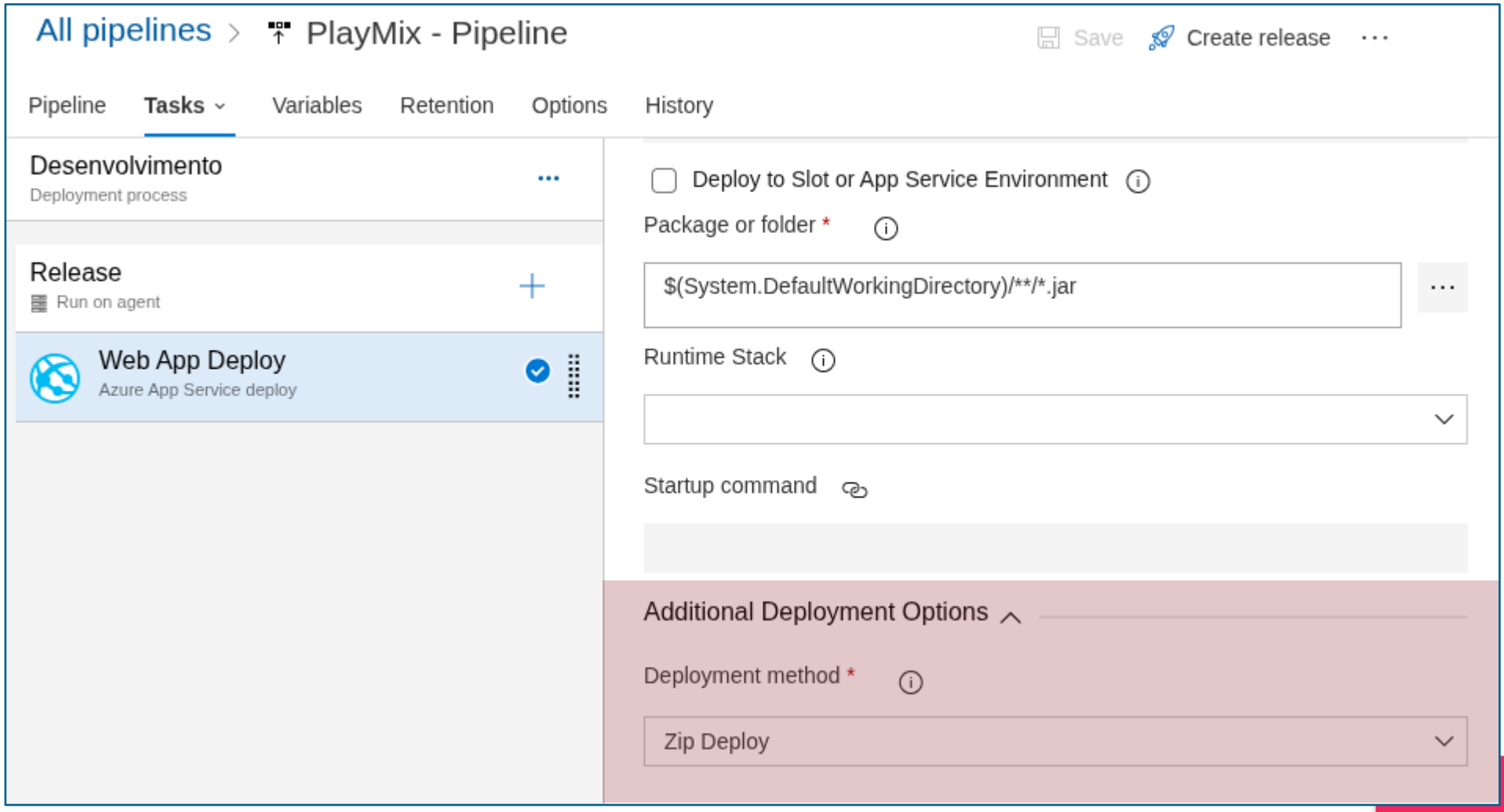
Azure Pipelines – Exercício Java x Oracle

Altere o tipo do pacote para **JAR**



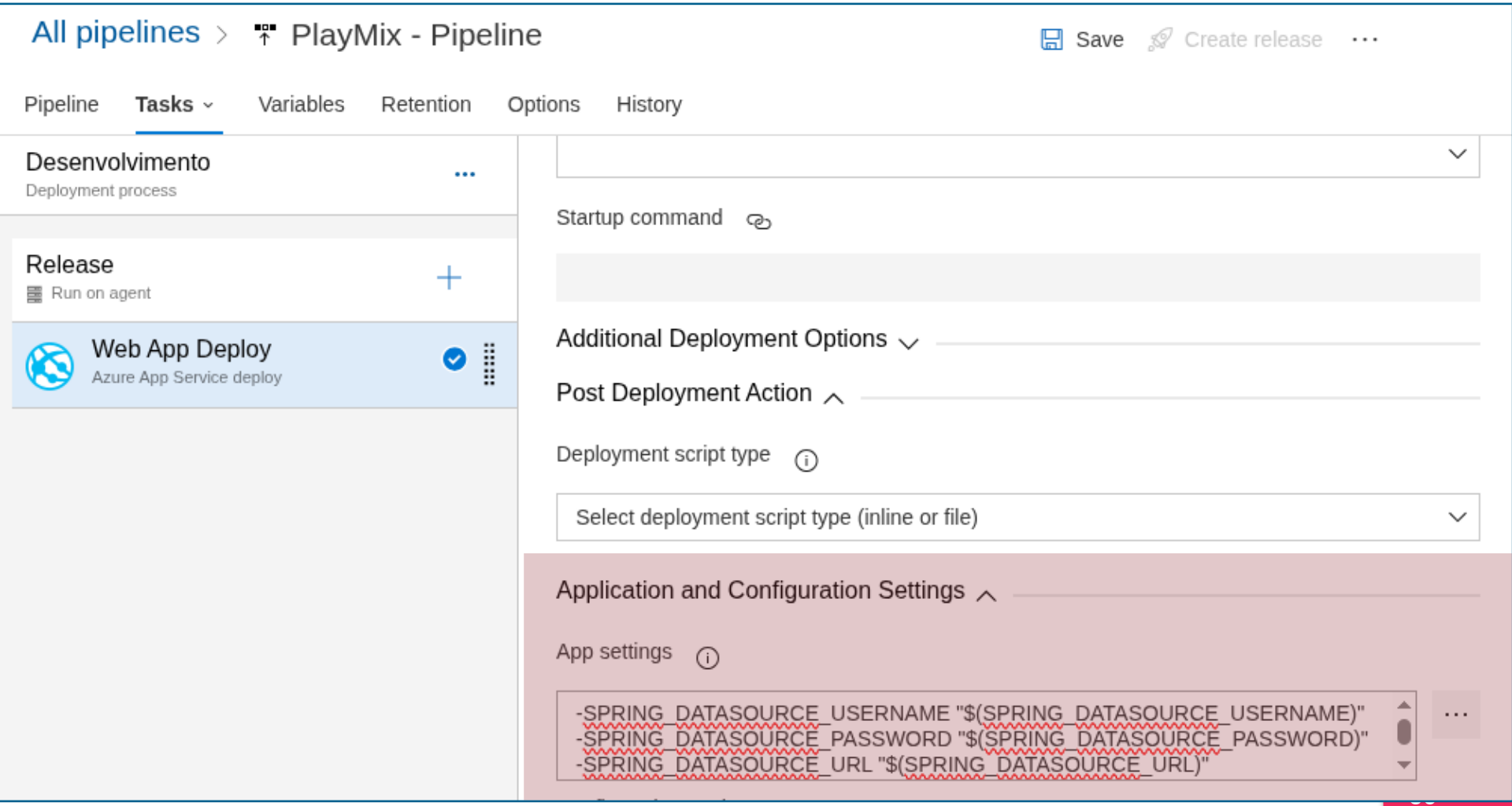
Azure Pipelines – Exercício Java x Oracle

Inclua o Método de Deploy



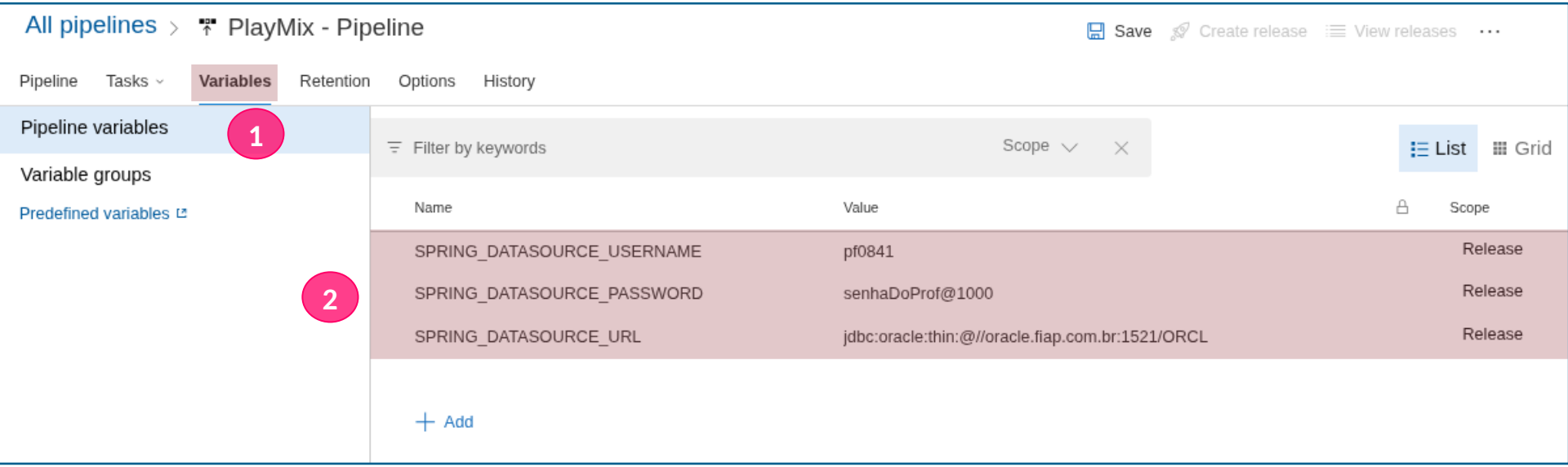
Azure Pipelines – Exercício Java x Oracle

Inclua as Variáveis de Ambiente do Serviço de Aplicativo



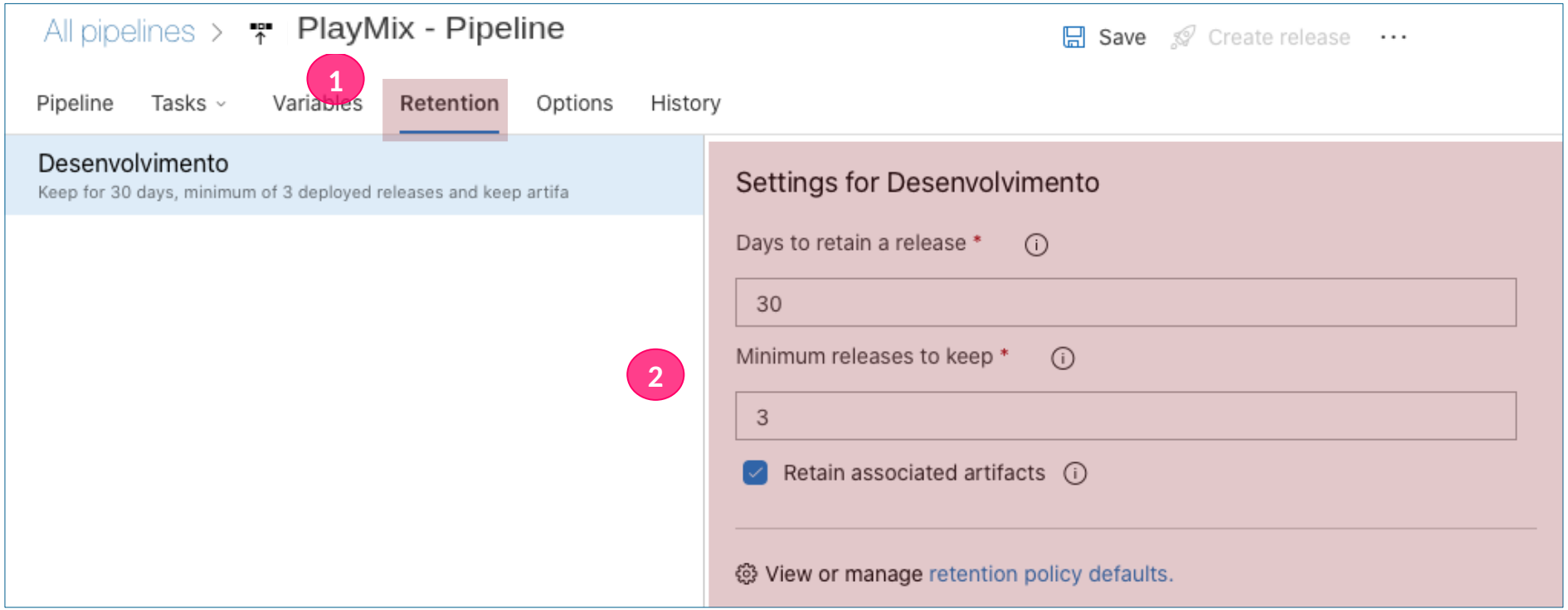
Azure Pipelines – Exercício Java x Oracle

Na aba **Variables** podemos criar variáveis específicas por Stage ou para toda a Release



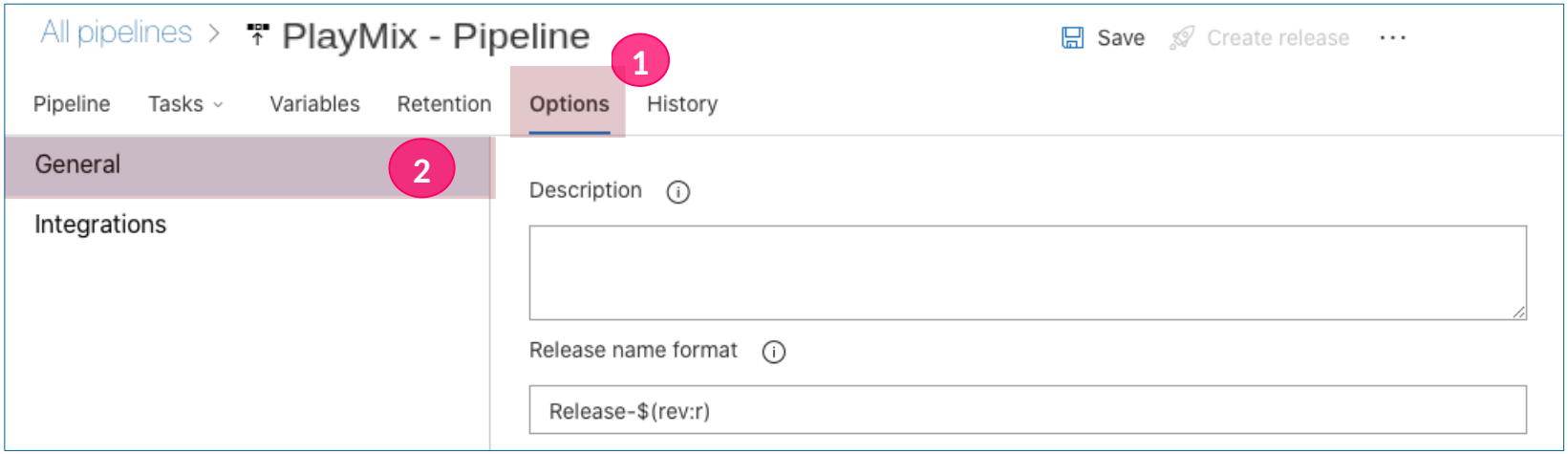
Azure Pipelines – Exercício Java x Oracle

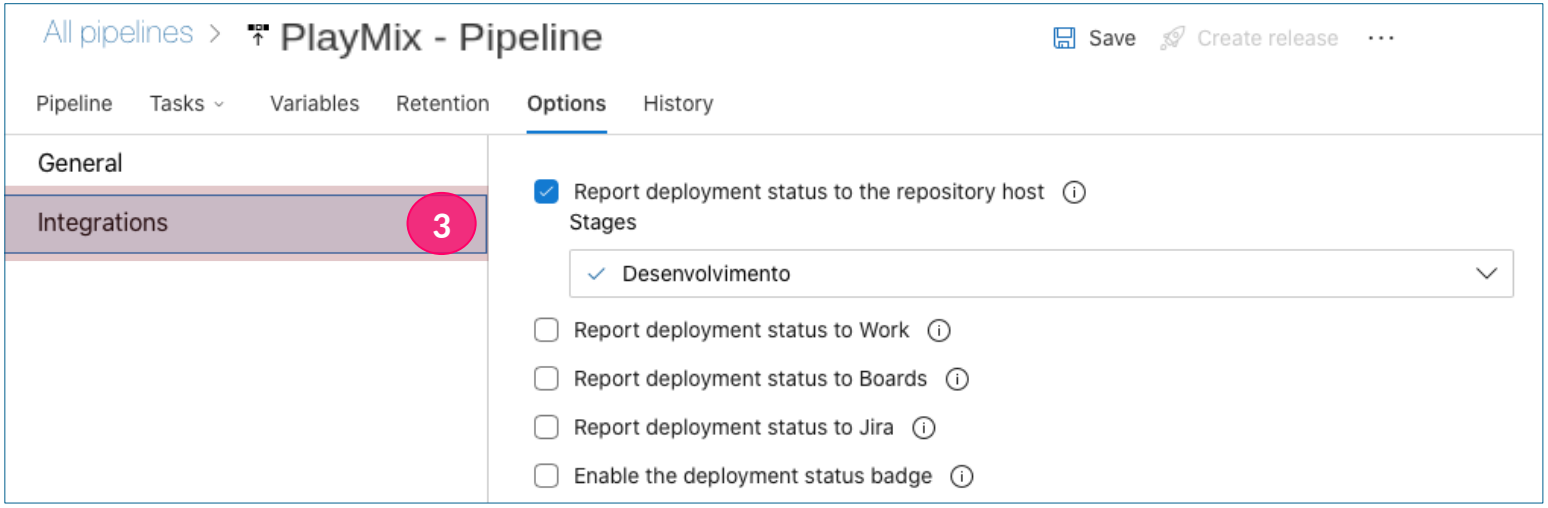
Na aba **Retention** podemos definir o tempo e a quantidade de Releases que serão retidos



Azure Pipelines – Exercício Java x Oracle

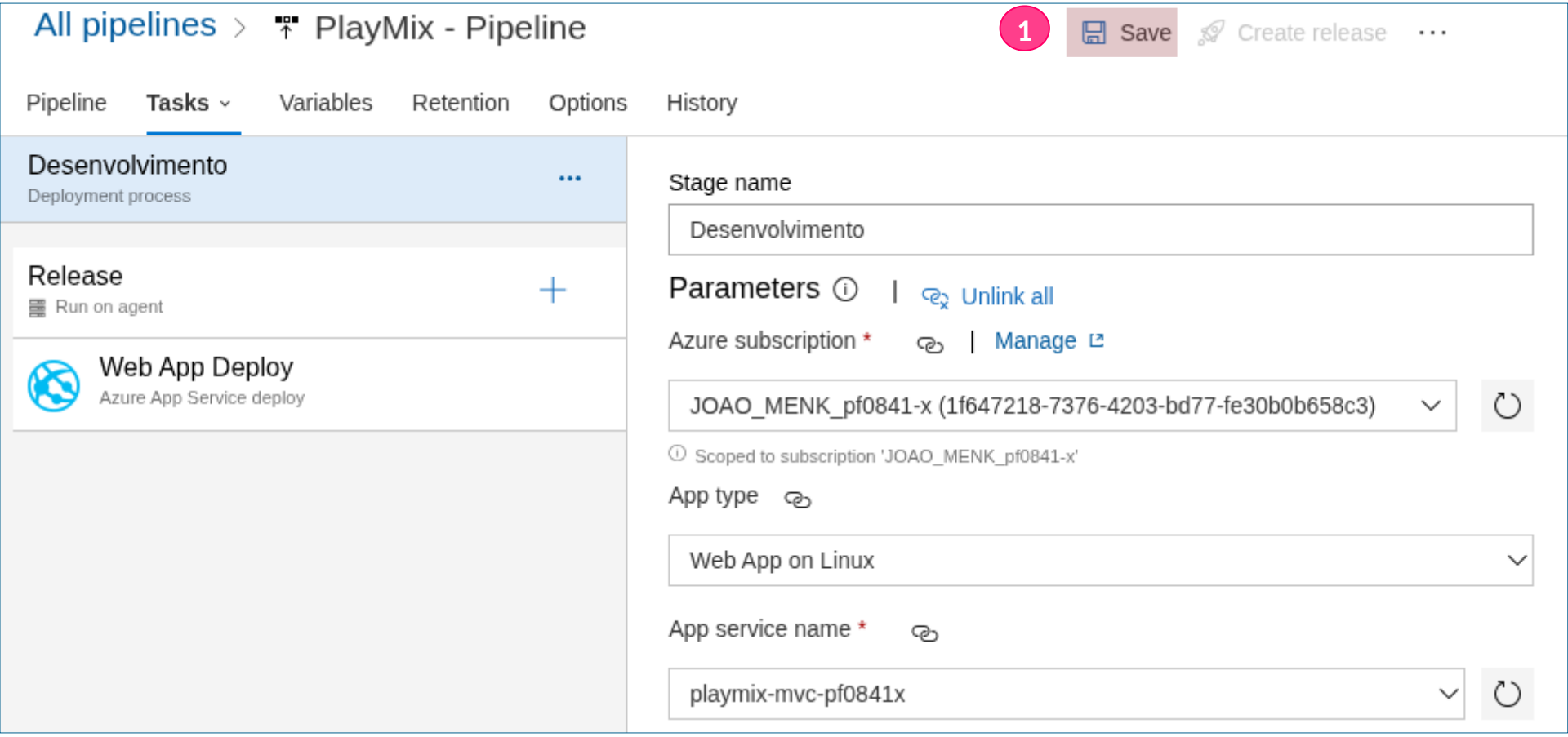
Na aba **Options** podemos incluir uma descrição para o Release e a formatação do nome





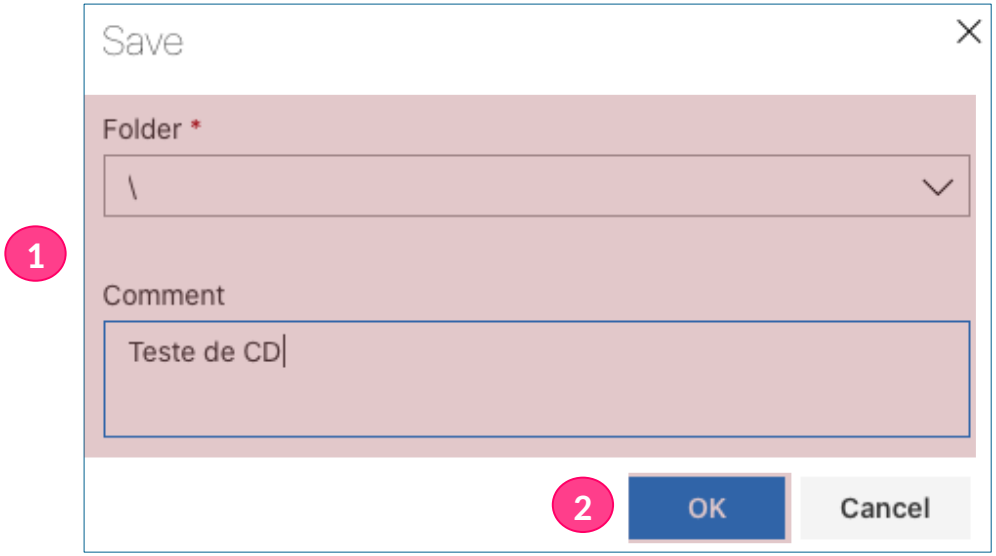
Azure Pipelines – Exercício Java x Oracle

Com a Pipeline de Release configurada, vamos salvar e executar



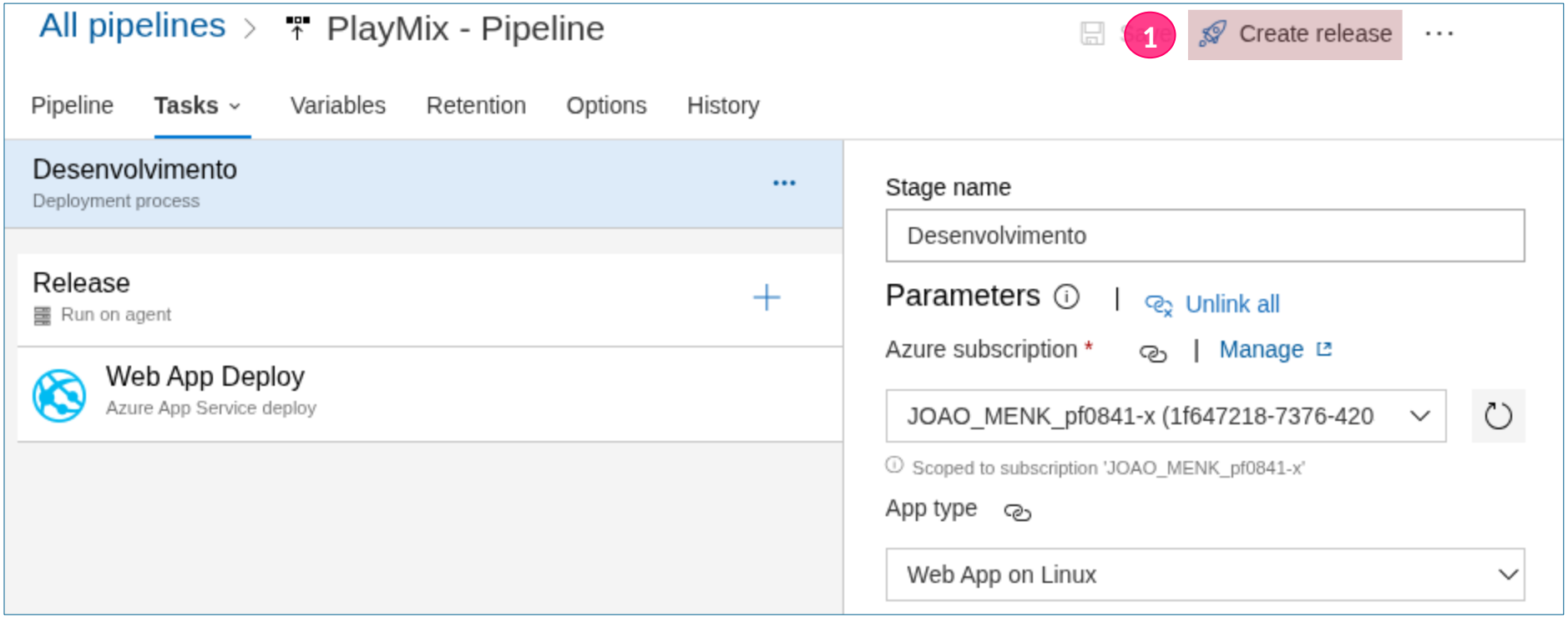
Azure Pipelines – Exercício Java x Oracle

Informe a pasta e um comentário



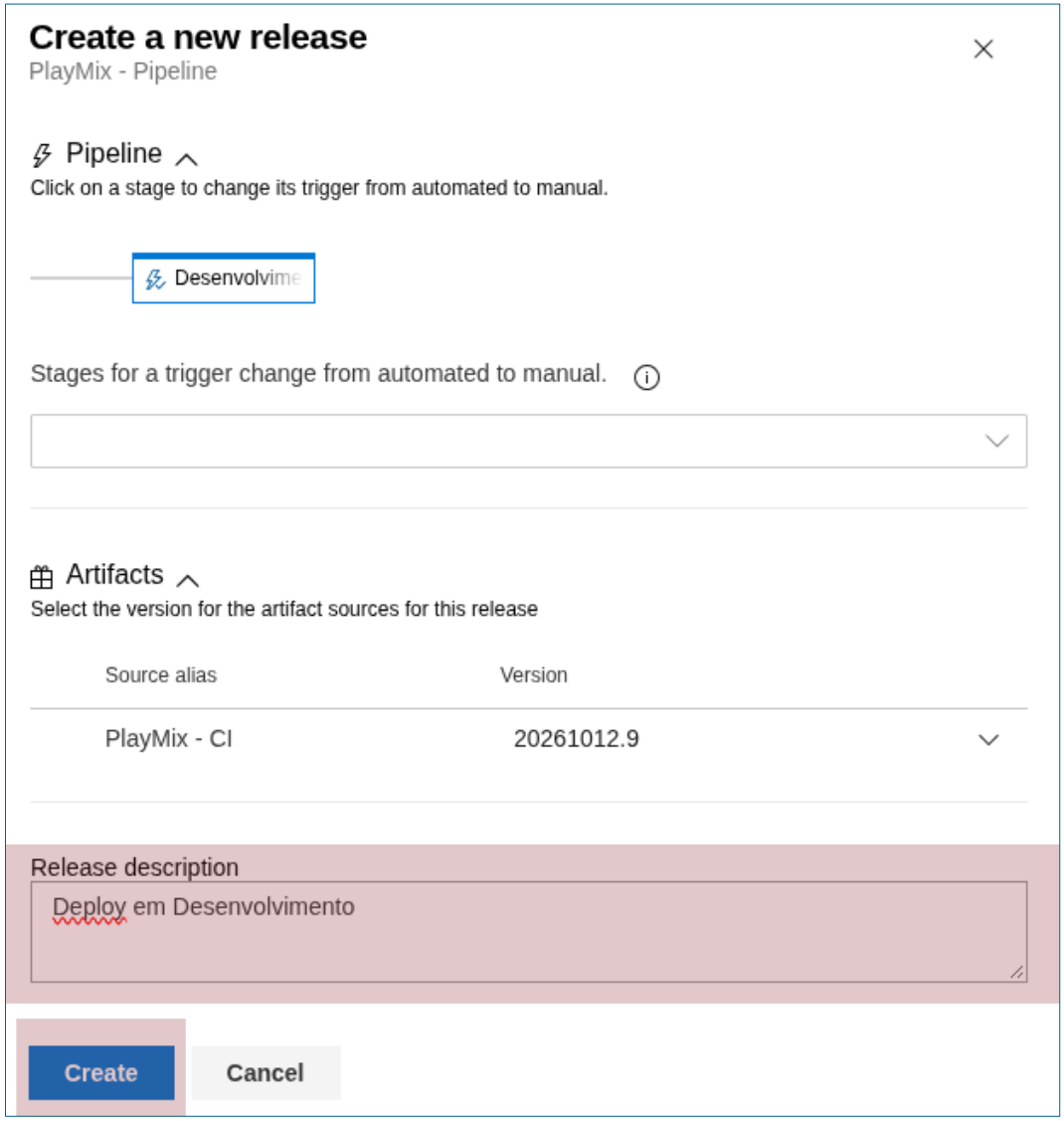
Azure Pipelines – Exercício Java x Oracle

Clique em **Create release**



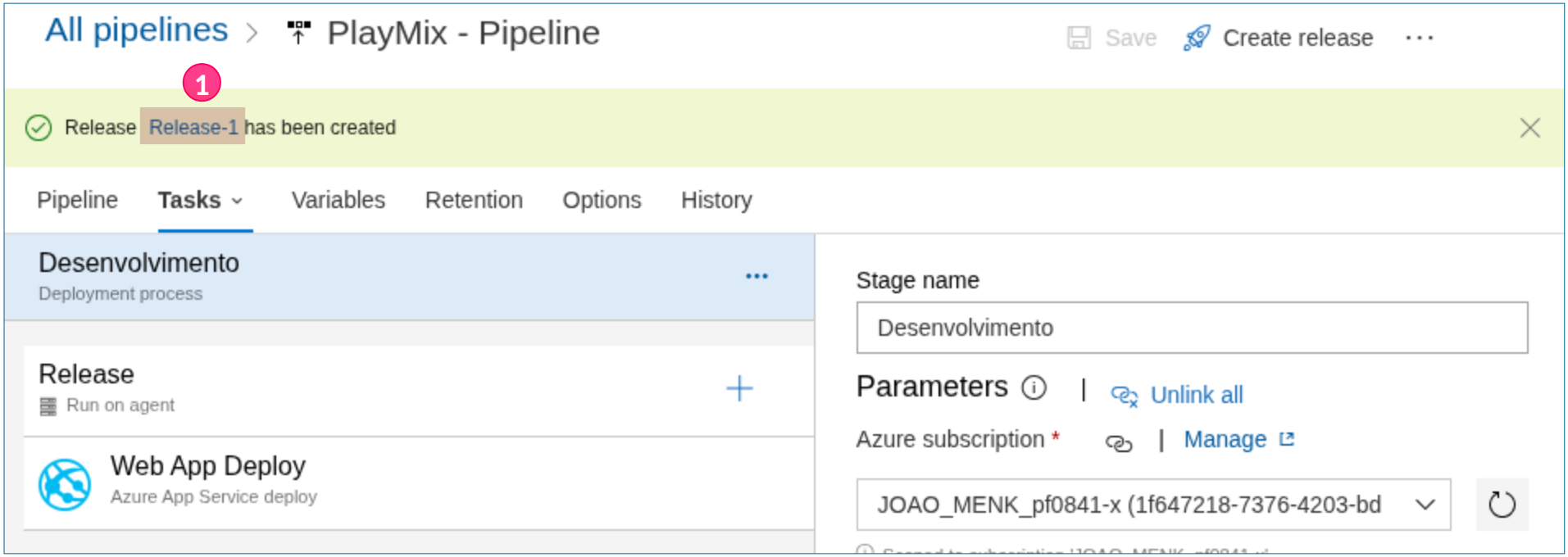
Azure Pipelines – Exercício Java x Oracle

Informe uma descrição para a Release e clique em **Create**



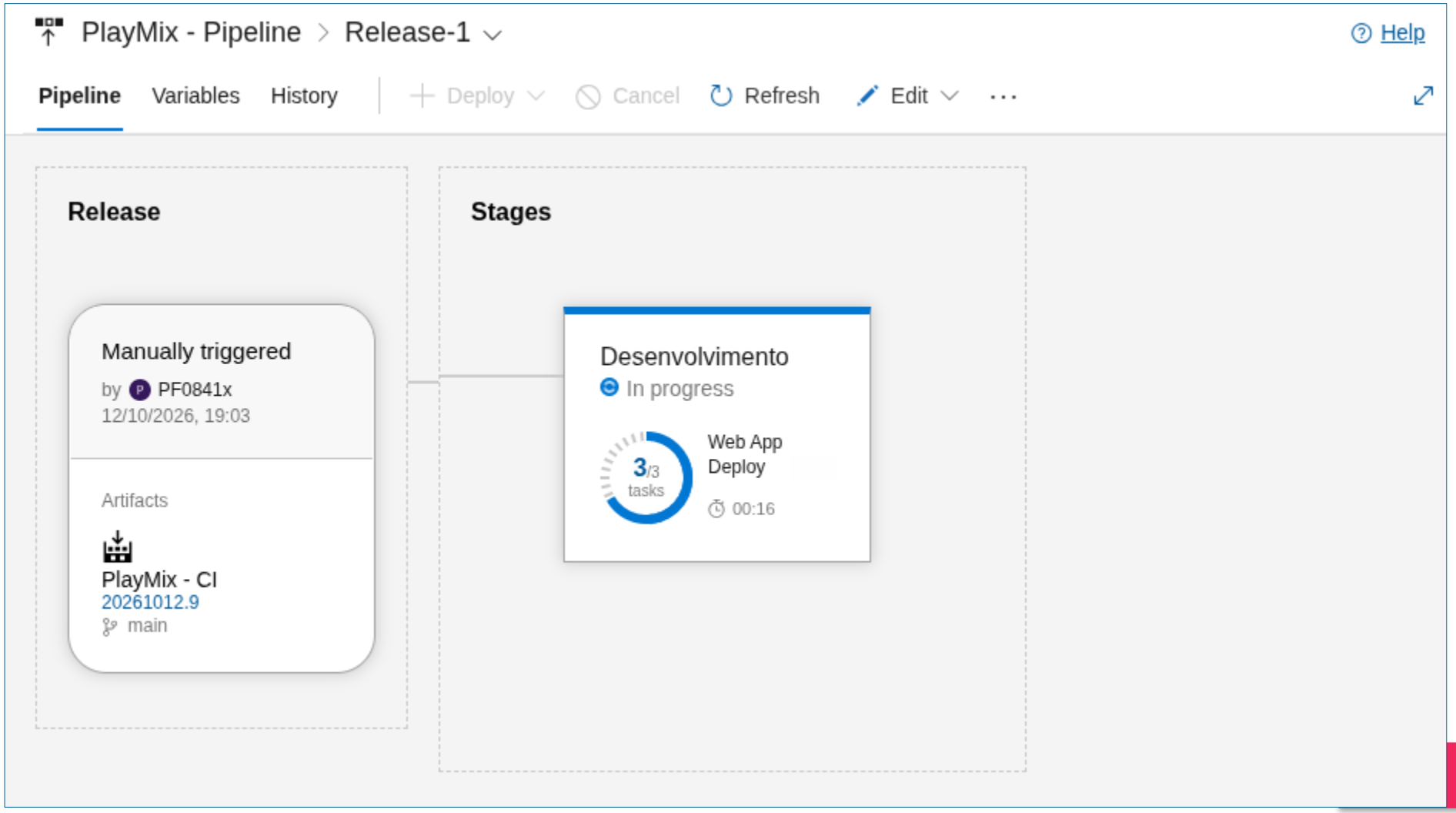
Azure Pipelines – Exercício Java x Oracle

Vamos acompanhar a Pipeline



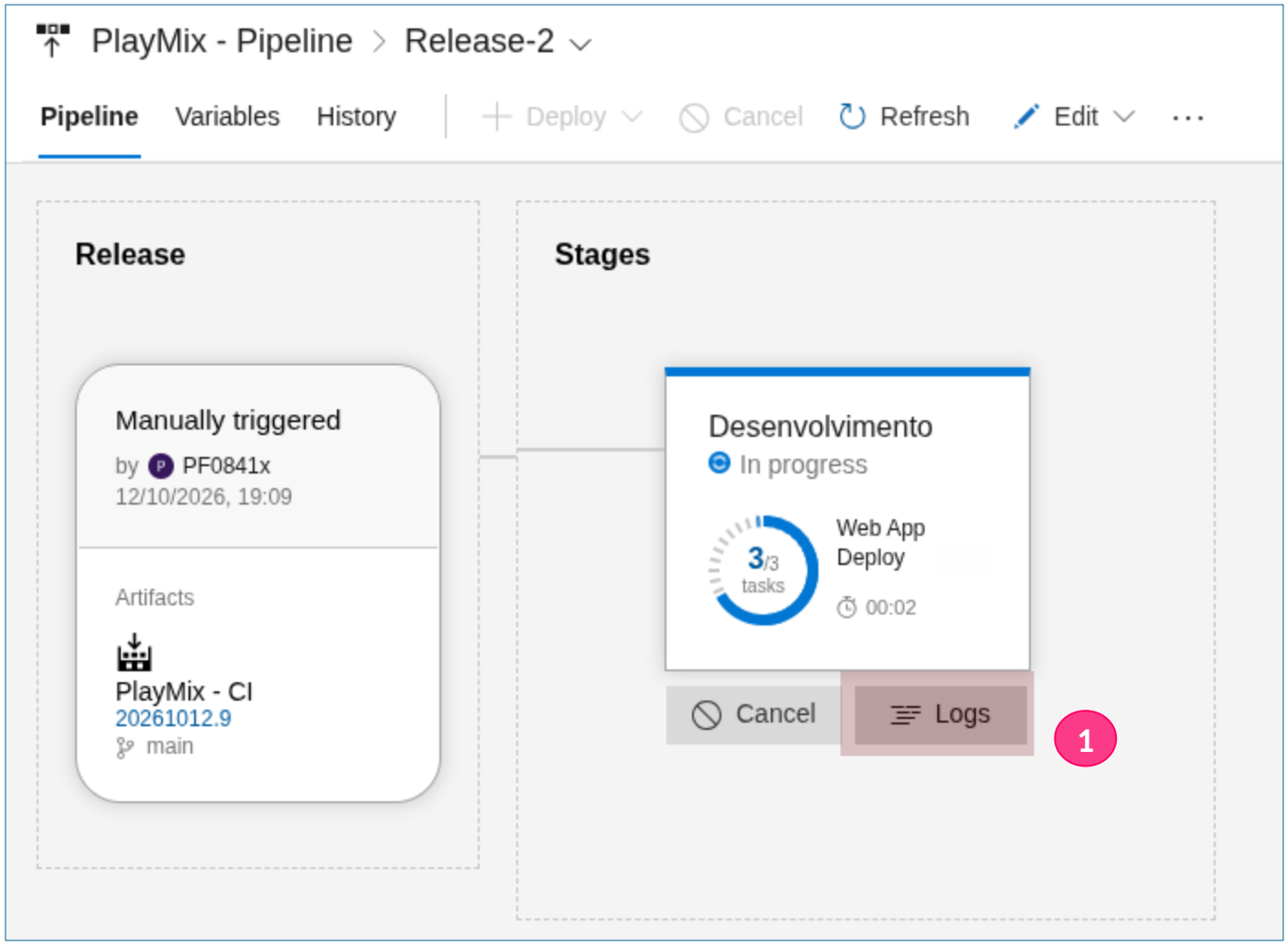
Azure Pipelines – Exercício Java x Oracle

Vamos acompanhar a Pipeline



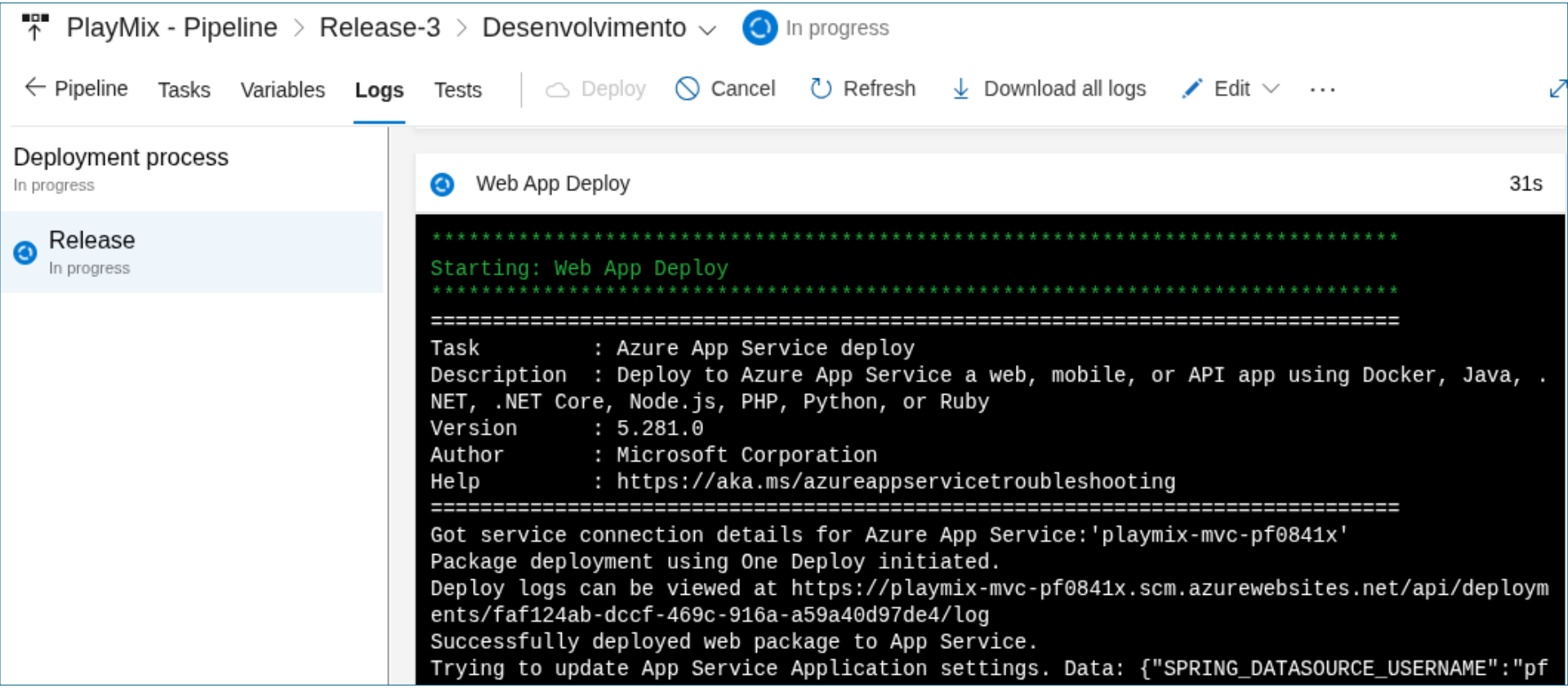
Azure Pipelines – Exercício Java x Oracle

Passe o mouse por cima do Stage Desenvolvimento e clique em **Logs**



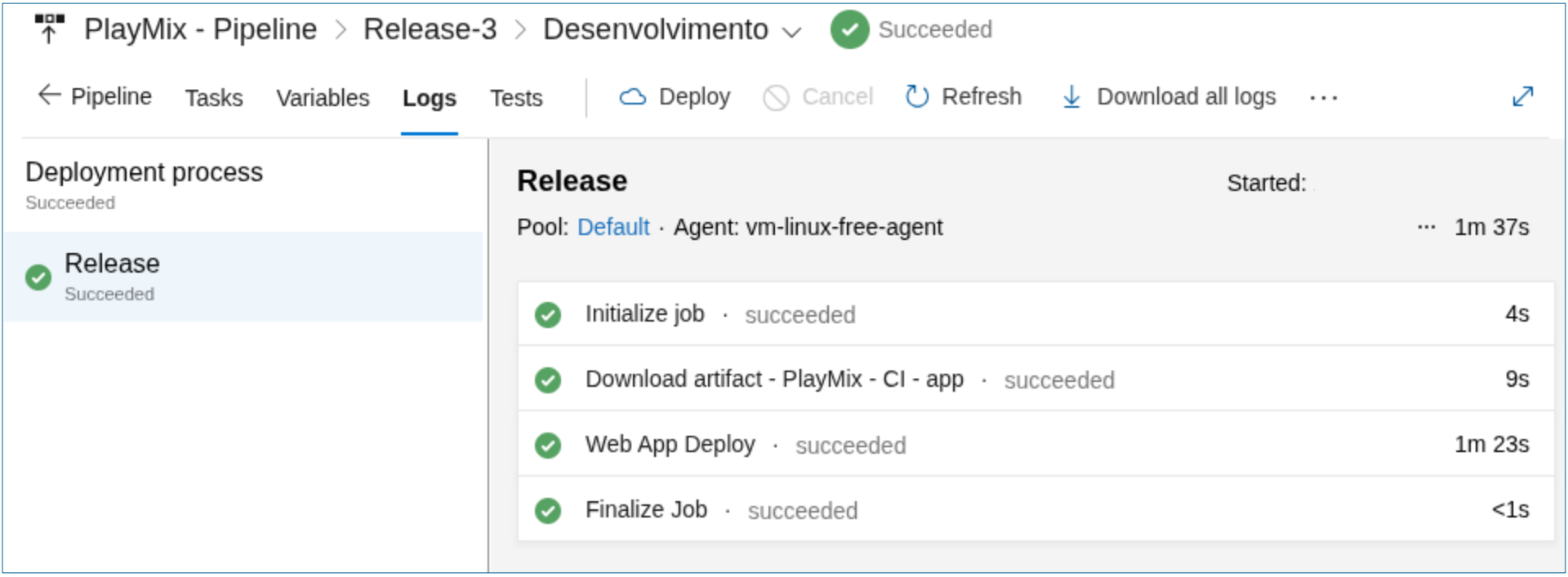
Azure Pipelines – Exercício Java x Oracle

Acompanhando os Logs



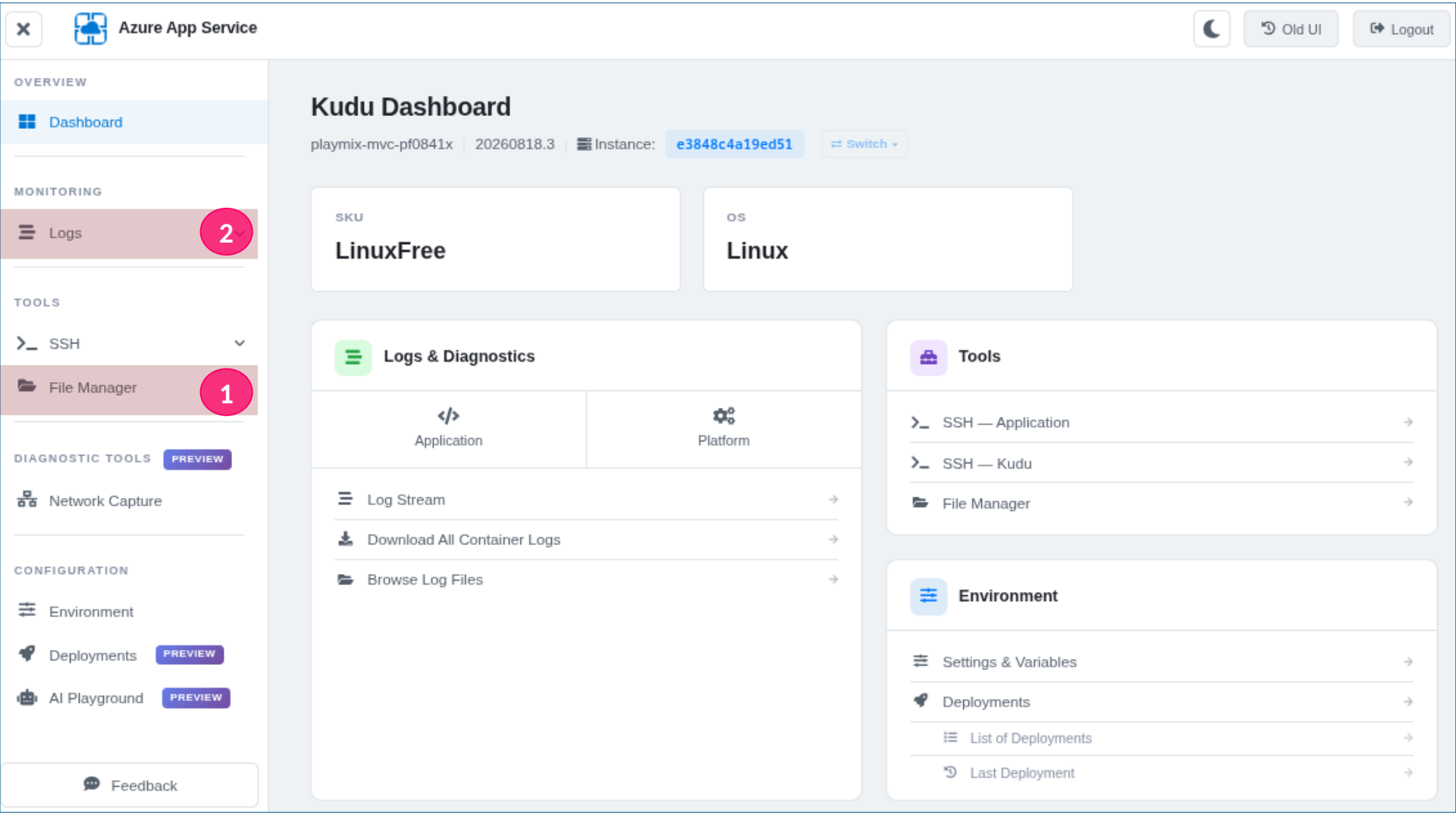
Azure Pipelines – Exercício Java x Oracle

Release executada



Azure Pipelines – Exercício Java x Oracle

No **Portal da Azure** verifique pelo *Kudo Service* a situação do Deploy na infra



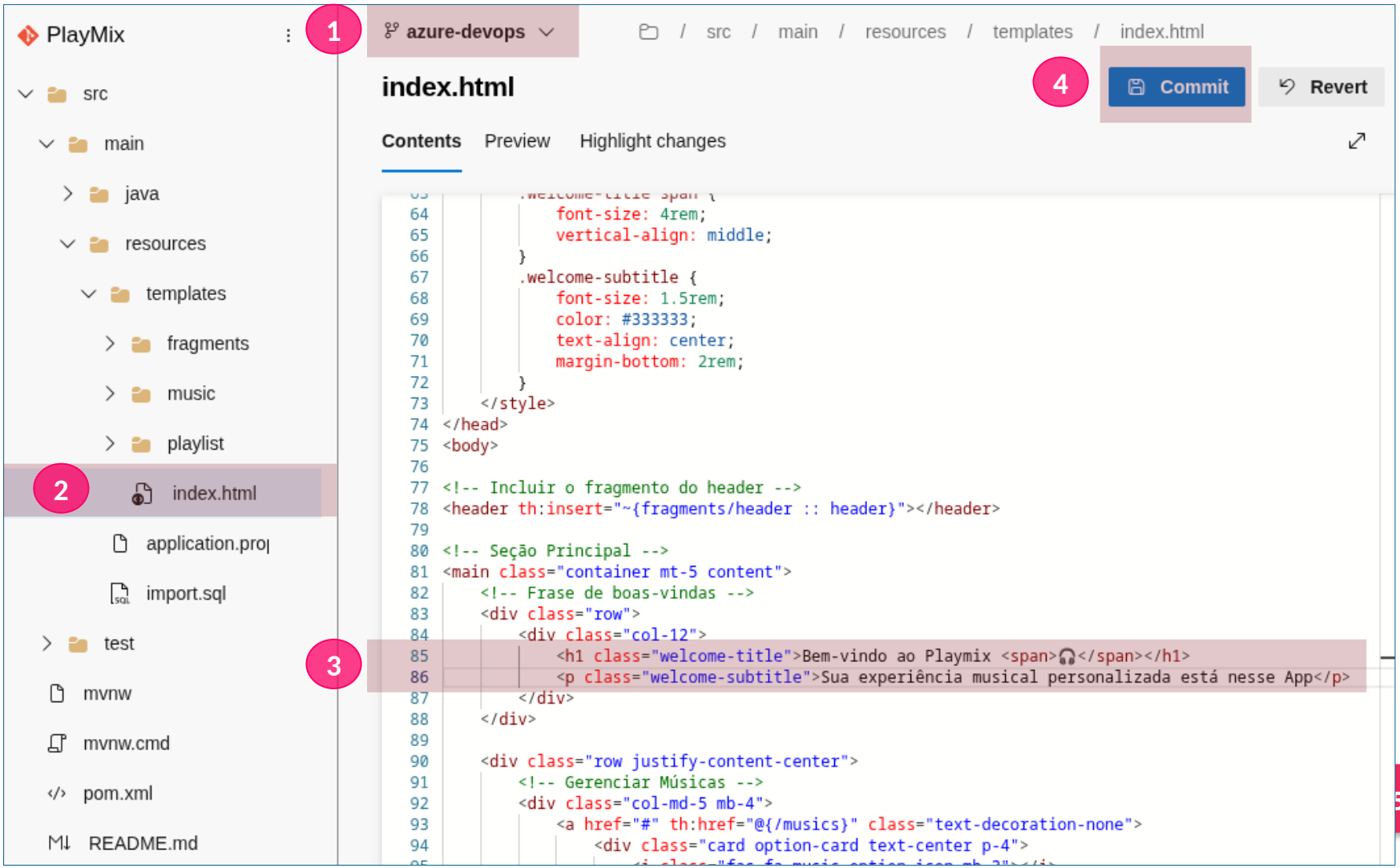
Azure Pipelines – Exercício Java x Oracle

Acesse o App disponibilizado na nuvem através da URL fornecida pelo Serviços de Aplicativo



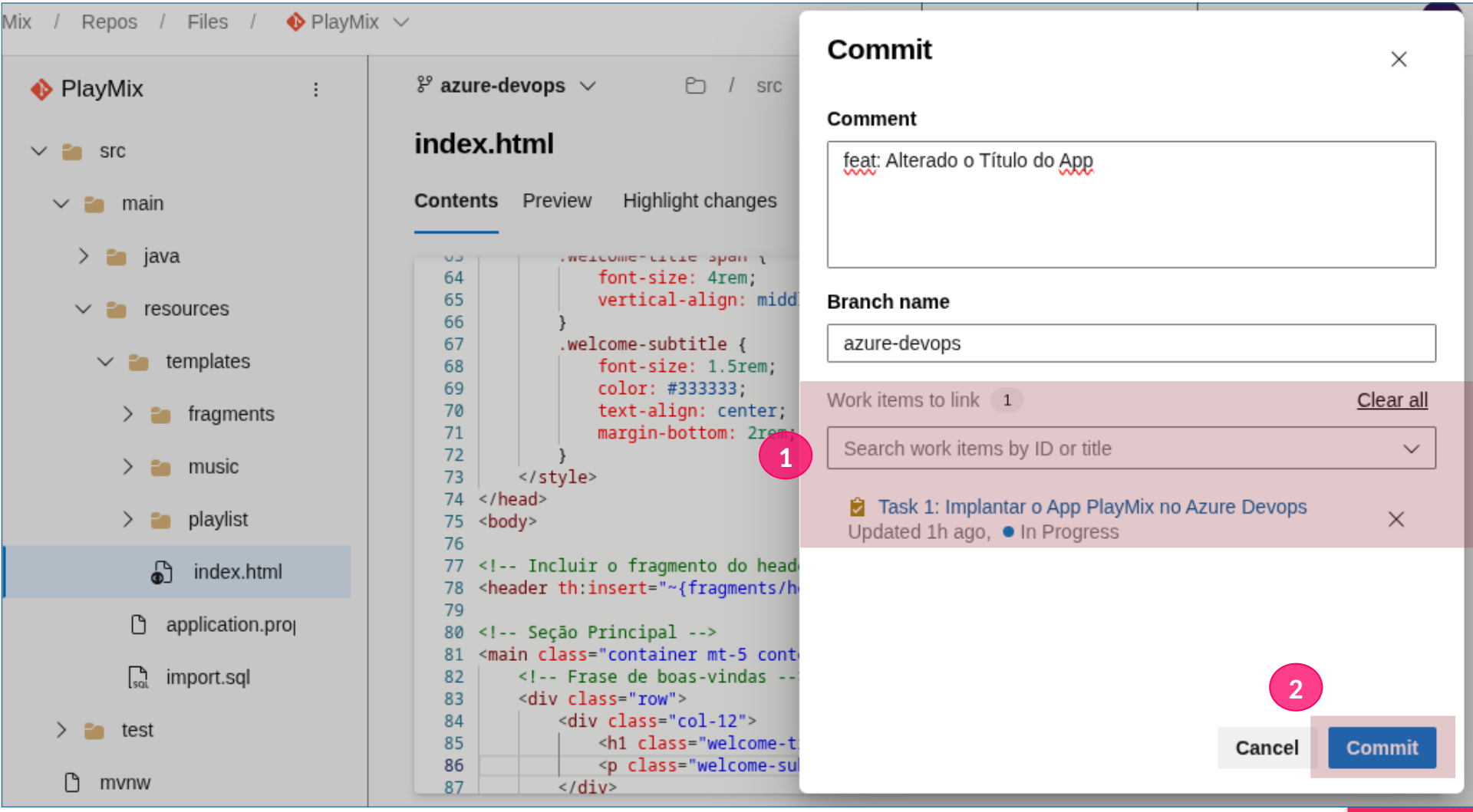
Azure Pipelines – Exercício Java x Oracle

Vamos testar o CI/CD completo. Realize a alteração abaixo no código em sua Branch criada



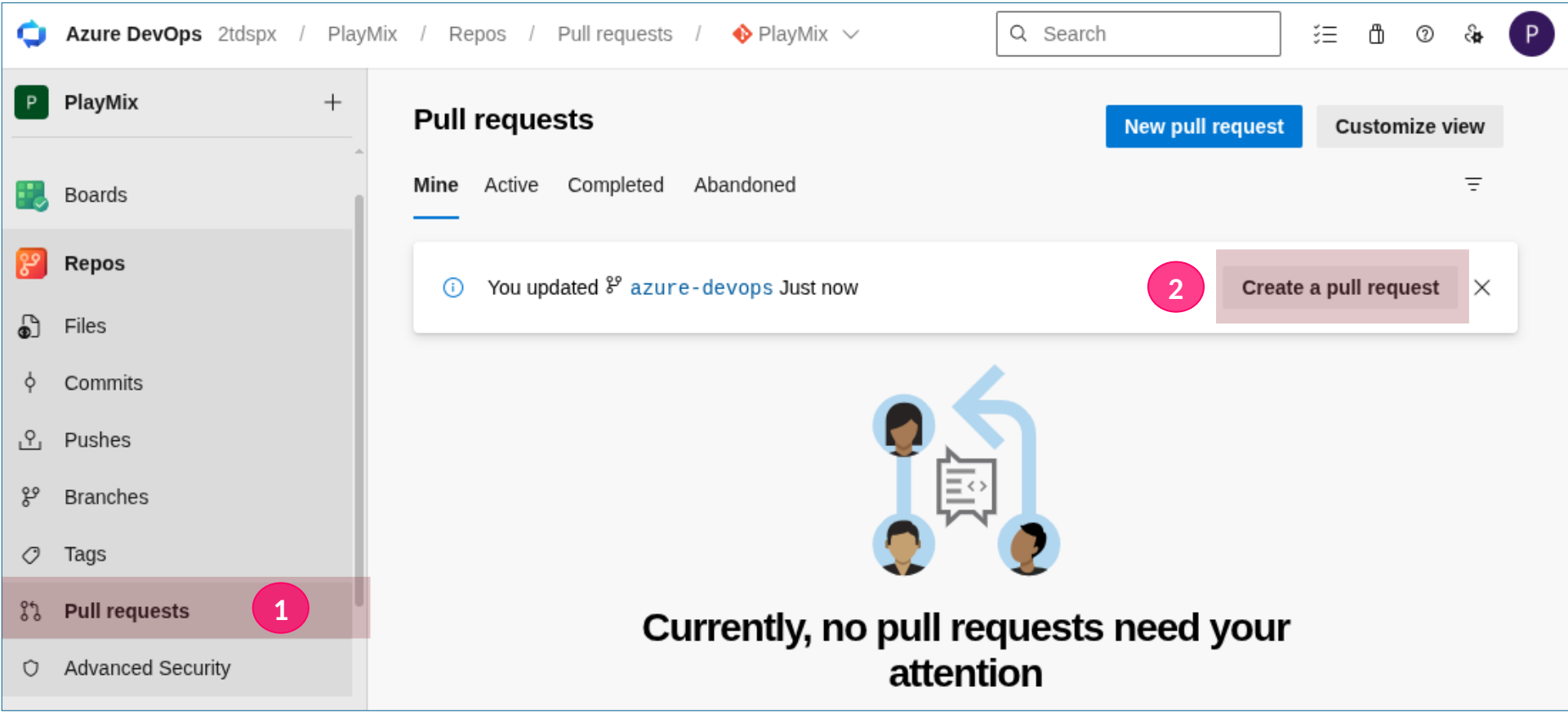
Azure Pipelines – Exercício Java x Oracle

Não esqueça de **incluir o Work Item que está trabalhando**



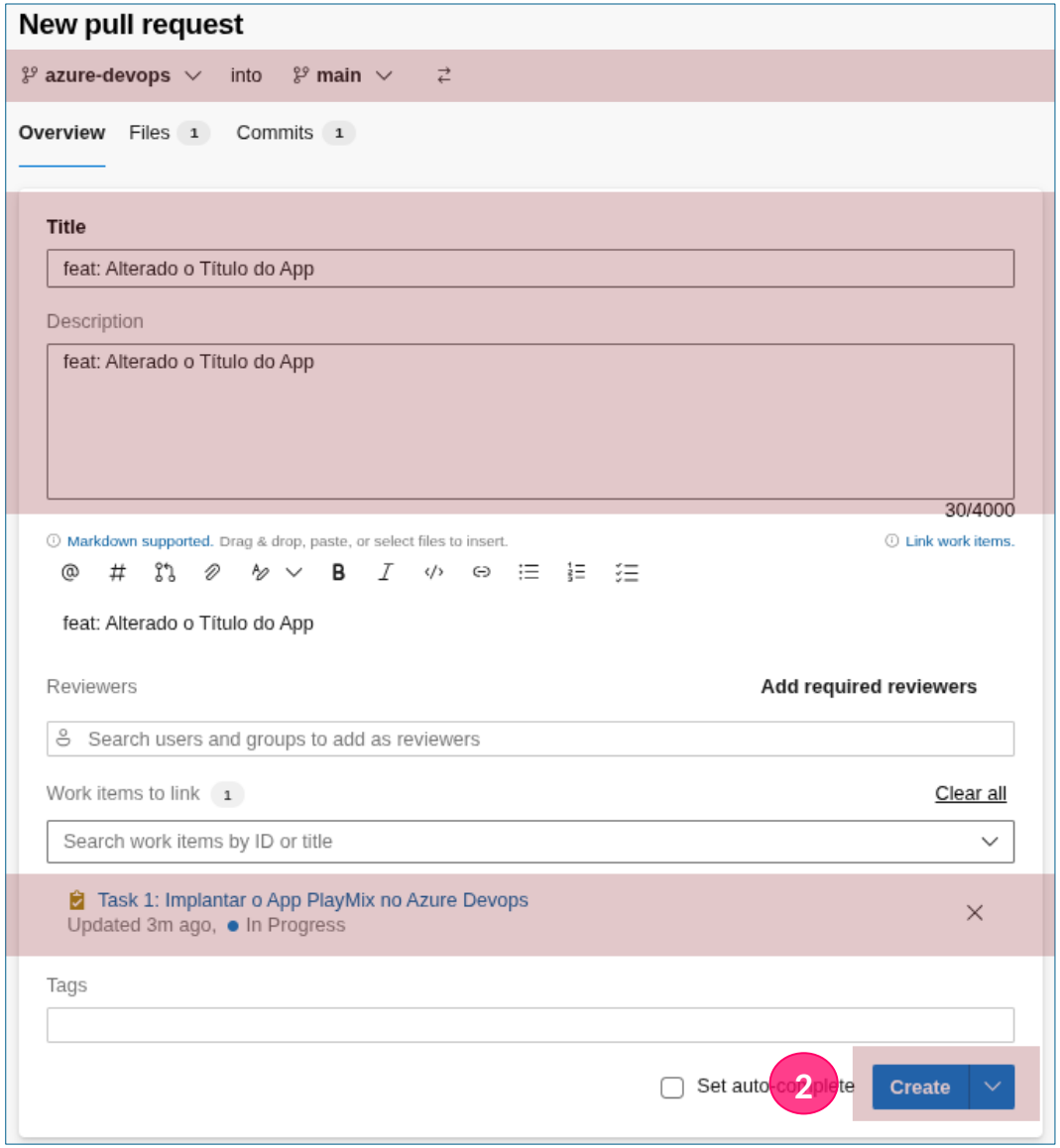
Azure Pipelines – Exercício Java x Oracle

Acesse a parte de Pull requests no Repos e clique em **Create a pull request**



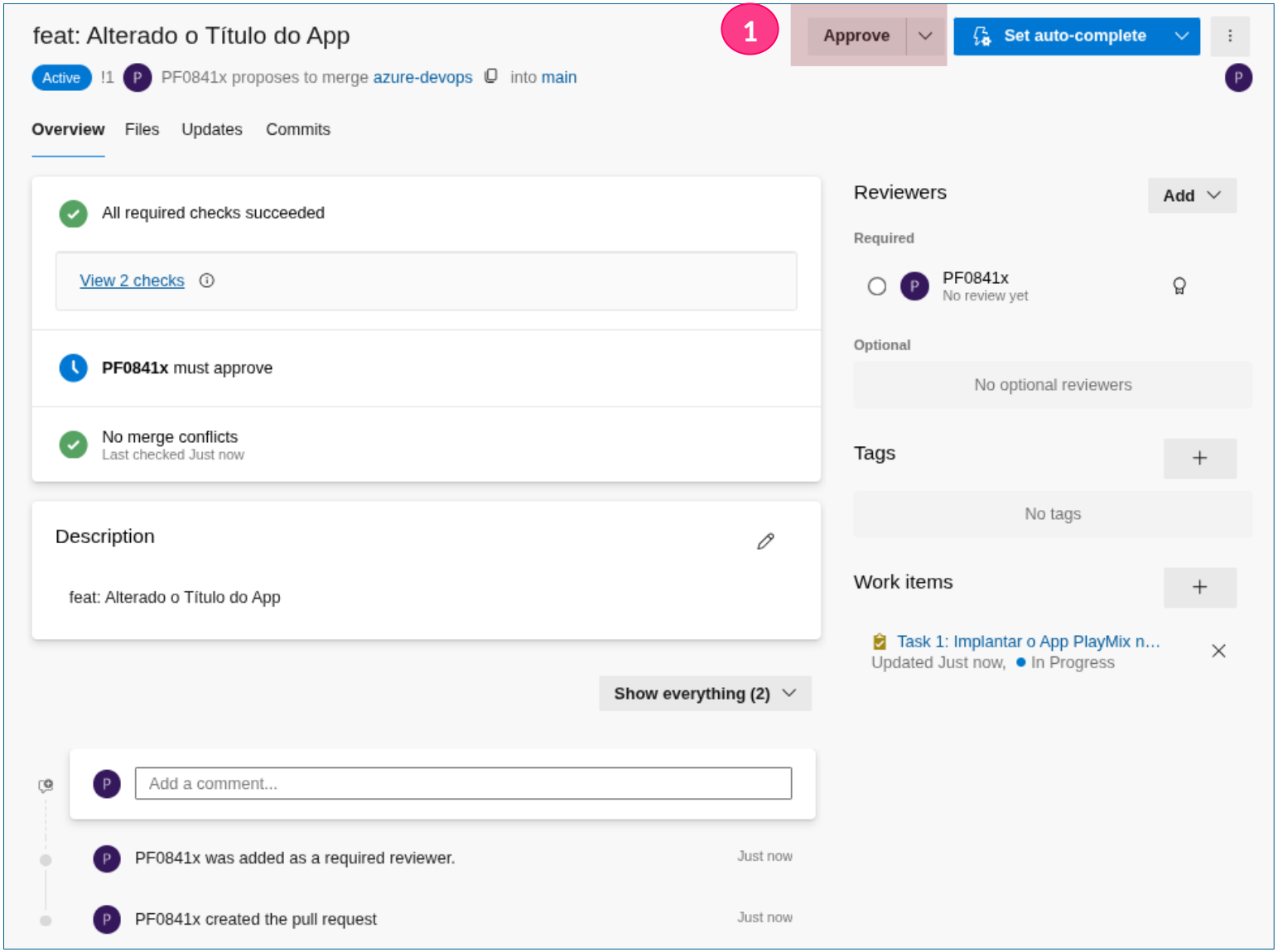
Azure Pipelines – Exercício Java x Oracle

Crie sua Pull Request



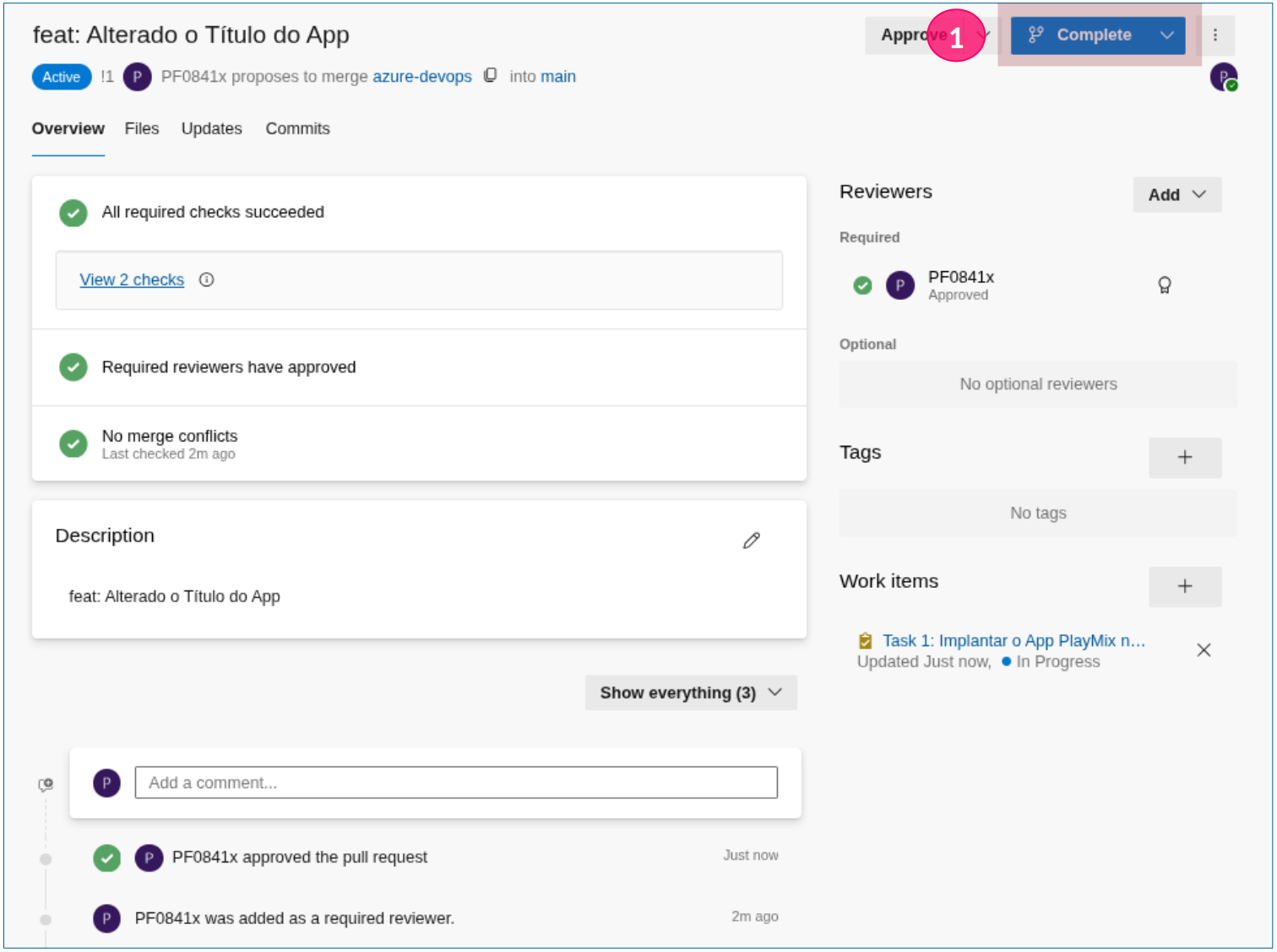
Azure Pipelines – Exercício Java x Oracle

Aprove sua Pull Request



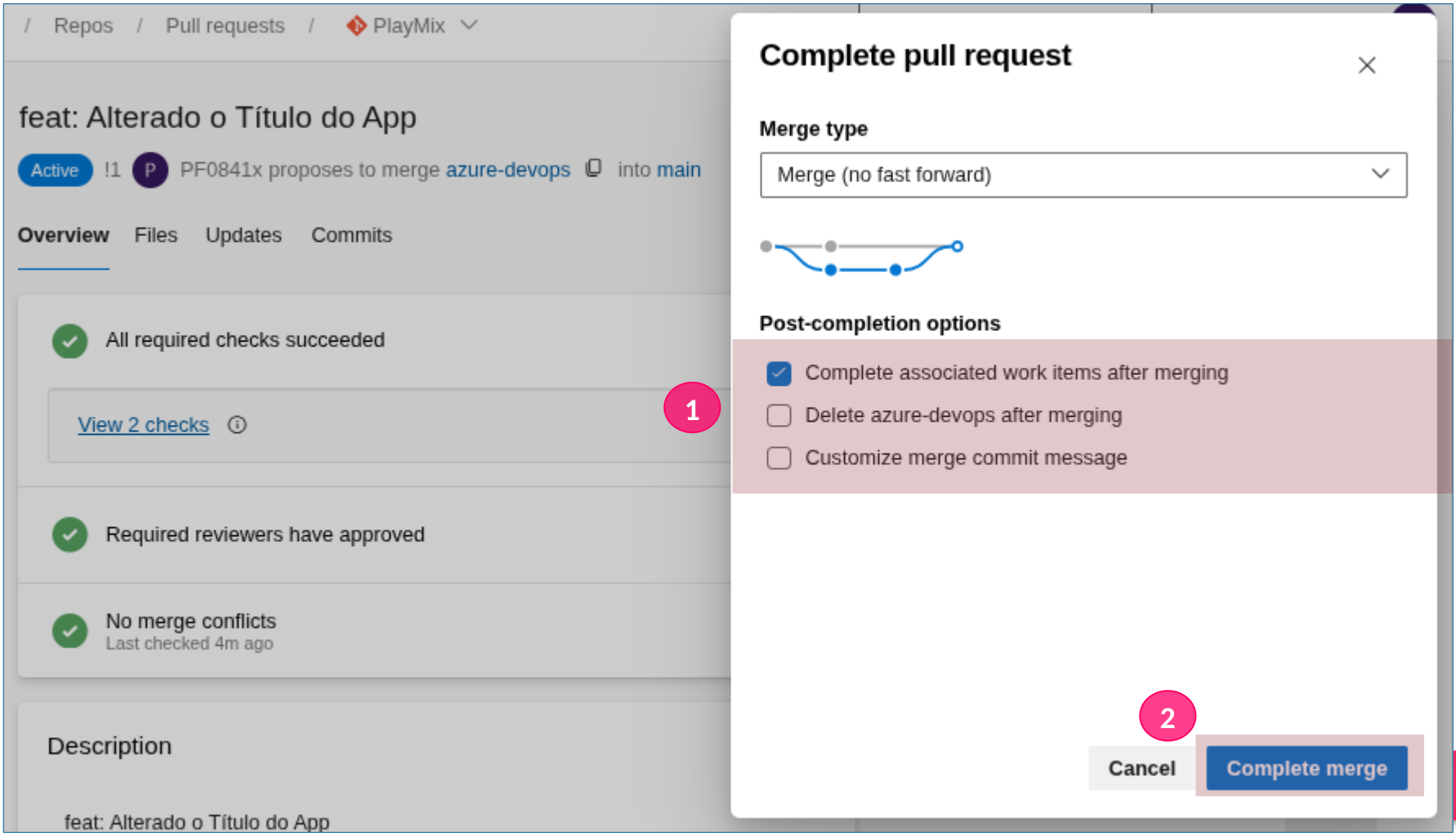
Azure Pipelines – Exercício Java x Oracle

Realize o Merge com a Branch Main



Azure Pipelines – Exercício Java x Oracle

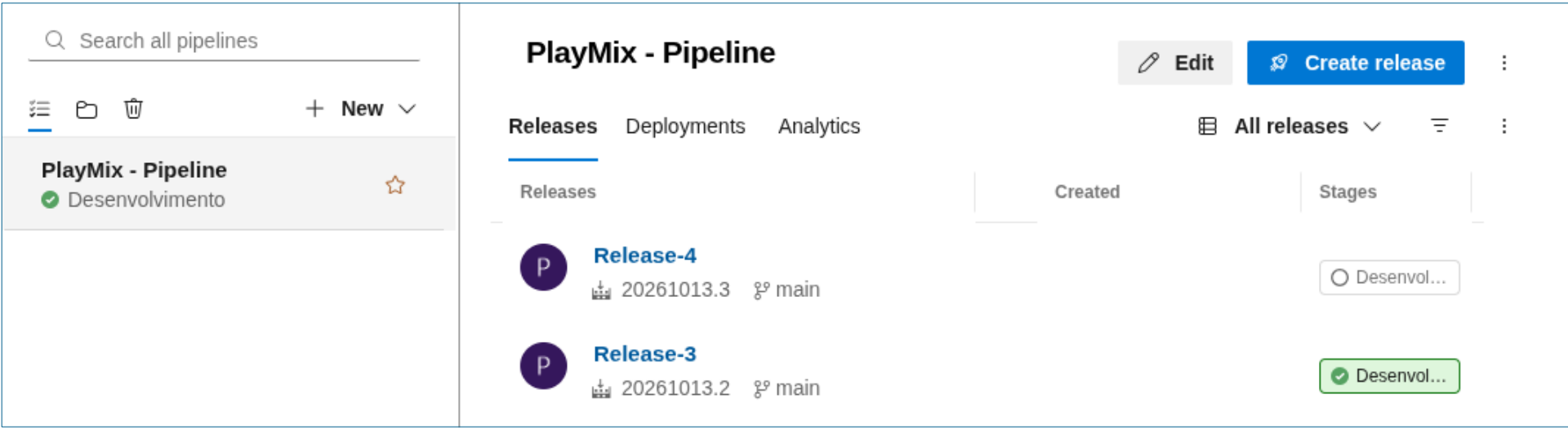
Deixe marcado para finalizar o Work Item de forma automática e desmarque a opção de deletar a Branch que criamos



Azure Pipelines – Exercício Java x Oracle

Acompanhe o CI/CD que será executado de forma automática

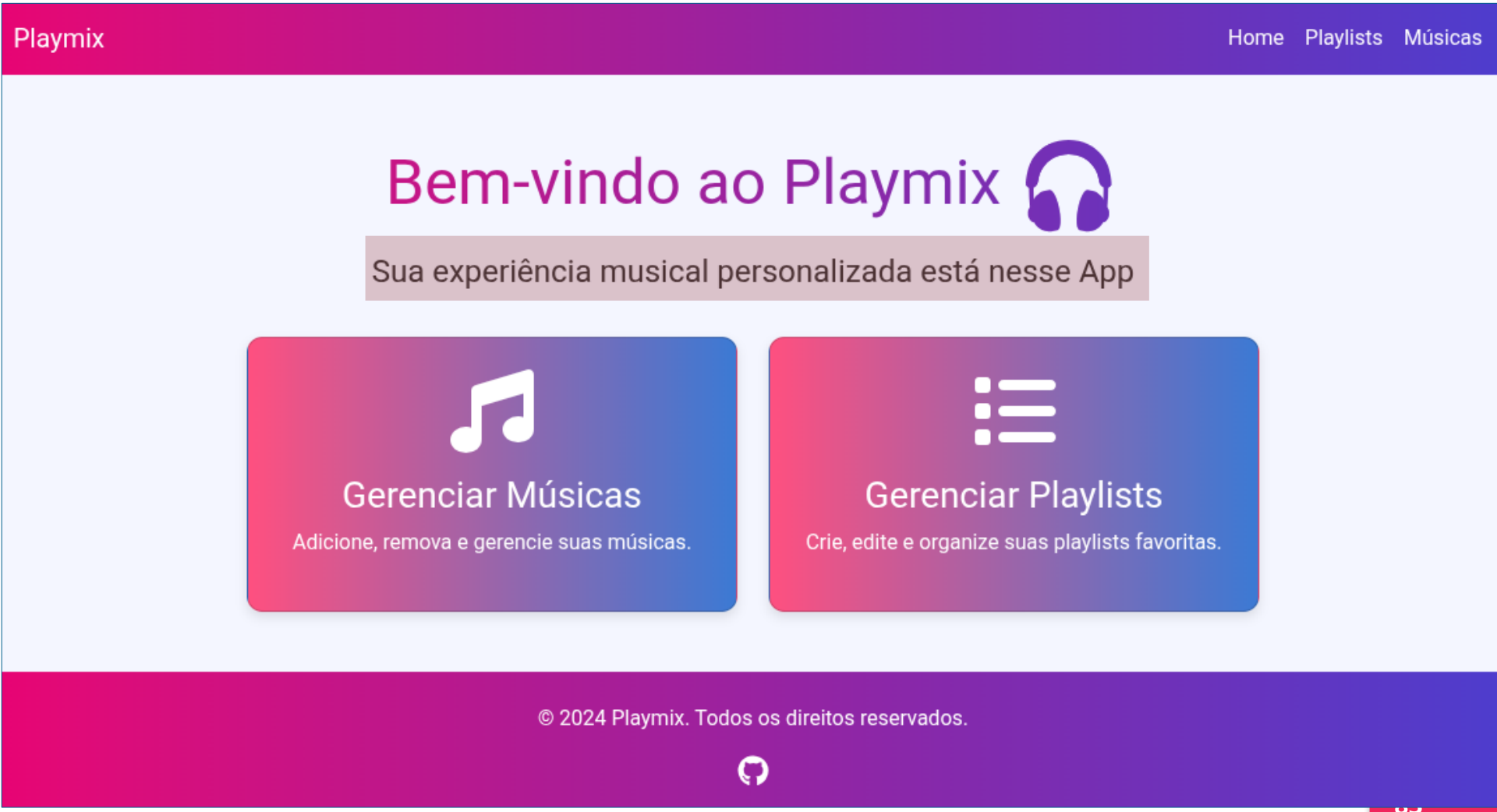




Azure Pipelines – Exercício Java x Oracle

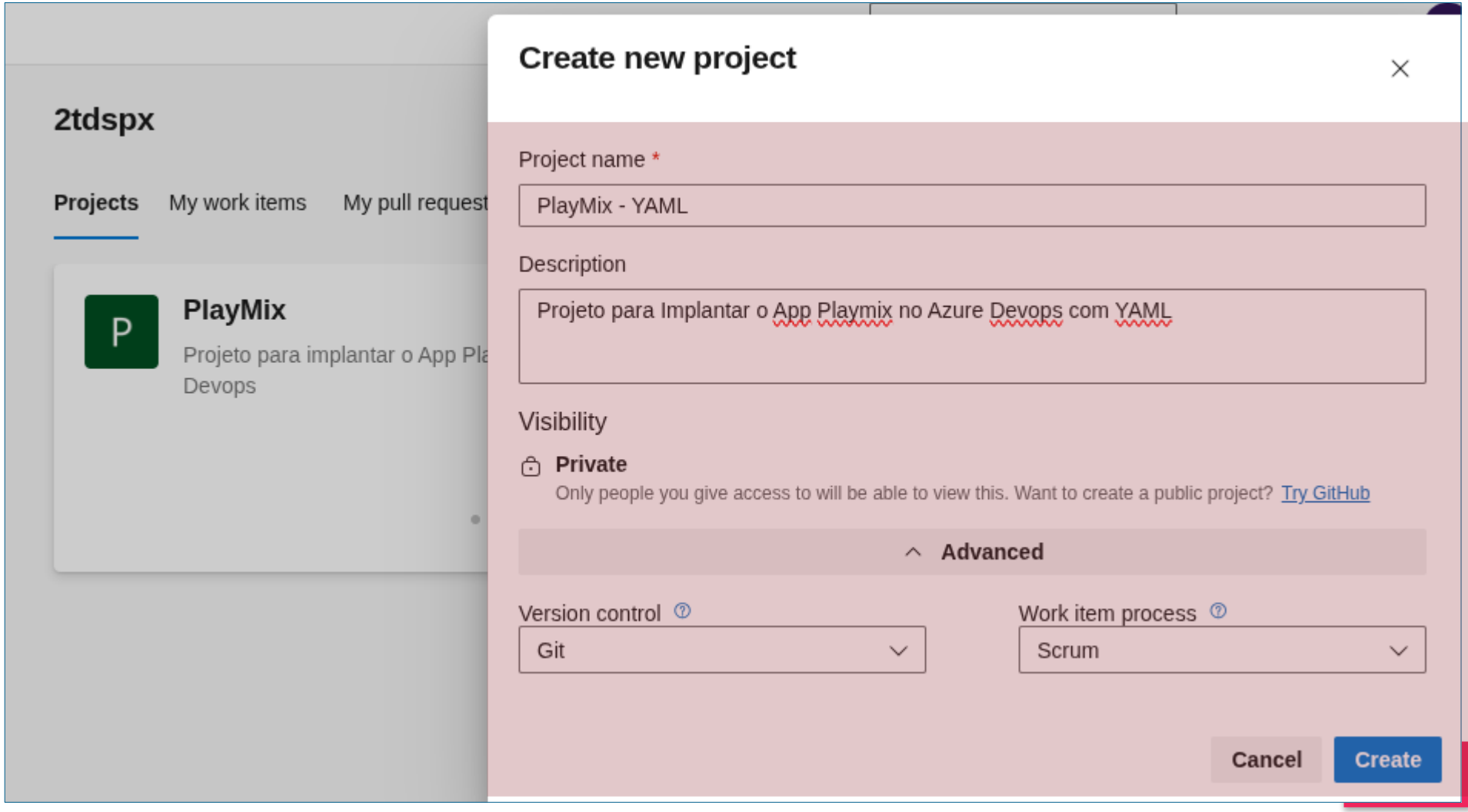
FIAP

Verifique a alteração efetuada



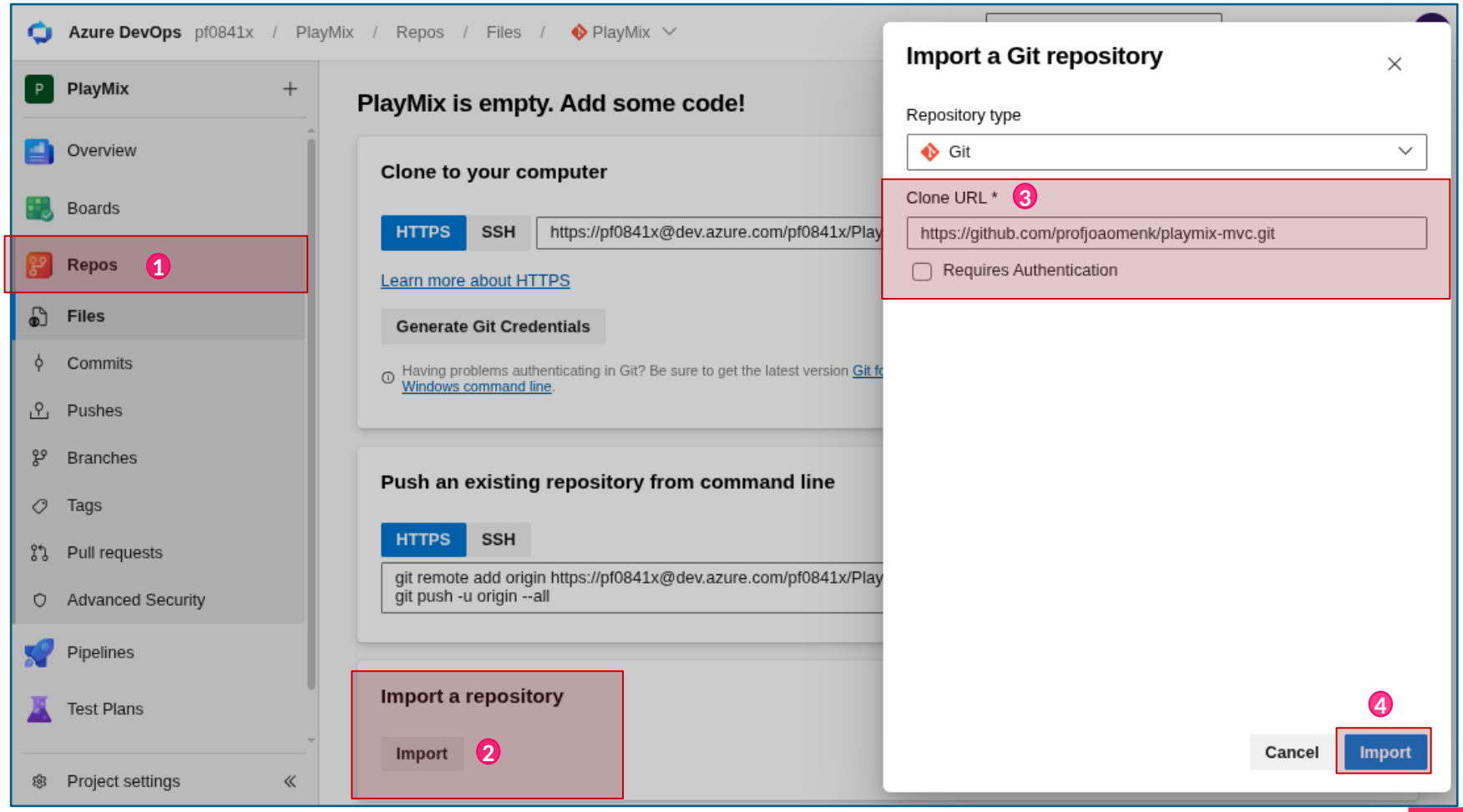
Azure Pipelines – Exercício Java x Oracle

Vamos criar outro Projeto para criar uma **Pipeline CI/CD com YAML**



Azure Pipelines – Exercício Java x Oracle

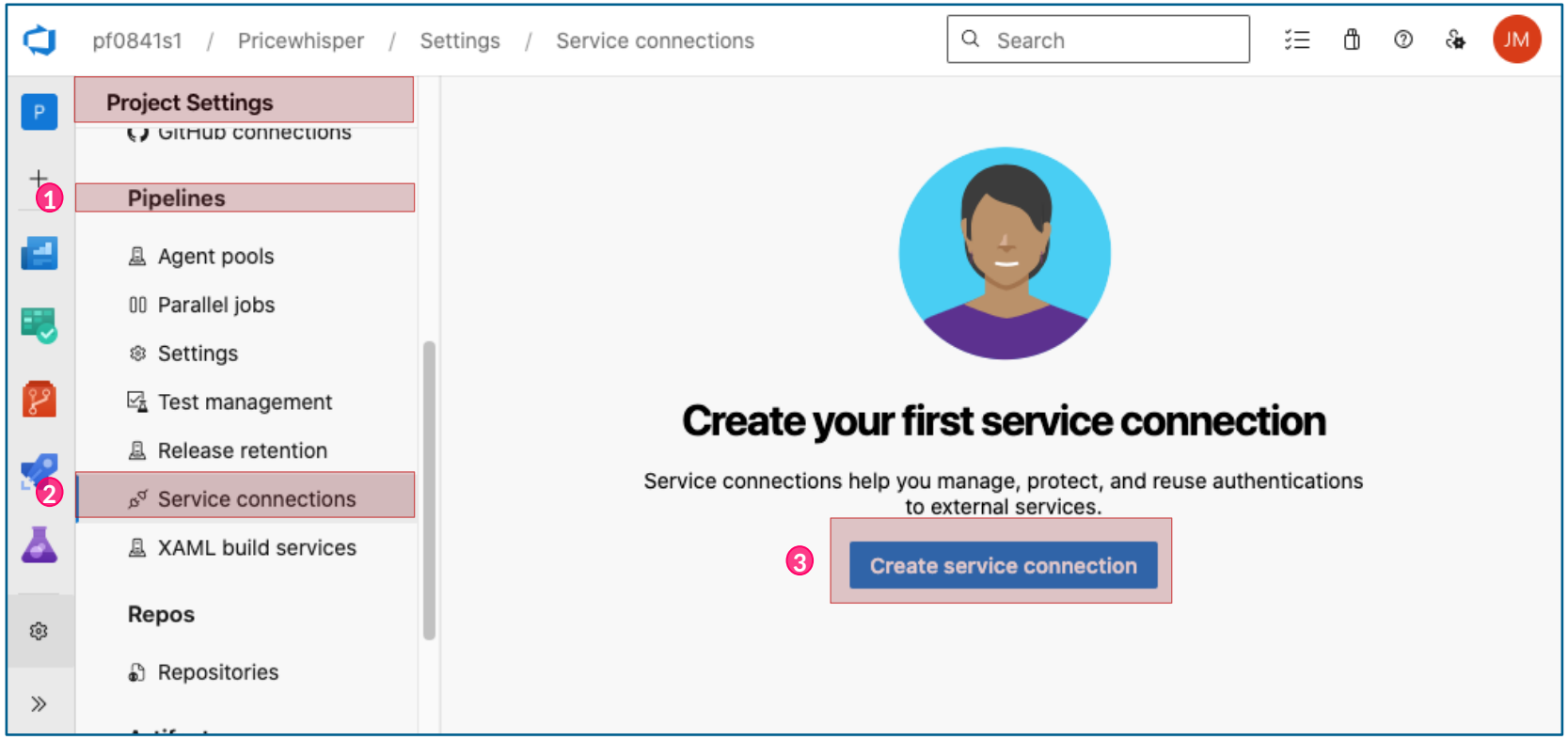
Importe o nosso Projeto do GitHub



https://github.com/profjoaomenk/playmix-mvc.git

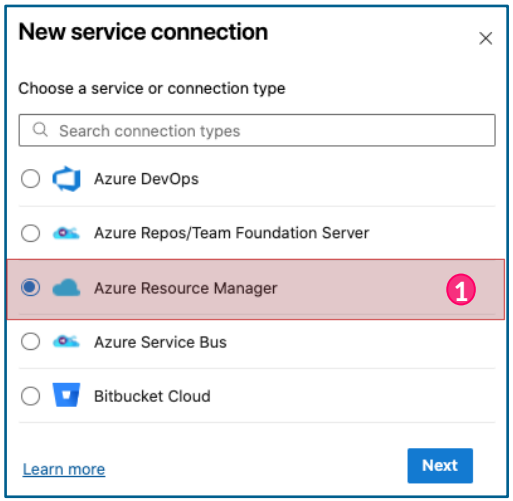
Azure Pipelines – Exercício Java x Oracle

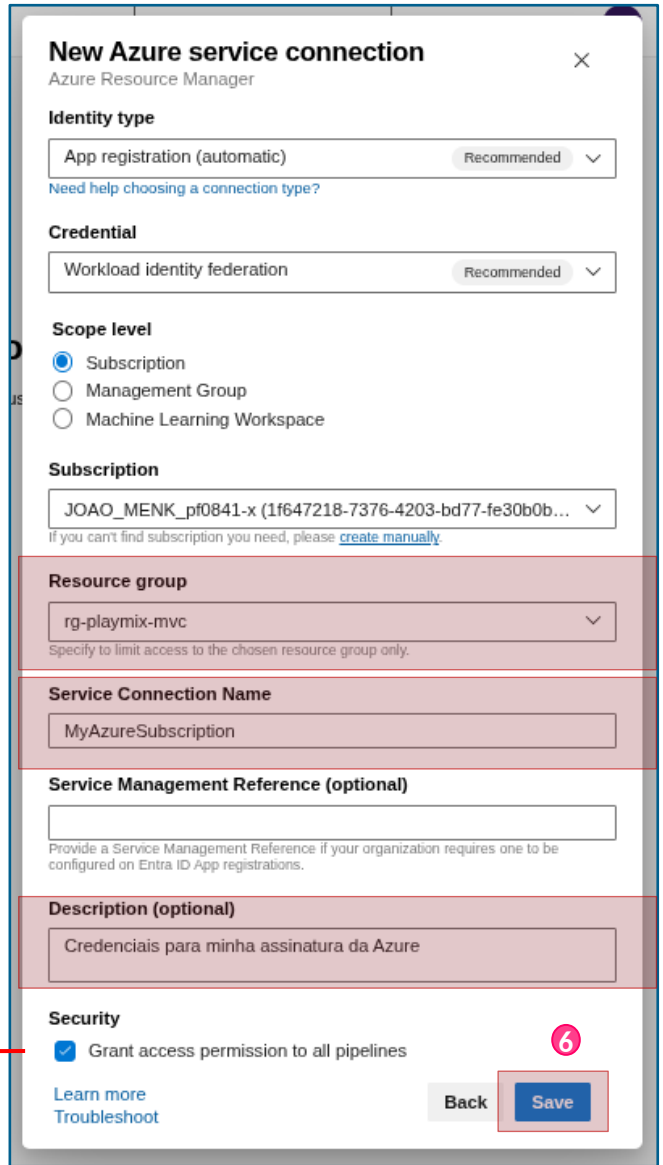
Para utilizar um aquivo YAML precisamos criar uma **Conexão de Serviço (Service Connection)** com a nossa assinatura do Azure



Azure Pipelines API .NET com Oracle

Escolha **Azure Resource Manager** como o Service connection e o configure para sua assinatura da Azure





→ **Deixe o padrão**

→ **Deixe o padrão**

→ **Nível do escopo: Subscription**

→ **Sua assinatura**

→ **Grupo de Recursos**

→ **Nome da Conexão de Serviço**

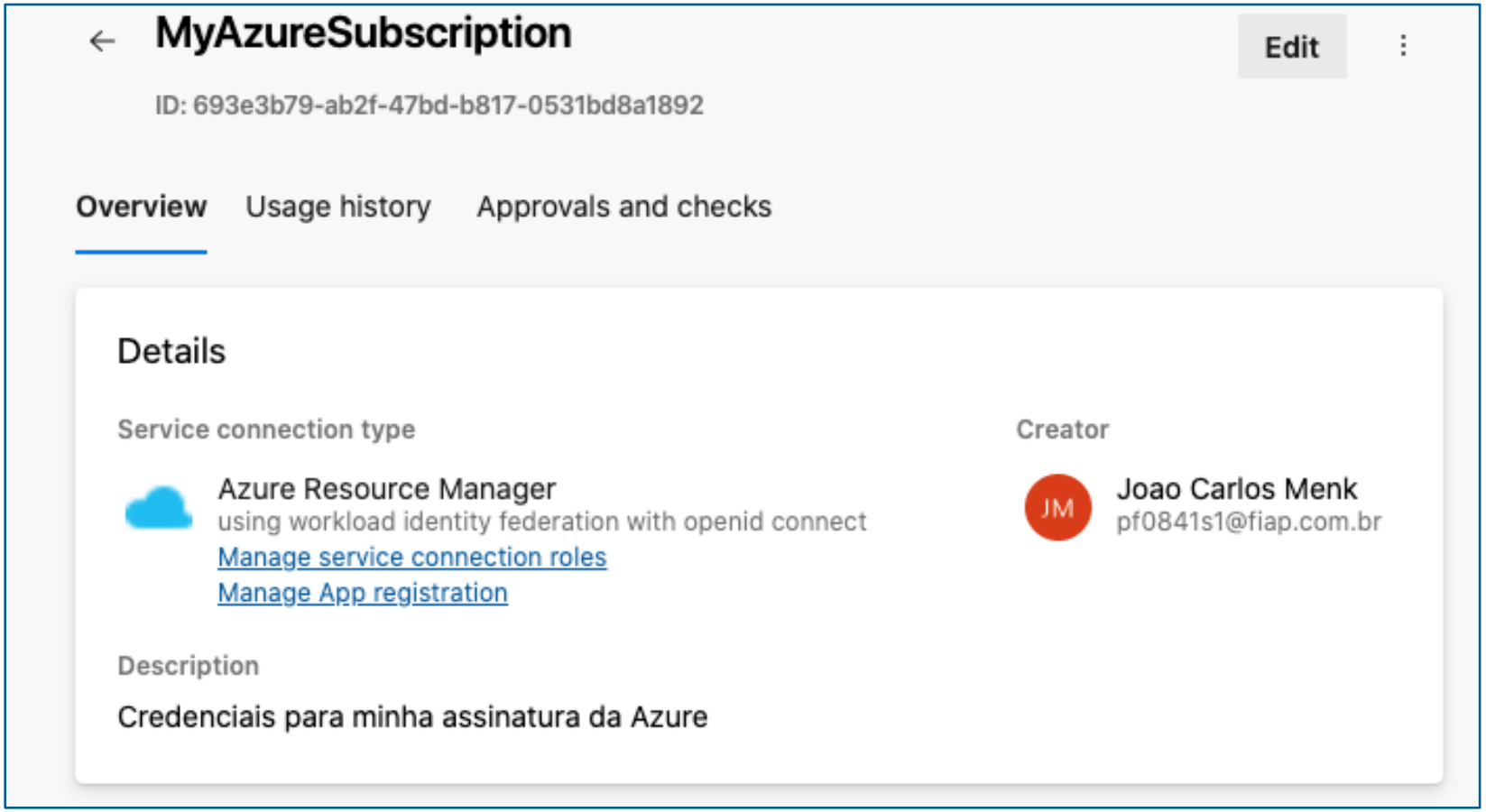
→ **Descrição**

Todas as Pipelines podem acessar esse Serviço ←

Azure Pipelines API .NET com Oracle

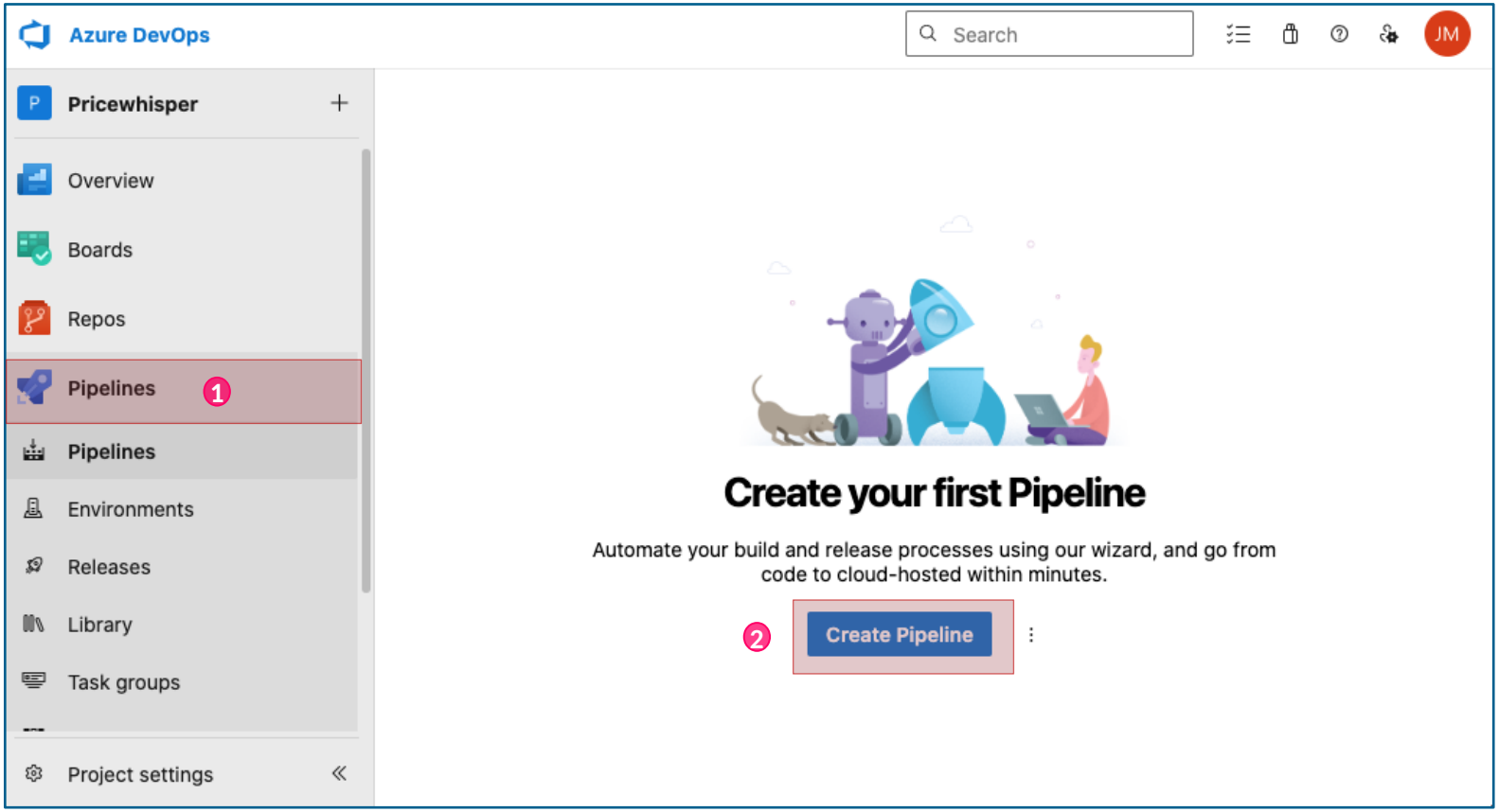
FIAP

Service connection criado



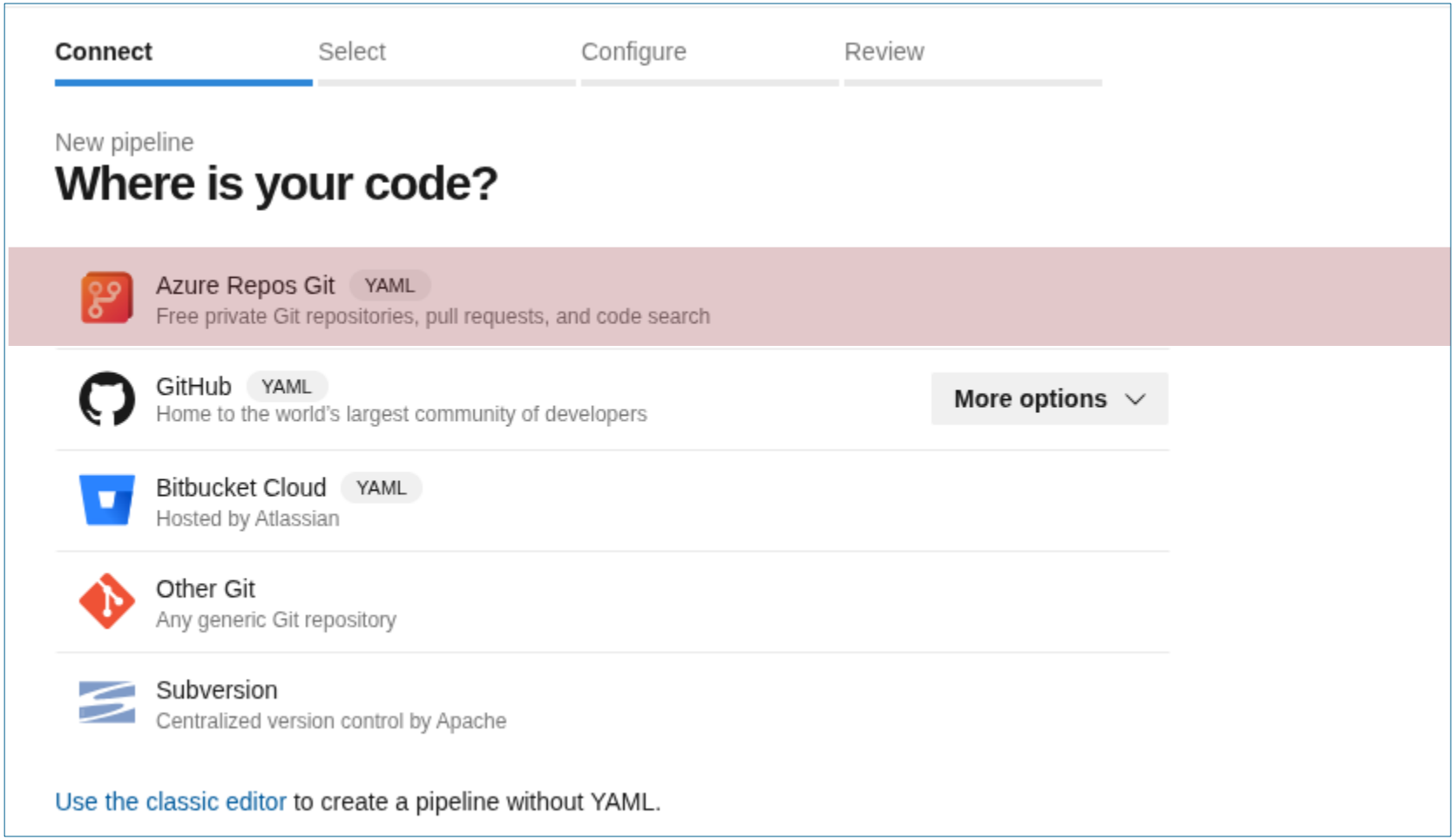
Azure Pipelines – Exercício Java x Oracle

Crie uma nova Pipeline



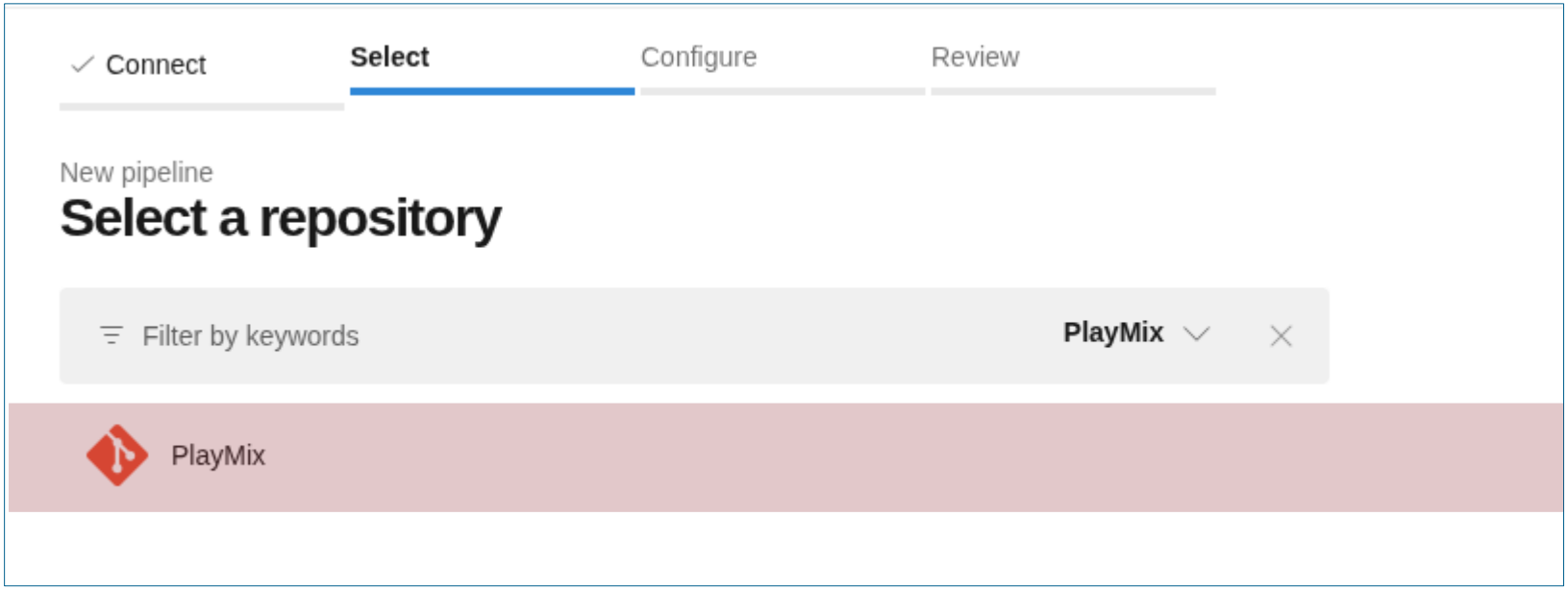
Azure Pipelines – Exercício Java x Oracle

Escolha **Azure Repos Git YAML**



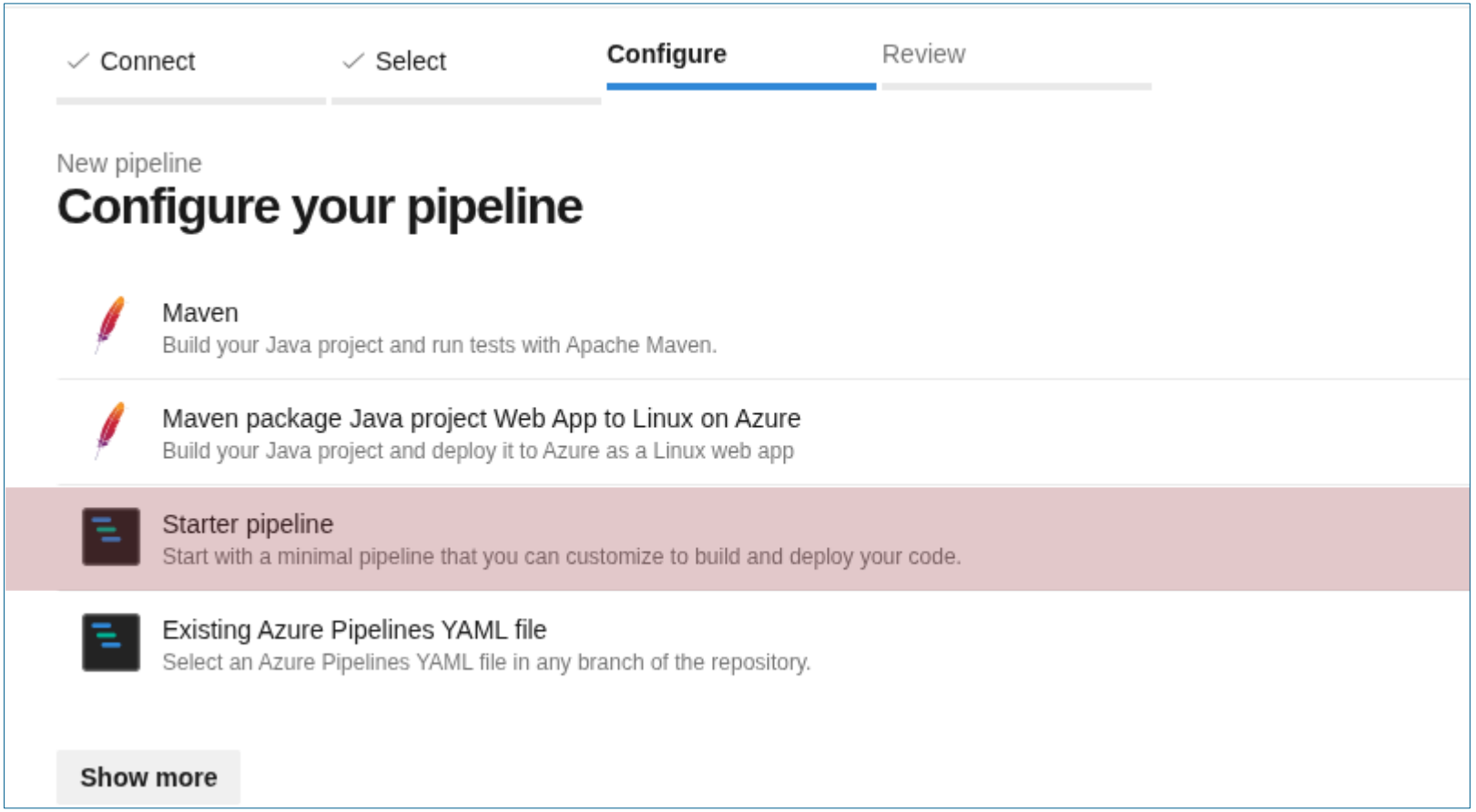
Azure Pipelines – Exercício Java x Oracle

Selecione o Repositório



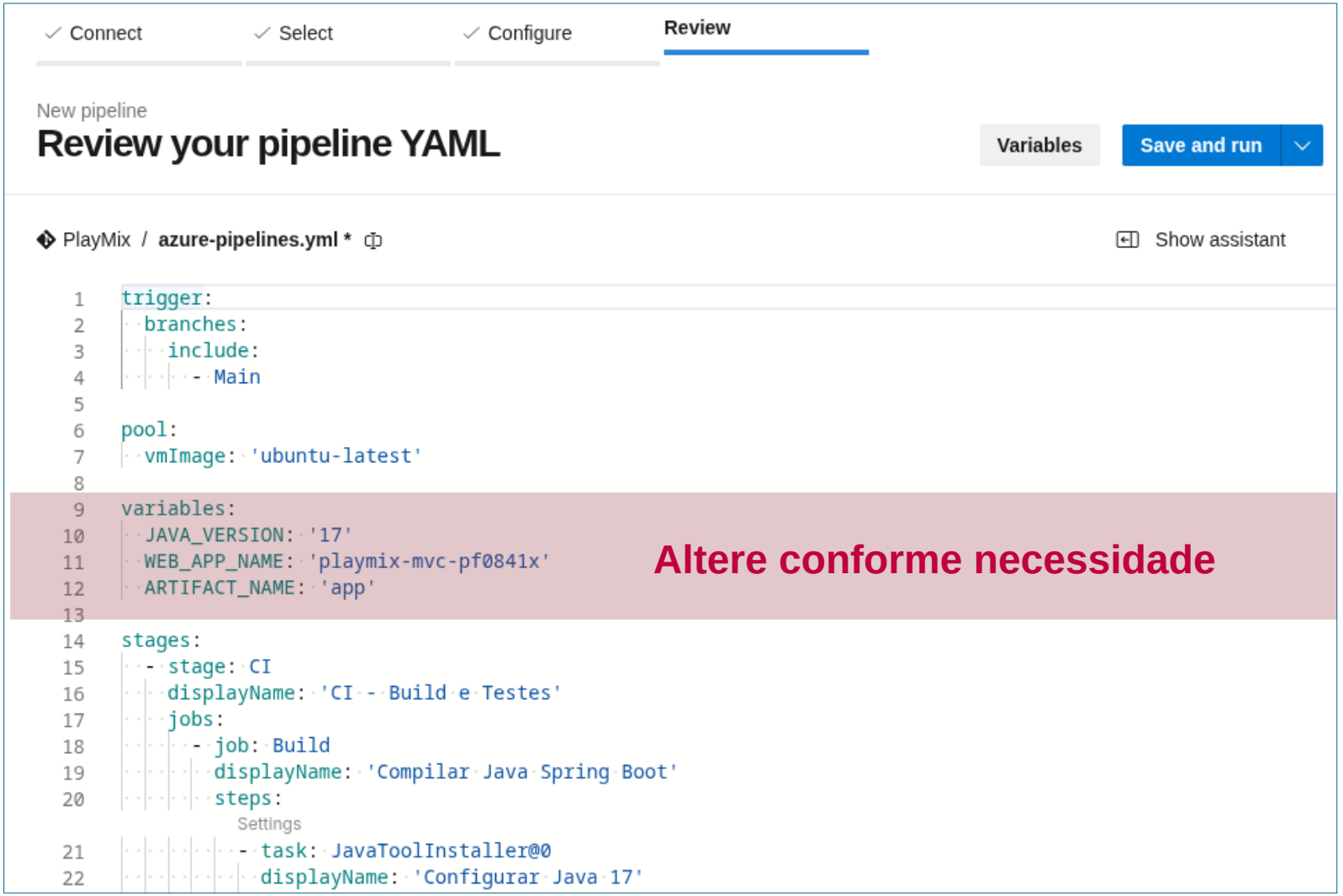
Azure Pipelines – Exercício Java x Oracle

Escolha **Starter pipeline**



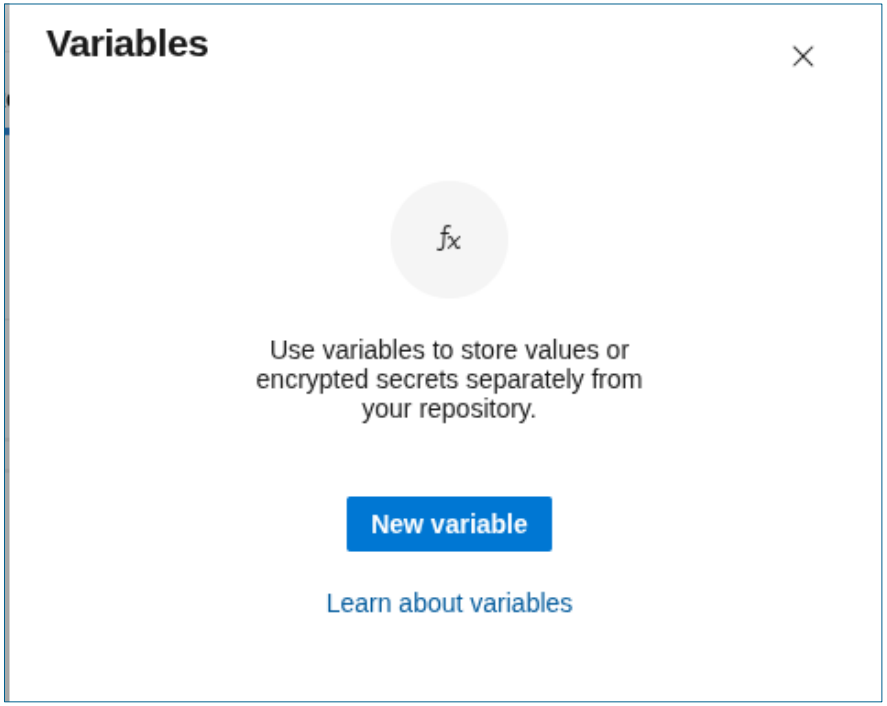
Azure Pipelines – Exercício Java x Oracle

Copie e cola o conteúdo do arquivo **Semana_19_02_Playmix-yaml.txt**



Azure Pipelines – Exercício Java x Oracle

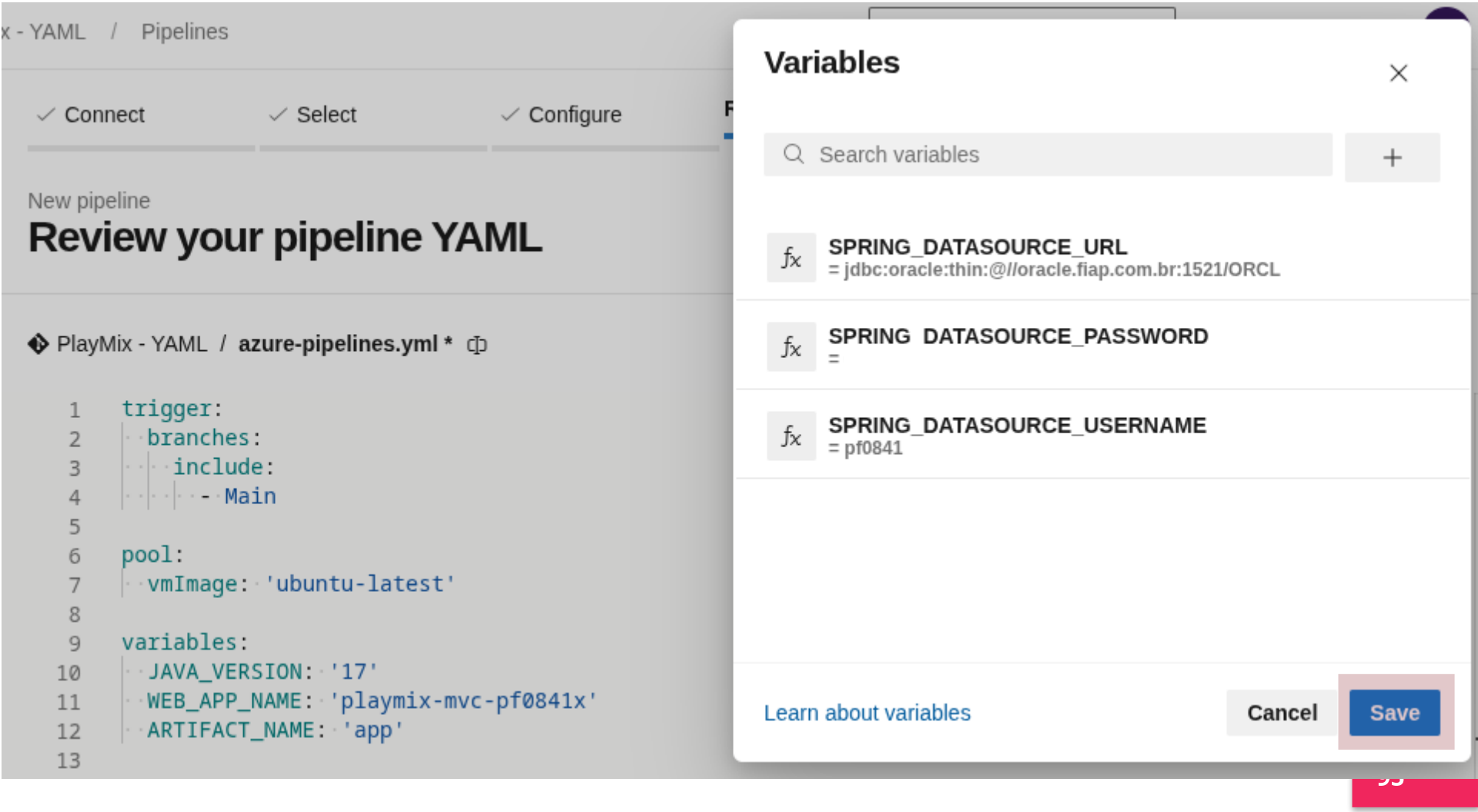
Clique em **Variables** e inclua as variáveis necessárias para o Projeto



Nome	**Valor**
SPRING_DATASOURCE_USERNAME	Usuário do banco
SPRING_DATASOURCE_PASSWORD	Senha do banco
SPRING_DATASOURCE_URL	URL JDBC do banco

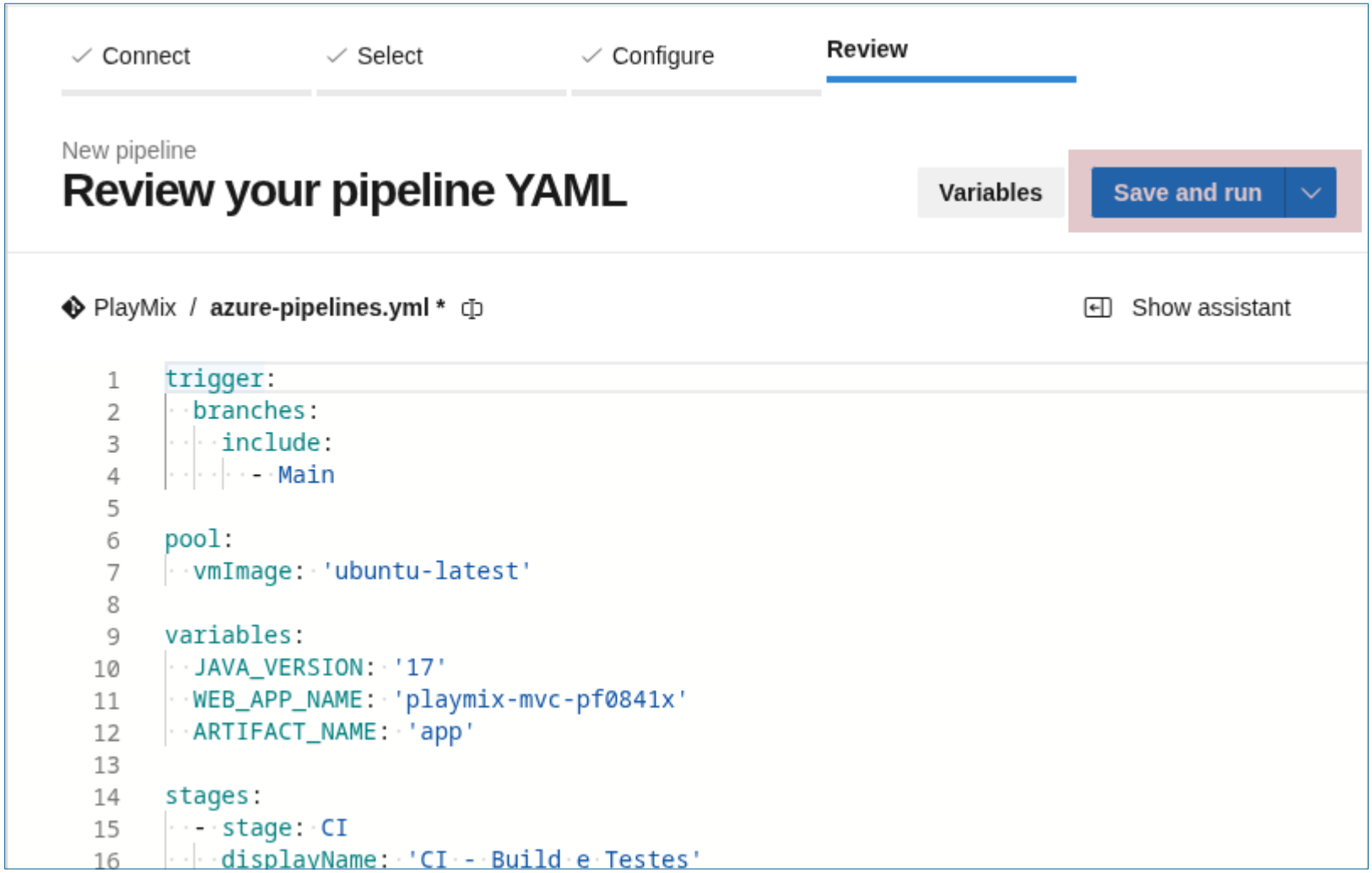
Azure Pipelines – Exercício Java x Oracle

Variáveis incluídas, clique em **Save**



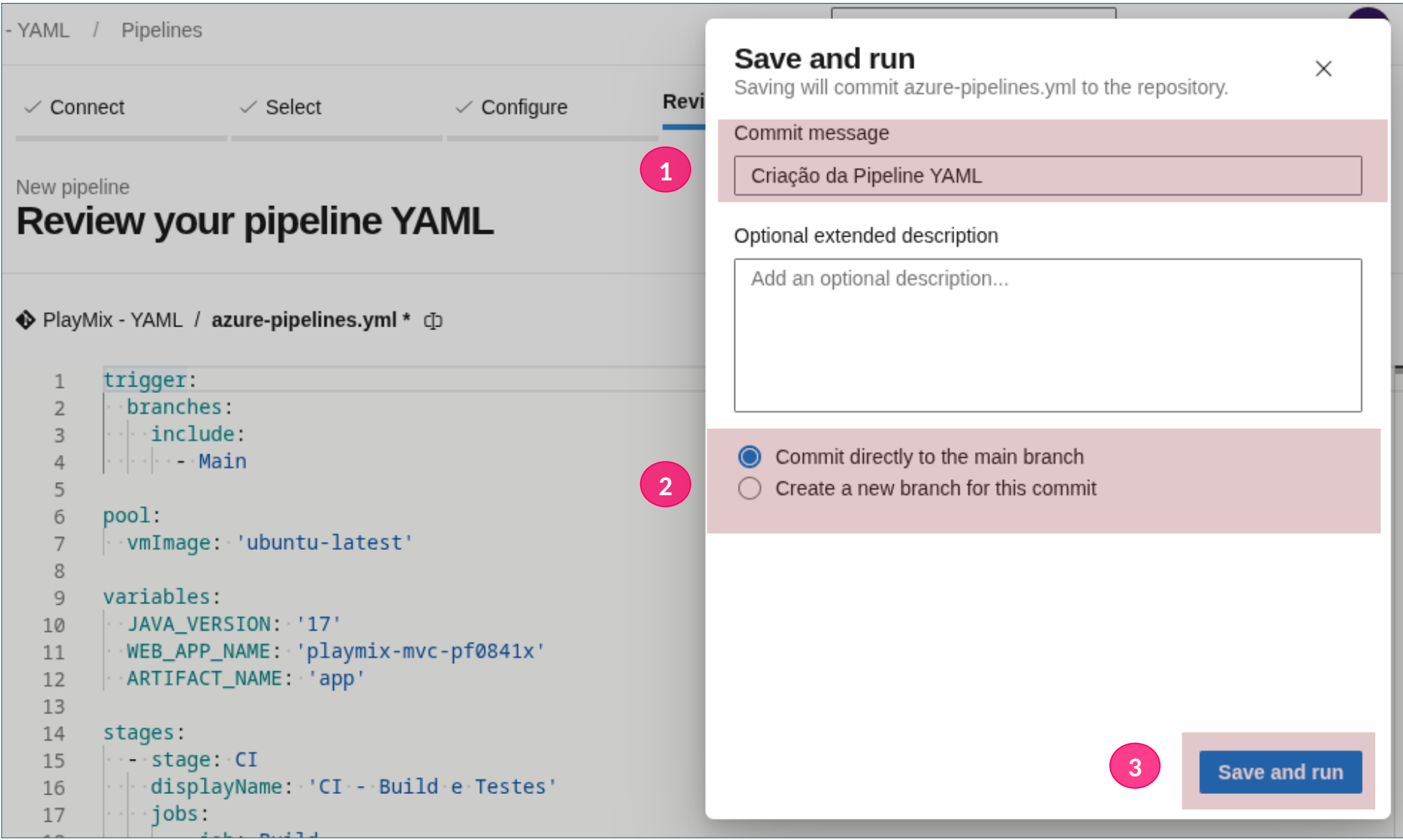
Azure Pipelines – Exercício Java x Oracle

Agora clique em Save and run



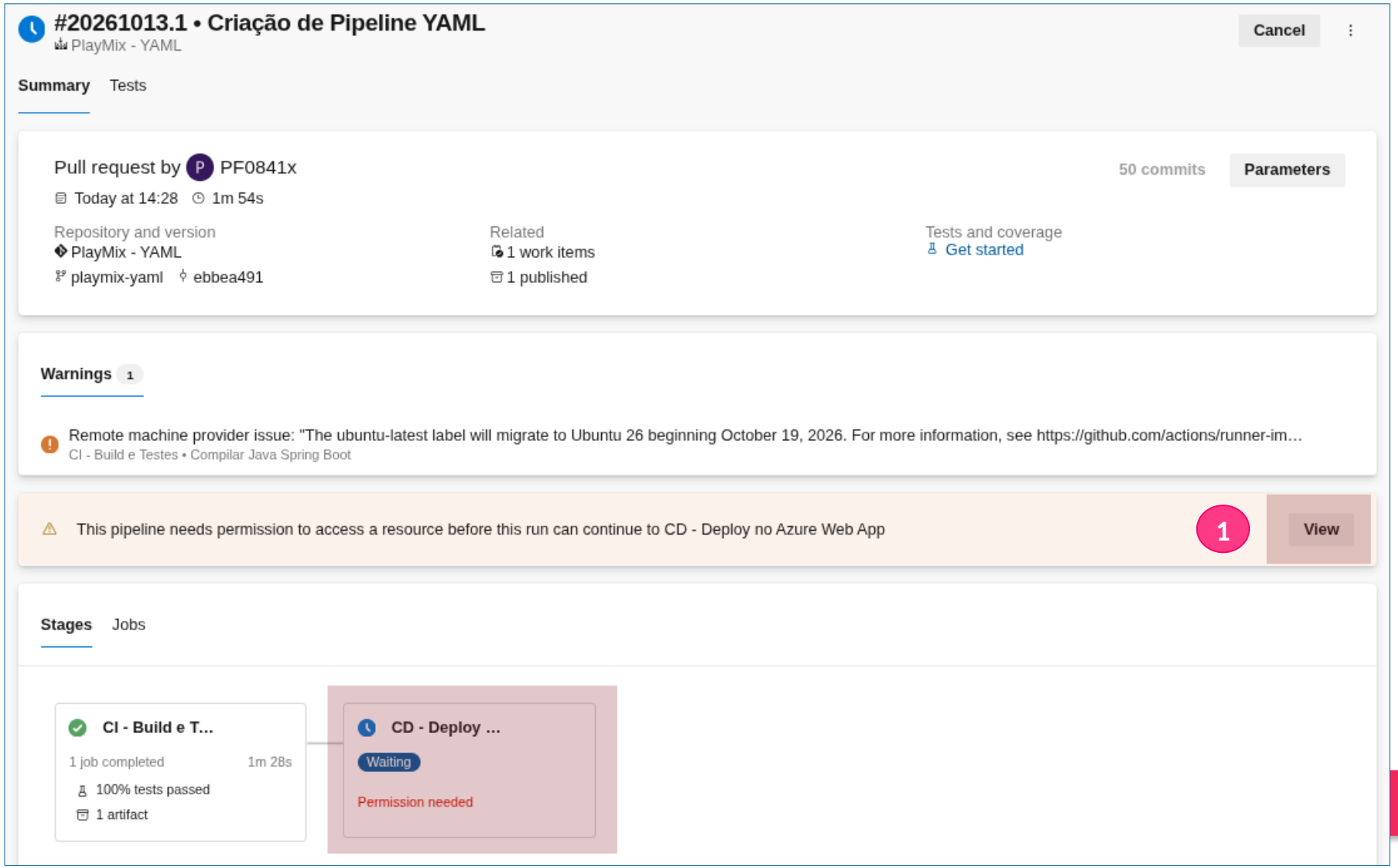
Azure Pipelines – Exercício Java x Oracle

Informe os parâmetros de execução



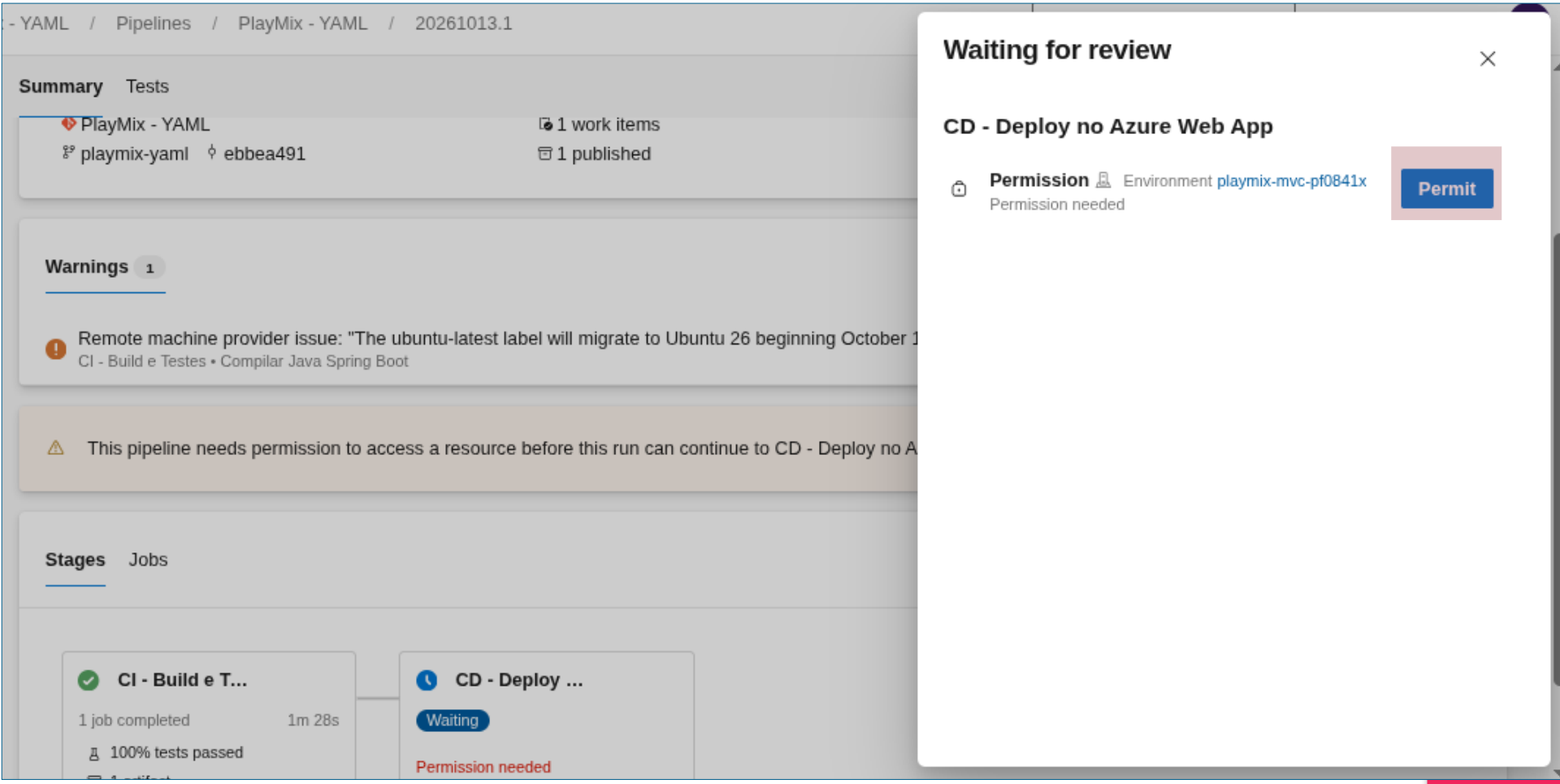
Azure Pipelines – Exercício Java x Oracle

Acompanhe a execução do CI/CD



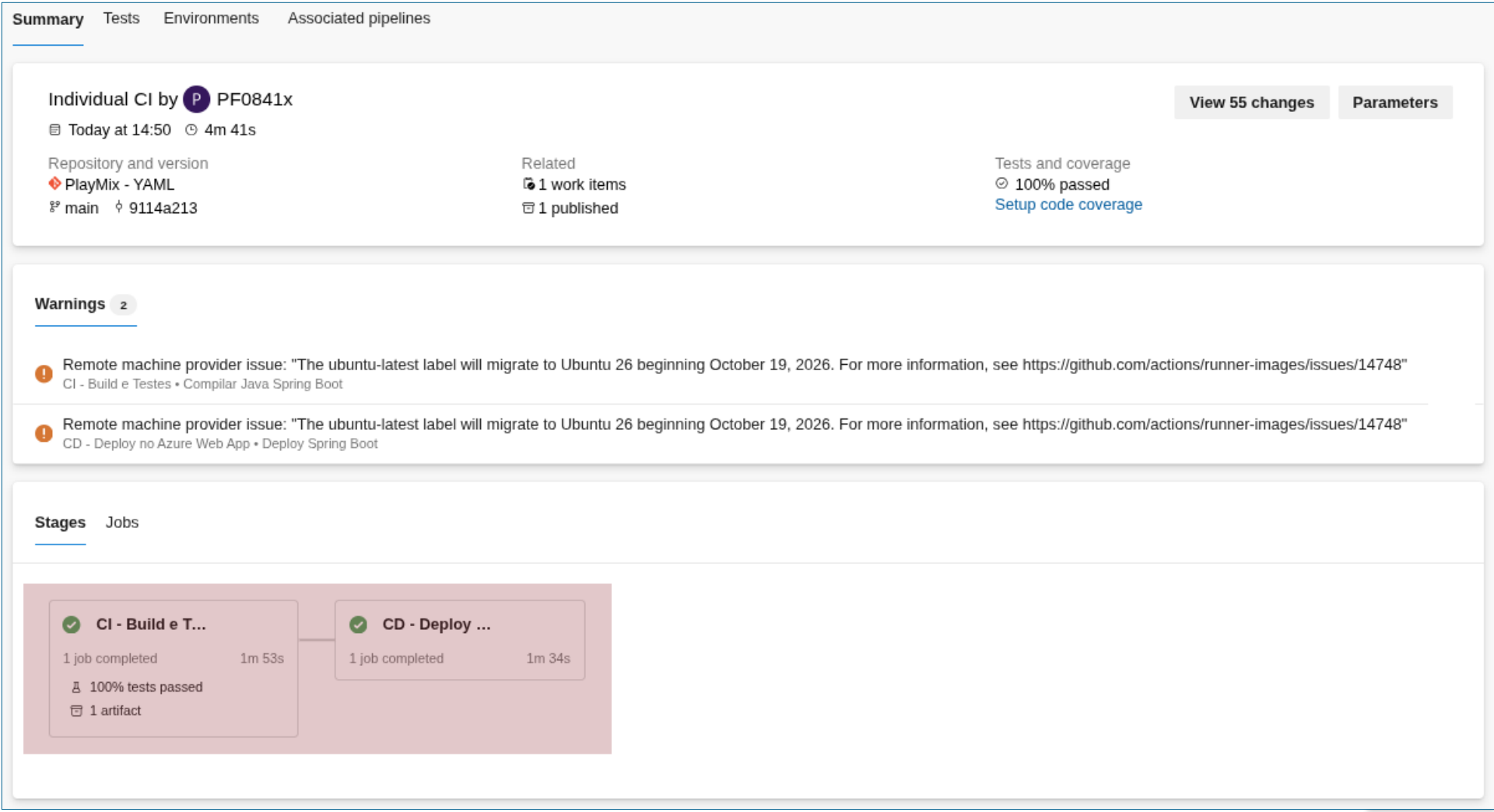
Azure Pipelines – Exercício Java x Oracle

Permita a execução do CD



Azure Pipelines – Exercício Java x Oracle

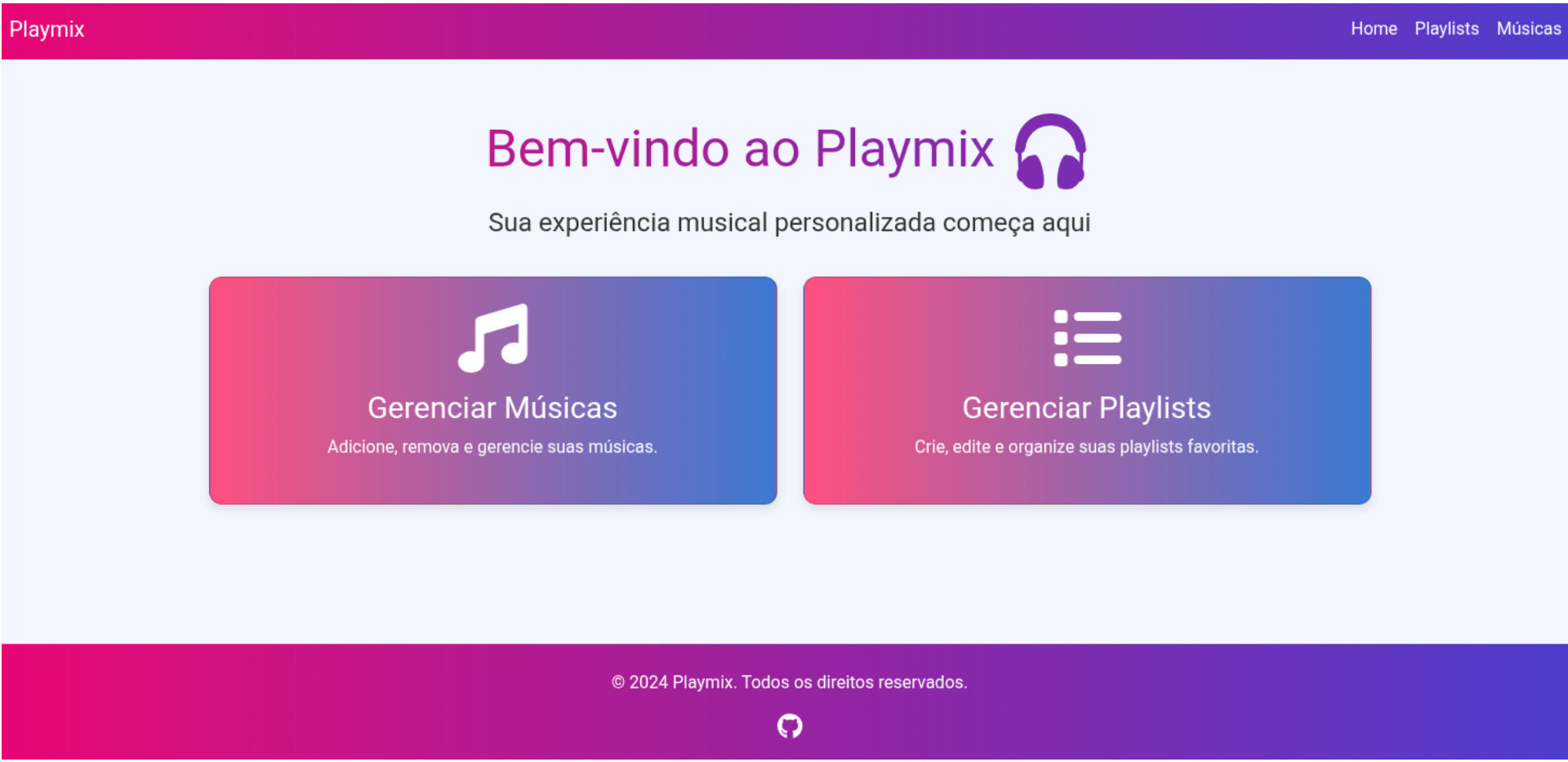
Acompanhe até o final do CI/CD



Azure Pipelines – Exercício Java x Oracle

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Verifique o App em funcionamento na nuvem



Azure Pipelines – Exercício Java x Oracle

Após os testes delete as tabelas do Oracle

```
DROP TABLE music CASCADE CONSTRAINTS;
DROP TABLE playlist CASCADE CONSTRAINTS;
DROP TABLE playlist_music CASCADE CONSTRAINTS;
```

Delete os Grupo de Recursos criados nesse exercício



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